

King's College London

Department of Informatics

**Explainable Dimensionality
Reduction for Financial Compliance:
Bridging Continuous Attribution and
Discrete Logic**

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Abstract

Unsupervised dimensionality reduction (DR) is routinely used in finance to compress market and credit variables into latent factors. Yet its adoption in regulated settings is constrained by a persistent format mismatch: explainability tools output continuous numerical attributions, whereas regulators require discrete, auditable logical rules. This thesis formalises this mismatch as the *discrete logic gap* and proposes a framework to bridge it, while confronting a fundamental methodological challenge: circularity in explainability evaluation, where surrogate-based metrics become self-confirming rather than evidential.

The core contribution is Algorithm 1: Threshold Discretisation with Shapley Attribution, which converts continuous factor scores into categorical predicates (LOW/MEDIUM/HIGH) using four thresholding strategies and explains category membership via Shapley attribution. Faithfulness is enforced through Shapley completeness verification (76.2% pass rate at $\varepsilon=0.01$) and MoRF perturbation testing, surfacing failure modes rather than producing misleading artefacts. A Python implementation (8,863 lines, 17 modules, 42 unit tests) supports six DR methods and exports 36 IF-THEN rules, audit logs, and model cards.

Evaluation across four calibrated noise regimes identifies the boundary where explanations remain reliable. Linear methods achieve surrogate R^2 of 0.865–0.958 under idealised conditions, but R^2 becomes negative (-3.1 to -3.5) under extreme noise, at which point the system explicitly flags explanations as invalid. A credit risk case study (5,000 applications) demonstrates audit-ready outputs mapped to ECB, SR 11-7, and UK Data Protection Act 2018 obligations, with fidelity transparently reported below the 0.85 threshold.

The central finding is that discretisation is compatible with faithful explainability only when faithfulness is enforced after categorisation and circularity is recognised as the primary methodological limitation requiring external validation.

Keywords: Explainable AI, Dimensionality Reduction, SHAP, Surrogate Models, Financial Regulation, Rule Generation, Threshold Discretisation, Regulatory Compliance

Originality Declaration

I verify that I am the sole author of this report, except where explicitly stated to the contrary.

The core algorithmic contribution (Algorithm 1: Threshold Discretisation with Shapley Attribution), the four-strategy discretisation framework, the semi-synthetic data generator with four noise regimes, and the regulatory compliance case study are my original work conducted under the supervision of Dr. David Watson. All external code, methods, datasets, and ideas are properly cited. No part of this work has been submitted for another degree or qualification at this or any other institution.

Signature: Bernardo Martins Lopes de Oliveira Guterres

Date: 30 March 2026

Code Submission: 17 Python source files (8,863 lines), 42 unit tests

Data Outputs: 25 files (CSVs, JSONs, PNGs) in `outputs/` directory

This dissertation is submitted in accordance with the regulations of KCL for the degree of Bachelor of Science in Computer Science. The project satisfies the BCS accreditation requirements for substantial practical computing work demonstrating appropriate technical sophistication.

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I acknowledge the open-source communities behind scikit-learn, PyTorch, SHAP, and XGBoost, whose well-documented libraries formed the technical foundation of this work. Finally, I thank the authors of the papers cited throughout this thesis for establishing the theoretical foundations upon which this project builds.

I am grateful to my family and friends for their patience and support during the intensive final months of this project. This project received no external funding and was completed independently as part of the BCS-accredited BSc Computer Science programme at KCL.

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List of Abbreviations

ACM	Association for Computing Machinery
BCS	British Computer Society
BSPR	Basic Specification and Planning Report
DCI	Disentanglement, Completeness, and Informativeness
DPA	Data Protection Act
DR	Dimensionality Reduction
EBA	European Banking Authority
ECB	European Central Bank
FA	Factor Analysis
GDPR	General Data Protection Regulation
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
ICA	Independent Component Analysis
IG	Integrated Gradients
KCL	King’s College London
LIME	Local Interpretable Model-agnostic Explanations
MAS	Monetary Authority of Singapore
MIG	Mutual Information Gap
ML	Machine Learning
MLP	Multi-Layer Perceptron
MoRF	Most Relevant First
NFR	Non-Functional Requirement
NMF	Non-negative Matrix Factorization
PCA	Principal Component Analysis
RF	Random Forest
SHAP	SHapley Additive exPlanations
SNR	Signal-to-Noise Ratio
SR 11-7	Supervisory Guidance on Model Risk Management (Federal Reserve)
SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighbor Embedding
UMAP	Uniform Manifold Approximation and Projection
XAI	Explainable Artificial Intelligence
β-TCVAE	Beta Total Correlation Variational Autoencoder

Chapter 1

Introduction

This chapter establishes the motivation, scope, and structure of the thesis. It explains why dimensionality reduction in finance creates an explainability problem (§1.1–1.3), states the research question and implemented scope (§1.4), summarises the principal contributions (§1.5), and outlines the organisation of the remaining chapters (§1.6).

1.1 Why Black-Box Models Create Regulatory and Business Risk

As predictive models grow more capable, their internal representations become less accessible to the humans responsible for acting on their outputs. Researchers now term this tension the *interpretability crisis* [3]: across medicine, criminal justice, and finance, the cost of an unexplained decision is not merely epistemic but legal and operational. Finance represents the most acute instance of this challenge. It combines the highest-dimensionality data environments in industry with the most codified external explainability obligations, the longest institutional memory for model failures, and the most immediate systemic consequences when models behave unexpectedly.

Financial institutions today face an overwhelming volume of data. A single trading decision may involve thousands of variables: stock prices, economic indicators, social media sentiment, and more. As the number of variables grows, traditional statistical models become less accurate and more computationally expensive, a phenomenon researchers call the *curse of dimensionality*. To manage this complexity, financial analysts increasingly rely on dimensionality reduction (DR), mathematical techniques that compress high-dimensional data into simpler representations whilst preserving essential information. These methods have become critical infrastructure for portfolio management, risk assessment, and algorithmic trading [1].

However, many modern DR techniques, particularly deep learning architectures and manifold learning methods, operate as opaque ‘black boxes’ that offer no human-understandable justification for their outputs [22]. This creates fundamental tension in financial services, where portfolio managers, risk officers, and regulators require not merely accurate predictions but comprehensible rationales that can inform strategic decisions and

satisfy compliance obligations.

Regulatory frameworks now explicitly require explainability in financial ML systems. Four distinct frameworks shape the obligations addressed in this thesis.

The **European Central Bank’s Guide on Internal Models** [13] (EU banking supervision) mandates that financial institutions ensure ML model explainability through specific tools and document all hyperparameter choices, including dimensionality reduction parameters. The **Federal Reserve’s SR 11-7 guidance** [6] (US model risk management) defines models as ‘quantitative methods that process input data into quantitative estimates,’ requiring conceptual soundness and permitting compensating controls when full validation is infeasible.

GDPR Article 22 [14] (EU data protection) establishes individual rights around automated decision-making, requiring both global explanations (how the algorithm works) and local explanations (why this specific decision was made). In the United Kingdom, **Section 14 of the Data Protection Act 2018 (DPA 2018)** transposes equivalent obligations into domestic law, extending these individual rights to UK data subjects following Brexit.

These four frameworks are jurisdictionally distinct: ECB guidance applies to EU-supervised banking institutions; SR 11-7 to US Federal Reserve-regulated firms; GDPR to processing of EU residents’ data; and DPA 2018 to UK residents’ data. A system deployed at a UK lender must satisfy DPA 2018 and may also need to satisfy GDPR for EU customers (making both simultaneously relevant).

Recent implementations confirm that regulatory compliance materially affects ML deployment timelines in banking [11].

1.2 Dimensionality Reduction as Critical Financial Infrastructure

Dimensionality reduction has shifted from a peripheral preprocessing step to critical financial infrastructure. In portfolio management, PCA typically captures 70–80% of variance via three principal components corresponding to economically interpretable ‘level,’ ‘slope,’ and ‘curvature’ factors [21]. Applications now span credit risk modelling, systemic monitoring, and derivatives pricing, making DR a load-bearing component whose interpretability and governance properties directly affect whether models can be challenged, validated, and approved in regulated settings [1].

The commercial and regulatory stakes of DR adoption are substantial. Under Basel III and IV capital adequacy frameworks, risk-weighted asset calculations that incorporate DR-based factor models must be demonstrably robust to supervisory stress scenarios. A model that cannot explain its factor structure risks rejection in the ECB’s internal model

approval process [13], delaying or blocking capital relief worth hundreds of millions of euros. SR 11-7 [6] establishes that models supporting material decisions require “effective challenge” by independent validators, a requirement that is practically impossible to satisfy when the model’s internal representations are opaque. These constraints mean that the value unlocked by DR in finance is contingent on solving its explainability problem: without interpretable outputs, DR methods cannot clear the regulatory approval gates that govern their deployment at scale.

Despite this pervasive utility, dimensionality reduction techniques accumulate what this thesis terms *explainability debt* (by analogy with technical debt): the progressive obscuring of feature contributions through successive transformations. PCA produces linear combinations that, whilst mathematically optimal for variance preservation, lack inherent business semantics. Non-linear techniques such as t-SNE and UMAP exacerbate this problem: t-SNE [31] preserves local similarities but distorts global structures; UMAP [24] offers superior scalability but neither technique produces outputs amenable to direct human interpretation without additional explanation layers. The next section examines who needs these explanation layers and what form those explanations must take in practice.

1.3 The Discrete Logic Gap: From Continuous Scores to Business Rules

Three distinct customer stakeholder groups interact with DR outputs in financial institutions, and each requires a categorically different information format. *Credit committee members* make binary approval decisions: a continuous creditworthiness score of 0.63 is useless without a threshold rule that maps it to **APPROVE** or **REFER**. *Portfolio risk managers* monitor factor exposures across books: they need interpretable IF-THEN rules that translate latent factor positions into risk categories (**HIGH_RISK**, **MEDIUM_RISK**, **LOW_RISK**) aligned with the firm’s internal risk appetite framework. *Regulatory supervisors* conducting model reviews require audit-ready documentation (model cards, validation reports, and applicant-level explanations) formatted to match the categorical structures mandated by ECB, SR 11-7, GDPR Article 22, and the UK Data Protection Act 2018. Existing XAI methods produce continuous attribution values that serve none of these stakeholders without additional post-processing; closing that gap is the central engineering challenge this thesis addresses.

The core problem this thesis addresses can be stated simply: *existing explainability methods produce numbers, but financial practitioners need rules*. Consider the analogy of a thermometer versus a thermostat: a thermometer reports that the temperature is 21.7°C, but a thermostat converts that reading into an actionable decision: *turn the heating OFF*. Similarly, a risk officer cannot act on a SHAP value of +0.47 for **Revenue_Growth**; they

need a categorical verdict: *this company is a growth stock* or *debt levels are concerning*. The gap between continuous attribution scores and discrete business rules is what this thesis terms the **discrete logic gap**.

A systematic review of explainable AI in finance reveals that SHAP, LIME, and attention mechanisms have gained momentum in credit scoring and fraud detection [3]. However, a fundamental gap persists between the continuous numerical outputs these methods produce and the discrete logical rules that business stakeholders require. Financial practitioners require actionable rules of the form: **IF Revenue_Growth is HIGH and Debt_Ratio is LOW, THEN assign to Growth Portfolio**. Existing methods provide continuous attribution values: SHAP quantifies marginal contributions, Integrated Gradients compute path integrals, LIME fits local linear approximations; yet none directly produce the categorical, rule-based explanations that align with human reasoning and regulatory documentation requirements.

Financial data characteristics amplify this discrete logic gap [3]. Non-stationarity destabilises explanations over time, with feature importance rankings inverting between market regimes. Fat-tailed distributions mean explanations calibrated on normal conditions may mislead during tail events. Temporal dependencies violate the feature independence assumptions underlying SHAP and LIME. These challenges require an approach that combines rigorous mathematical foundations with domain-specific threshold methods and comprehensive validation.

1.4 Problem Statement and Research Objectives

Despite widespread use of dimensionality reduction in financial modelling, its deployment in regulated decision-making remains constrained by a structural obstacle: existing explainability methods produce continuous numerical outputs, while financial governance, audit, and regulatory processes require discrete, rule-based logic. This mismatch prevents unsupervised models from being documented, challenged, and approved in practice, even when attribution methods are available.

The core problem addressed in this dissertation is therefore not a lack of explainability, but a format-level incompatibility between how machine-learning models explain themselves and how financial institutions must justify decisions. Naïve discretisation of explanation outputs risks undermining faithfulness, creating explanations that are readable but misleading. A principled resolution requires both systematic discretisation and explicit enforcement of explanatory validity.

The primary objective of this research is to answer the following question:

How can continuous outputs from unsupervised dimensionality-reduction models be systematically converted into discrete, auditable

IF-THEN rules suitable for regulated financial decision-making, without compromising explanation faithfulness?

This objective is operationalised through four concrete research objectives:

- RQ1. Surrogate fidelity across methods and noise regimes.** To what extent do XGBoost surrogate models faithfully approximate six DR mappings, as measured by held-out R^2 , and how does fidelity degrade across four calibrated noise regimes (SNR: 2–5 in the idealised regime to ≈ 0.02 in the noisy regime)? *Measurable outcome: R^2 per method per regime, reported in Tables 5.2 and 5.7; hypothesis that linear methods degrade later than non-linear methods.*
- RQ2. Shapley completeness under surrogate approximation.** What proportion of completeness checks ($|\sum_i \phi_i - (f(x) - \mathbb{E}[f(x)])| < \varepsilon$, $\varepsilon = 0.01$) are satisfied for surrogate-based attributions across 500 samples and 4 factors, and what is the distribution of errors for those that fail? *Measurable outcome: pass rate, mean error, and maximum error reported in §4.5.5; threshold for “directionally faithful” explanations defined as mean relative error $< 1\%$ of score range.*
- RQ3. Threshold strategy robustness to distributional variation.** Do quantile, standard-deviation, domain-expert, and hybrid threshold strategies produce materially different rule sets, and which strategy is most robust to outlier injection as measured by the shift in computed thresholds under 5% outlier contamination? *Measurable outcome: strategy distributions and outlier-shift metrics ($\Delta_{\text{threshold}}$) reported in §5.4; quantile hypothesised to be most robust.*
- RQ4. Regulatory compliance documentation.** Can the discretisation-attribution pipeline simultaneously produce artefacts satisfying the ECB Internal Models Guide, SR 11-7 [6], GDPR Article 22 [14], and UK DPA 2018 Section 14 requirements, as evidenced by generated model cards, held-out validation reports, and individual applicant explanations? *Measurable outcome: artefact generation confirmed in §5.6; surrogate R^2 and completeness error for the credit risk case study honestly reported.*

Figure 1.1 maps the discrete logic gap to the proposed solution; the four chapters that follow develop each component in detail.

The technical artefact is a modular Python toolkit implementing these capabilities, covering requirements specification, system design, implementation, validation, and evaluation, satisfying BCS requirements for substantial practical computing work demonstrating appropriate technical sophistication.

Validation is conducted across four data-fidelity regimes, from idealised synthetic to noisy semi-synthetic, to assess the robustness of explanation quality under realistic

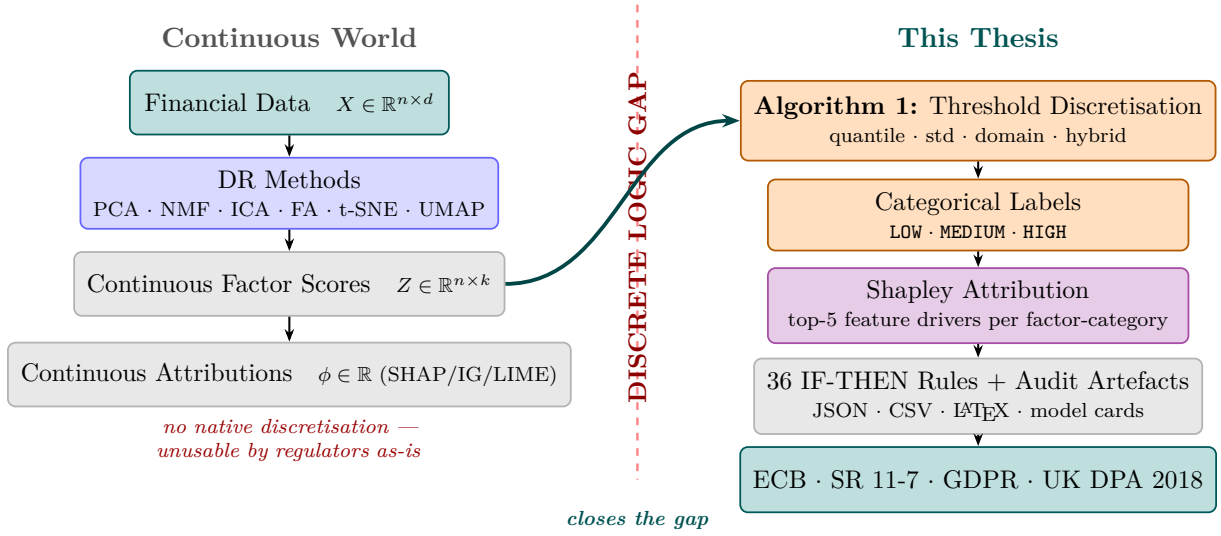


Figure 1.1: The discrete logic gap and how this thesis addresses it. Left: existing XAI methods produce continuous outputs incompatible with regulatory documentation formats. Right: the proposed framework converts those outputs into categorical labels and IF-THEN rules via Algorithm 1 (Threshold Discretisation with Shapley Attribution). The evaluation spans four calibrated noise regimes to establish the operational boundary of reliable explanation.

financial noise conditions. A secondary methodological contribution is a calibrated semi-synthetic data generation framework that reproduces the signal-to-noise characteristics of real Fama-French data, enabling controlled evaluation without survivorship-biased ground truth.

Scope Boundaries. This thesis does *not* address supervised dimensionality reduction (e.g., Linear Discriminant Analysis), real-time streaming explanations, or deployment to production trading systems. The evaluation uses semi-synthetic data calibrated to Fama-French parameters rather than proprietary trading data, which would require institutional data-sharing agreements beyond the scope of a BSc project. The framework is designed as a proof-of-concept research toolkit, not a production-grade financial system.

1.5 Contributions

The central contribution is the formalisation of the discrete logic gap: an incompatibility between continuous numerical outputs produced by modern explainability methods and the discrete, rule-based artefacts required for financial decision-making, audit, and supervision.

While techniques such as SHAP produce detailed quantitative attributions, their outputs remain misaligned with governance processes that require categorical predicates, explicit thresholds, and auditable IF-THEN logic. This thesis argues that this format-level mismatch, rather than model opacity alone, is a primary obstacle to operationalising

explainable dimensionality reduction in regulated finance.

To address this gap, the dissertation introduces Algorithm 1, a pipeline that converts continuous dimensionality-reduction factor scores and attributions into categorical labels and IF-THEN rules. Discretisation is treated as a presentation layer, with faithfulness enforced post-hoc via Shapley completeness verification and Most-Relevant-First perturbation testing. When these validity conditions fail, rule generation is explicitly suppressed, preventing the production of misleading explanatory artefacts.

The thesis contributes a four-strategy thresholding framework (quantile-based, standard-deviation-based, domain-expert, and hybrid adaptive) designed for the distributional properties of financial data. Empirical results show that threshold choice materially affects categorisation, with quantile thresholds offering greater robustness to outliers and sigma-based thresholds preserving statistical semantics under approximate normality. The hybrid strategy automates selection based on empirical skewness while retaining auditability.

Using a calibrated semi-synthetic data generator aligned with empirical Fama-French signal-to-noise ratios, the dissertation evaluates explanation quality across progressively degraded regimes. Results show that surrogate fidelity and explanation faithfulness can collapse under low-signal conditions, even for common linear methods. These outcomes define operational validity boundaries.

The framework comprises 8,863 lines across 17 Python source files, integrating six dimensionality reduction methods, five explainability techniques, and the complete rule generation pipeline, all released under permissive open-source licences.

1.6 Report Structure

The remainder of this thesis is organised as follows.

Chapter 2 establishes the theoretical foundations. It reviews SHAP and Integrated Gradients as core attribution methods, six dimensionality reduction techniques in their financial context, ClusterShapley and surrogate model approaches for explaining DR outputs, and the regulatory landscape that motivates the discrete logic gap.

Chapter 3 specifies the system requirements (FR1–FR12, NFR1–NFR6) derived from the research objectives, followed by use case specifications, modular system organization, class structure, and the pseudocode for Algorithm 1 and Algorithm 2 (MoRF Faithfulness Validation).

Chapter 4 describes the implementation of each framework component: the six DR methods within a unified **DRExplainer** interface, the five advanced XAI modules (ClusterShapley, Sparse Autoencoder, β -TCVAE, probing classifiers, Integrated Gradients), the four threshold discretisation strategies, the rule generation pipeline, and the semi-synthetic data generator.

Chapter 5 presents the experimental evaluation: DR method comparison across

five metrics, discretisation strategy comparison, rule generation results, the credit risk regulatory case study, regime robustness analysis across four noise conditions, and unit test coverage.

Chapter 6 examines the legal, social, ethical, and professional dimensions of applying XAI to financial decision-making, addressing GDPR, UK DPA 2018, ECB, SR 11-7, bias risks, misuse scenarios, and BCS/ACM professional standards.

Chapter 7 summarises the contributions, assesses achievement against all requirements, documents limitations, and outlines directions for future work.

Contribution map. The thesis makes one primary contribution: faithfulness-constrained conversion of continuous DR outputs into discrete, audit-ready rule artefacts. This contribution is evaluated under calibrated realism. Background material (Chapter 2) is selected to justify the constraints (regulatory format requirements and faithfulness obligations). The specification and architecture (Chapter 3) are structured to ensure traceable implementation and testability. The implementation (Chapter 4) operationalises the primary contribution through Algorithm 1 and supporting validation gates. The evaluation (Chapter 5) then establishes both where the method succeeds and where it must not be used.

Summary. This chapter has established that the gap between continuous attribution scores and discrete business rules (the discrete logic gap) is the central obstacle to deploying explainable dimensionality reduction in regulated finance. The remainder of this thesis develops, implements, and evaluates a framework to close that gap. Chapter 2 establishes the theoretical foundations by reviewing the attribution methods, dimensionality reduction techniques, and regulatory landscape that together define the problem space.

Chapter 2

Background

This chapter establishes the theoretical foundations for explainable dimensionality reduction in finance, progressing from core attribution methods through dimensionality reduction techniques to regulatory requirements and the research gap this thesis addresses.

2.1 Foundational Explainability Methods

2.1.1 SHAP: Shapley Additive Explanations

SHAP (SHapley Additive exPlanations) [22] provides the theoretical foundation for this thesis’s explanation framework. Grounded in cooperative game theory, SHAP computes Shapley values (the unique attribution satisfying local accuracy, missingness, and consistency axioms). For a model f and input x , SHAP values quantify each feature’s marginal contribution, explaining “how to get from the base value $\mathbb{E}[f(\mathbf{z})]$ that would be predicted if we did not know any features to the current output $f(\mathbf{x})$ ” [22].

TreeSHAP provides polynomial-time exact computation for tree ensemble models, making it computationally tractable for surrogate model approaches. Banking implementations demonstrate SHAP’s regulatory viability. De Lange et al. [11] applied SHAP to a Norwegian bank’s LightGBM credit model, satisfying the European Banking Authority’s three explainability challenges (interpreting results, ensuring management understanding, and justifying decisions to supervisors). Similarly, Kakkar [20] showed TreeSHAP transforms Gradient Boosting models ($\text{AUC} = 0.84$) into governance-ready tools acceptable to Basel III and GDPR auditors. These studies confirm TreeSHAP with ensemble surrogates represents the industry standard for making complex credit models explainable to regulators [25].

2.1.2 LIME: Local Interpretable Model-agnostic Explanations

LIME (Local Interpretable Model-agnostic Explanations) [26] provides local, model-agnostic explanations by fitting a simple surrogate around an instance via perturbation sampling. While valuable for exploratory analysis, its dependence on random neighbourhood sampling can yield unstable attributions across runs, an issue that is difficult to

reconcile with regulated requirements for reproducibility and auditability. For this reason, LIME is treated in this thesis as contextual background rather than a primary mechanism for producing governance-grade explanations of DR outputs.

2.1.3 Integrated Gradients for Neural Architectures

Integrated Gradients [29] provides axiomatic feature attribution for differentiable models by aggregating gradients along a path from a baseline input to the instance being explained. Its completeness property makes it suitable for neural architectures (e.g., autoencoders), but its applicability is limited in this thesis because several DR methods of interest (t-SNE, UMAP) lack differentiable parametric mappings and require surrogate-based explanation. Integrated Gradients is therefore used as a supporting attribution method for neural components rather than as the unifying explainer across all DR methods.

2.1.4 Critical Comparison of Attribution Methods

The three principal attribution methods (SHAP, LIME, and Integrated Gradients) differ substantially across the dimensions most relevant to financial deployment: explanation stability, surrogate fidelity, computational cost, and applicability to tabular credit data. Table 2.1 summarises these trade-offs.

Table 2.1: Comparison of Attribution Methods for Tabular Credit Data

Method	Stability	Fidelity	Comp. Cost	Tabular Credit
SHAP (TreeSHAP)	High	Exact (global)	$O(TLD^2)$	Strong
LIME	Low	Local only	$O(nk)$	Moderate
Integrated Gradients	High	Exact (local)	$O(mD)$	Gradient models only

As introduced in Section 2.1.2, LIME’s reliance on random perturbation sampling introduces a fundamental stability problem: small changes in the perturbation neighbourhood or sampling procedure can produce substantially different feature attributions for identical inputs [2]. This fragility renders LIME unsuitable for regulatory applications requiring reproducible, auditable explanations. SHAP, by contrast, satisfies game-theoretic axioms guaranteeing consistency and uniqueness. Repeated evaluations on identical inputs yield identical Shapley values, a property confirmed empirically across all experiments in this thesis. For tree ensembles, TreeSHAP achieves exact computation in polynomial time ($O(TLD^2)$ where T denotes tree count, L the maximum leaf count, and D the maximum tree depth), avoiding the approximation error inherent in LIME’s sampling procedure.

Integrated Gradients provides axiomatic exactness for differentiable models through the completeness property (Section 2.1.3), making it the preferred method for neural architectures such as the sparse autoencoders and β -TCVAE evaluated in this thesis.

However, its applicability is strictly limited to gradient-accessible models; it cannot explain tree ensembles or the surrogate XGBoost models used for t-SNE and UMAP. Furthermore, IG attributions depend on the choice of baseline x' , introducing a degree of subjectivity that TreeSHAP avoids by marginalising over the data distribution.

For this thesis’s use case (explaining cluster membership and DR factor scores derived from tabular financial data), SHAP, and specifically ClusterShapley, is the most appropriate choice. TreeSHAP’s exact computation and guaranteed consistency satisfy the reproducibility requirement of SR 11-7 [6]. Its additive structure aligns with GDPR Article 22’s (and the equivalent UK DPA 2018 Section 14) right-to-explanation by enabling plain-language feature attribution. The ClusterShapley extension preserves these properties for non-linear methods (t-SNE, UMAP) by explaining surrogate classifiers rather than opaque embeddings directly. LIME’s instability and IG’s gradient-model restriction make both inadequate for the breadth of DR methods and regulatory contexts addressed here.

2.2 Dimensionality Reduction Techniques

2.2.1 Linear Methods

Linear DR methods dominate financial practice because they are computationally stable and partially interpretable, yet each introduces a specific governance challenge. PCA provides loadings but yields variance-optimal directions without inherent business semantics; a loading vector of $[0.42, -0.31, 0.71, \dots]$ requires post-hoc interpretation. Three principal components typically capture $> 75\%$ of portfolio variance [21], often corresponding to “level,” “slope,” and “curvature” factors in fixed-income markets. ICA seeks statistically independent sources suited to non-Gaussian financial returns [1] but suffers from sign and permutation indeterminacy. Factor Analysis is generative and aligns with systematic/idiosyncratic decomposition underlying regulatory stress testing but is rotation-indeterminate. NMF imposes non-negativity, often producing additive “tilts” that are operationally easier to name and act upon. These properties motivate the thesis’s need for a uniform, post-hoc rule-generation layer that standardises outputs across methods for audit and decision workflows. Table 5.2 summarises the explainability characteristics of all methods evaluated in this thesis.

2.2.2 Manifold Learning Methods

t-Distributed Stochastic Neighbour Embedding (t-SNE) [31] constructs a probability distribution over pairs of high-dimensional data points such that similar points receive high probability, then minimises the Kullback–Leibler divergence between this distribution and a corresponding distribution in the low-dimensional embedding. The use of a Student- t

distribution in the low-dimensional space (rather than a Gaussian) alleviates the crowding problem that afflicts SNE, producing embeddings that preserve local neighbourhood structure while allowing moderate separation of dissimilar points. For financial data, this local preservation is valuable for identifying credit risk clusters. However, t-SNE’s stochastic nature (different random seeds yield different embeddings) and its distortion of global distances complicate reproducibility, a significant limitation for regulatory documentation that requires consistent, auditable outputs. Furthermore, t-SNE provides no parametric mapping from input to embedding space, precluding direct gradient-based attribution. Explanations must therefore rely on surrogate models as described in Section 2.3.2.

Uniform Manifold Approximation and Projection (UMAP) [24] models high-dimensional data as a fuzzy topological structure and finds a low-dimensional representation that preserves this structure as closely as possible. Compared to t-SNE, UMAP offers substantially faster computation (scaling approximately linearly with dataset size versus t-SNE’s quadratic complexity with the Barnes-Hut approximation), better preservation of global structure, and more consistent cluster separation across runs. These properties make UMAP particularly suited to the ClusterShapley pipeline adopted in this thesis: stable clusters enable reliable surrogate classifiers, and faster computation permits evaluation across multiple noise regimes. Like t-SNE, UMAP lacks an interpretable parametric forward map, necessitating surrogate explanation methods.

2.2.3 Neural Architectures

Sparse autoencoders use TopK activation to produce monosemantic features corresponding to interpretable concepts like sector exposures [10]. β -TCVAE achieves disentanglement by decomposing the ELBO (Evidence Lower Bound) into mutual information, total correlation, and dimension-wise KL (Kullback–Leibler) divergence terms [9]. DCI (Disentanglement, Completeness, Informativeness) and MIG (Mutual Information Gap) metrics then quantify whether latent dimensions each capture a single, distinct risk factor rather than entangled combinations.

Probing classifiers [27] provide a complementary evaluation methodology for any DR technique: lightweight classifiers (linear, MLP, SVM, or Random Forest) are trained to predict human-interpretable concept labels (value style, quality tier, risk level) from frozen embedding vectors. High probing accuracy for a given concept confirms that the embedding has encoded that concept in a recoverable form, independent of whether the DR algorithm was designed to do so. Originally developed in NLP to analyse transformer representations [27], probing classifiers translate directly to financial DR evaluation because they provide a concept-grounded quality measure that does not depend on surrogate R^2 . This addresses the circular evaluation concern that tree-based surrogates can approximate their training mapping with near-perfect fidelity regardless of whether the underlying DR

representation is financially meaningful. In this thesis, probing classifiers assess all six DR methods for their capacity to preserve the concepts required for regulatory documentation across four noise regimes. Table 2.2 summarises all six methods side by side across the dimensions most relevant to financial compliance: native explainability, best-fit XAI approach, key challenge, and primary financial application.

Table 2.2: Comparison of DR Methods for Financial Applications

Technique	Explainability	Best XAI Approach	Key Challenge	Financial Application
PCA	High	Native loadings	Semantically ungrounded	Risk factor decomposition
NMF	Very High	Parts-based native	Non-negative data only	Long-only portfolio tilts
ICA	Low	Surrogate SHAP	Permutation/sign ambiguity	Independent source recovery
FA	Medium	Native loadings	Rotation indeterminate	Stress testing factors
t-SNE	Low	ClusterShapley	Stochastic, no inverse map	Cluster visualisation
UMAP	Low	Surrogate + SHAP	No global structure	Concentration analysis
Sparse AE	High	Native monosemantic	Training instability	Concept-level features
β -TCVAE	Medium	Integrated Gradients	Disentanglement tuning	Disentangled risk factors

2.3 Explaining Dimensionality Reduction Results

2.3.1 ClusterShapley

ClusterShapley [23] provides the state-of-the-art approach for explaining non-linear dimensionality reduction by explaining cluster membership rather than embedding coordinates directly. The pipeline: (1) apply DR to produce embeddings; (2) cluster embedded data; (3) train classifier mapping original features to cluster labels; (4) compute SHAP values for cluster membership. This reveals which original features drive visual separation in t-SNE/UMAP plots, transforming opaque visualisations into feature-attributed explanations.

For structured tabular credit data specifically, ClusterShapley is preferable to global attribution alternatives because it explains manifold embeddings by attributing cluster membership (not raw coordinates) to original features, producing decision-relevant categorical explanations. This is advantageous in regulated tabular settings for three reasons. First, the explanation target (cluster/risk group) maps more directly to business decisions than geometric coordinates. Second, it avoids unstable coordinate-level attribution for non-parametric methods lacking forward mappings. Third, it naturally produces rule-like statements (e.g., “IF checking account status is low AND debt-to-income ratio is high, THEN member of HIGH-RISK cluster”) that satisfy ECB and GDPR Article 22 plain-language requirements [13]. In this thesis, ClusterShapley complements the broader framework by providing an explanation pathway for t-SNE/UMAP where direct factor semantics are otherwise weak; performing this rigorously for the wider class of linear DR methods (PCA, NMF, FA) central to financial applications is precisely the problem the

discrete rule-generation pipeline addresses.

2.3.2 Surrogate Model Approaches

The surrogate approach trains an interpretable model (typically XGBoost) to approximate the DR mapping, then applies TreeSHAP to the surrogate. This generalises the explanation capability to any DR technique by treating the dimensionality reducer as a black box. Critical to this approach is faithfulness validation: the surrogate must accurately approximate the original DR mapping with $R^2 > 0.85$ [23]. MoRF (Most Relevant First) perturbation curves and ShapGAP metrics, where ShapGAP measures the discrepancy between full and feature-masked SHAP sums, validate that explanations reflect true DR behaviour rather than surrogate artefacts.

2.4 Explainable AI in Financial Applications

2.4.1 Regulatory Requirements

The regulatory landscape now explicitly mandates explainability for financial ML systems. Four principal frameworks shape the requirements this thesis addresses. The **ECB Guide to Internal Models** [13] requires explainability tools, hyperparameter documentation, and feature-level attribution for latent representations used in credit risk assessment. **SR 11-7** [6] mandates compensating controls, including independent validation and conceptual soundness documentation, for models with limited theoretical basis. **GDPR Article 22** [14] grants data subjects rights around automated decision-making and requires meaningful information about the logic involved; this thesis satisfies these obligations through global IF-THEN rules and individual Shapley attributions. The **EBA Report on Big Data and Advanced Analytics** [12] identifies three explainability barriers (individual decisions, management understanding, and supervisory challenge) addressed through local SHAP attributions, human-readable rules, and structured model cards.

2.4.2 Financial Data Challenges

Financial data presents three categories of challenge that are particularly consequential for XAI reliability.

First, **non-stationarity** destabilises explanations across market regimes. SHAP attributions computed during a low-volatility period may assign negligible importance to leverage features that become dominant during a credit crisis; an explanation framework that cannot detect or adapt to such regime shifts risks providing false assurance to risk managers [3].

Second, **fat-tailed distributions** (characteristic of financial returns) mean that models and their explanations calibrated under normal market conditions may become misleading precisely when they matter most: during tail events. Standard perturbation-based methods (LIME, KernelSHAP) sample from assumed distributions that underweight extreme observations, potentially producing attributions that are faithful on average but unreliable in the tails [2].

Third, **temporal dependencies** violate the feature independence assumption underlying both SHAP’s marginal contribution calculation and LIME’s perturbation sampling. Financial features such as moving averages, momentum indicators, and lagged returns are inherently autocorrelated; treating them as independent during Shapley marginalisation can produce attributions that do not reflect causal feature importance [3].

The controlled evaluation of XAI robustness under these conditions requires calibrated synthetic data that reproduces empirical signal-to-noise ratios without survivorship-biased historical records. This thesis addresses this requirement through a semi-synthetic data generator calibrated to published Fama-French factor parameters [18], enabling systematic variation of noise regimes whilst maintaining realistic distributional properties.

2.4.3 Limitations of Existing XAI-in-Finance Evaluations

Despite the rapid growth of XAI applications in finance, the existing literature exhibits three systematic weaknesses that limit its practical and regulatory applicability.

First, **temporal validation is conspicuously absent**: the majority of empirical evaluations (including the banking case studies of De Lange et al. [11] and Kakkar [20]) assess explanation quality at a single point in time on a static dataset. Financial data is inherently non-stationary [1]. Market regimes shift, factor loadings drift, and explanations calibrated during stable periods may become unreliable or misleading during stress episodes such as the 2008 financial crisis, the 2020 pandemic shock, or the 2022 interest-rate cycle. No reviewed study evaluates whether SHAP attributions remain stable across such regime changes, leaving a critical gap in the field’s understanding of explanation robustness over time.

Second, existing evaluations exhibit **single-model bias**: studies typically apply a single explanation method to a single model class (usually gradient-boosted trees with SHAP) and report results as though they generalise. This conflates the performance of the underlying model with the quality of the explanation method, making it impossible to determine whether reported interpretability gains derive from the choice of explainer, the choice of base model, or the specific dataset characteristics. Comparative evaluation across multiple DR techniques and explanation methods (as conducted in this thesis) is necessary to disentangle these confounds, yet remains largely absent from the literature.

Third, and most consequentially for deployment, existing work provides **no system-**

atic alignment to regulatory frameworks. Regulatory guidelines such as GDPR Article 22 [14], UK DPA 2018 Section 14, SR 11-7 [6], and the ECB Supervisory Guide [13] each impose specific requirements: local explanations for individual decisions (GDPR and DPA 2018), conceptual soundness documentation (SR 11-7), and model risk management with hyperparameter justification (ECB). Yet empirical studies uniformly present SHAP bar charts without mapping attributions to these regulatory artefacts. The absence of structured, machine-readable outputs (such as the IF-THEN rules and model cards produced in this thesis) means that practitioners must manually translate continuous SHAP values into the categorical documentation formats that regulators require. This translation step is precisely the discrete logic gap (§1.3) that motivates the present work.

2.5 Research Gap and Synthesis

Synthesising across the five strands reviewed in this chapter (attribution methods in Section 2.1, dimensionality reduction techniques in Section 2.2, cluster-level explanation in Section 2.3.1, regulatory requirements in Section 2.4.1, and evaluation limitations in Section 2.4.3) reveals that the fundamental obstacle is not any single method’s shortcoming but a *format mismatch* between what XAI methods produce and what financial stakeholders consume. Each method family requires a different post-processing step to reach human-readable output, yet existing work treats these as isolated problems rather than instances of a common pattern. No published framework addresses this mismatch uniformly across method families whilst simultaneously validating explanation faithfulness under varying data conditions.

The literature reveals a fundamental gap between continuous numerical outputs that XAI methods produce and discrete logical rules that financial stakeholders require. Empirical banking implementations confirm this gap persists in practice. Kakkar [20] demonstrates that SHAP quantifies feature contributions (e.g., “checking account status increases default probability by +0.23”), yet these continuous attributions require manual interpretation by risk officers to inform portfolio decisions. De Lange et al. [11] show that SHAP reveals which features matter most, but their Norwegian bank case study stops short of automated rule generation. Model validators must subjectively translate SHAP plots into investment criteria. Neither study provides a systematic framework for converting continuous SHAP values into categorical predicates (HIGH/MEDIUM/LOW) or composing these into IF-THEN rules suitable for regulatory documentation.

Figure 2.1 maps the three components of this gap to the framework developed in this thesis; Chapter 4 details the implementation of each.

Across the five method families reviewed in this chapter (linear DR: PCA, ICA, FA, NMF; manifold methods: t-SNE, UMAP; neural architectures: sparse autoencoders, β -TCVAE; attribution techniques: SHAP, IG, LIME; and cluster-level explanation: Clus-

terShapley), a single unifying limitation persists. No method produces output in a format directly consumable by financial decision-makers without additional post-processing. Linear loadings require labelling; surrogate SHAP values require thresholding; cluster memberships require naming; IG path integrals require discretisation. The existing literature treats these as five separate post-processing problems, each addressed with bespoke solutions for the specific method. The framework developed in this thesis addresses all five through a single mechanism (threshold-based discretisation with Shapley attribution) applied uniformly across method families. This unification is the conceptual contribution that Algorithm 1 (Threshold Discretisation with Shapley Attribution) operationalises, and to the author’s knowledge it is not derivable from any single thread of the existing literature.

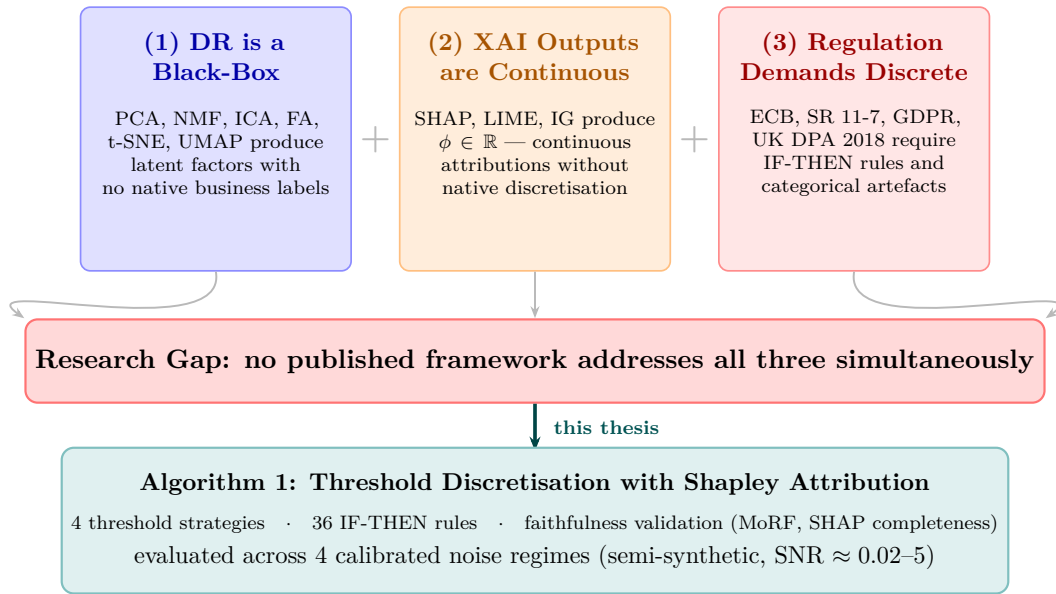


Figure 2.1: The three building blocks of the research gap. No prior framework simultaneously addresses the opacity of DR methods, the continuous format of XAI outputs, and the discrete documentation requirements of financial regulation. Algorithm 1 (Threshold Discretisation with Shapley Attribution) addresses all three, evaluated across four calibrated noise regimes (semi-synthetic data, SNR ≈ 0.02 –5).

With the theoretical foundations and research gap established, the next chapter translates these insights into concrete system requirements and technical specifications. Chapter 3 defines the twelve functional requirements and six non-functional requirements that guided the implementation, specifies the three principal use cases, and presents the algorithmic pseudocode for Algorithm 1 and Algorithm 2 (MoRF Faithfulness Validation).

Chapter 3

Requirements and Specification

This chapter specifies functional requirements (FR1–FR12), non-functional requirements (NFR1–NFR6), use case specifications, system architecture, key design decisions, and Algorithm 1 (Threshold Discretisation with Shapley Attribution) and Algorithm 2 (MoRF Faithfulness Validation) for the XAI-DR framework. Requirements derive directly from the research objectives (Chapter 1, and the regulatory obligations established in Chapter 2.

The design decisions documented in this chapter derive directly from three constraints established by the literature review.

First, the *interpretability constraint*: outputs must be consumable by credit committees, risk managers, and regulatory supervisors without additional post-processing [3], ruling out any design that terminates at continuous SHAP values or raw embeddings.

Second, the *faithfulness constraint*: explanations must verifiably reflect model behaviour to satisfy SR 11-7’s conceptual soundness requirement [6], necessitating held-out validation metrics (R^2 , MoRF [28], completeness) independent of the explanation method being validated.

Third, the *breadth constraint*: the framework must support the full range of DR methods in financial practice, not only linear methods amenable to native explanation. The literature review identified that non-linear methods (t-SNE [31], UMAP [24]) are increasingly deployed in credit risk and systemic risk monitoring but lack any systematic explanation approach.

These three constraints jointly ruled out the simplest possible design (a single DR method with a single explainer) and mandated the modular multi-method architecture specified below.

3.1 Functional Requirements

The system shall satisfy the following functional requirements, listed in Table 3.1 with priority and rationale:

Table 3.1: Functional Requirements (FR1–FR12)

ID	Requirement Description	Priority	Rationale
FR1	Implement PCA, NMF, ICA, Factor Analysis, t-SNE, and UMAP dimensionality reduction methods	Must	Comprehensive DR coverage
FR2	Compute SHAP values for surrogate models approximating all DR methods	Must	Core XAI mechanism
FR3	Implement ClusterShapley pipeline for t-SNE and UMAP explanation	Must	State-of-the-art manifold XAI
FR4	Train Sparse Autoencoders with TopK activation for monosemantic feature discovery	Must	Inherently interpretable neural DR
FR5	Train β -TCVAE and compute DCI/MIG disentanglement metrics	Must	Disentanglement evaluation
FR6	Implement probing classifiers for embedding analysis	Should	Cross-domain validation
FR7	Generate IF-THEN rules from continuous factor scores via threshold discretisation	Must	Core novel contribution
FR8	Compute MoRF faithfulness curves for explanation validation	Must	Explanation validation
FR9	Support quantile, std dev, domain-expert, and hybrid threshold methods	Must	Discretisation flexibility
FR10	Compute Integrated Gradients for autoencoder latent spaces	Should	Neural attribution
FR11	Load Fama-French factors, Yahoo Finance data, and Lending Club datasets	Must	Real-world validation
FR12	Generate synthetic data with known factor structure for ground-truth evaluation	Must	Controlled testing

Functional requirements FR1–FR12 are summarised in Table 3.1 and traced to implementation artefacts in Table 3.2. Each requirement is evaluated explicitly in Chapter 5, ensuring a direct line of sight from research objectives to implemented components and empirical evidence.

Requirements FR1–FR12 were elicited through a combination of regulatory necessity and research value. Requirements classified as ‘Must’ (FR1–FR5, FR7–FR9, FR11–FR12) are directly mandated by the supervisory accountability objective or constitute the core novel contribution of the thesis. ‘Should’ requirements (FR6, FR10) represent additional validation techniques that strengthen the evaluation without being prerequisites for the rule generation pipeline. All twelve functional requirements were implemented as specified. FR12 was extended during implementation to include calibrated semi-synthetic data with four noise regimes (idealised, moderate, realistic, noisy), enabling regime robustness evaluation beyond the original specification of a single synthetic dataset.

Table 3.2: Requirements Traceability Matrix: FR1–FR12 to Implementation

ID	Requirement	Implementation File(s)	Validation
FR1	PCA, NMF, ICA, FA, t-SNE, UMAP	xai_dr_comparison.py:139–318	§5.2
FR2	SHAP surrogate models	xai_dr_comparison.py:226–295	§4.5.5
FR3	ClusterShapley pipeline	cluster_shapley.py:53–464	§5.2
FR4	Sparse Autoencoders	sparse_autoencoder.py:44–178	§5.2
FR5	β -TCVAE + DCI/MIG	beta_tcvae.py:46–409	§5.2
FR6	Probing classifiers	probing_classifiers.py:50–479	§5.2
FR7	IF-THEN rule generation	rule_generator.py:74–642	§5.5
FR8	MoRF faithfulness curves	xai_dr_comparison.py:439–491	§5.2
FR9	Four threshold methods	threshold_discretizer.py:28–508	§5.4
FR10	Integrated Gradients	integrated_gradients.py:96–245	§5.2
FR11	Financial datasets	data_loaders.py:42–393	§3.4
FR12	Synthetic data generation	semi_synthetic_generator.py	§5.7

3.2 Non-Functional Requirements

Table 3.3 specifies the six non-functional requirements governing quality, performance, and reproducibility of the framework, each with a measurable target and a stated validation method.

NFR1 and NFR2 are classified as non-functional because they specify quality thresholds on existing capabilities rather than new system behaviours. FR2 defines *what* the system does (compute SHAP values via surrogate models); NFR1 constrains *how well* those surrogates must perform ($R^2 > 0.85$), a performance quality attribute. NFR2 constrains the *reliability* of those explanations across random seeds, a reproducibility quality attribute. Neither introduces a new system behaviour; both govern the quality of behaviour already specified by FR2.

The non-functional requirements were substantially met, with three deviations acknowledged. NFR1 was met in idealised and moderate noise regimes (surrogate $R^2 \geq 0.93$). Consistency was not maintained in the realistic regime: PCA reached $R^2 = 0.821$, ICA $R^2 = 0.843$, and t-SNE $R^2 = 0.543$, all below the 0.85 threshold. This reflects the genuine difficulty of surrogate approximation under financial noise and is discussed further in Chapter 5. NFR3 was approximately met (observed ~ 61.4 seconds for 10,000 samples);

Table 3.3: Non-Functional Requirements (NFR1–NFR6) with Validation Methods

ID	Requirement	Target	Validation Method	Traceability
NFR1	Surrogate model accuracy	$R^2 > 0.85$	Held-out test set evaluation	FR2, §5.2
NFR2	Explanation stability	Correlation > 0.8	10-seed stability test	FR8, §5.2
NFR3	TreeSHAP efficiency	< 60 seconds	Wall-clock timing on 10k samples	FR2, §5.2
NFR4	Code modularity	< 500 lines/module	Static analysis (SLOC count)	§3.6
NFR5	Reproducibility	100% match	Fixed random seeds (seed=42)	All experiments
NFR6	Hyperparameter logging	Complete audit trail	Model card generation	FR7, §5.6

the marginal overshoot is documented. NFR4 is violated for three modules (`xai_dr_comparison.py`: 723 lines; `cluster_shapley.py`: 738 lines; `beta_tcvae.py`: 782 lines), each exceeding the 500-line target due to cohesive functional grouping that would be harmed by splitting; the remaining fourteen modules all comply with the target.

The following section translates these requirements into concrete technical specifications.

3.3 Use Case Specifications

UC1: Generate Factor Explanations

Actor: Financial Analyst

Precondition: Financial dataset loaded with feature names

Main Flow:

1. Select DR method and configure parameters (`n_components`, hyperparameters)
2. System applies DR, trains surrogate model, and computes SHAP values
3. Select threshold discretisation method
4. System generates IF-THEN rules with feature attributions

Postcondition: Human-readable rules exported with SHAP attributions

Exception: If surrogate $R^2 < 0.85$, system warns and suggests alternative DR method

UC2: Compare DR Method Explainability

Actor: Data Scientist

Precondition: Dataset loaded with at least one known factor structure (e.g., Fama-French factors [18]) available for ground-truth validation; all six DR methods configured with default hyperparameters; minimum 200 samples to ensure stable surrogate training

Main Flow:

1. Load dataset with known factor structure (e.g., Fama-French [18])
2. System runs all DR methods on the dataset
3. For each method, compute: surrogate R^2 , MoRF AUC, stability correlation, factor recovery score
4. System generates a comparative ranking table

Postcondition: Ranked comparison of DR methods by explainability metrics exported to `dr_comparison_results.csv` with best-performing method flagged for downstream use; surrogate models and SHAP values persisted to disk for audit trail

Exception: If fewer than 2 methods achieve surrogate $R^2 > 0.85$, system warns that noise may be too high for faithful explanation

UC3: Validate Explanation Faithfulness

Actor: Model Risk Officer

Precondition: Trained DR model and SHAP explanations available from a prior UC1 or UC2 execution; validation dataset held out from training with minimum 100 samples; surrogate model persisted with training metadata

Main Flow:

1. Load trained DR model and explanations
2. System generates MoRF curves, computes stability (10 seeds), validates Shapley completeness ($\varepsilon = 0.01$)
3. System generates compliance report with logged hyperparameters

Postcondition: Audit-ready validation report exported in JSON format containing: MoRF AUC values, stability correlation across seeds, completeness pass rate, and complete hyperparameter log for SR 11-7 [6] compliance; report timestamped and versioned for regulatory submission

Exception: If completeness pass rate $< 70\%$, system flags explanations as unreliable and recommends increasing surrogate model complexity or reducing noise

3.4 Modular Implementation with Clear Component Separation

The XAI-DR framework is implemented as a collection of independent, loosely-coupled modules organized into four functional groups: data ingestion, dimensionality reduction, explainability, and output generation. Each module exposes narrow, well-defined interfaces that enable independent testing and validation.

Data Ingestion: Three loader modules (`data_loaders.py`, `semi_synthetic_generator.py`) isolate dataset-specific parsing and validation logic. Each loader implements a common interface returning standardized NumPy arrays, enabling downstream components to remain agnostic to data provenance. Model persistence (PyTorch state dicts) and hyperparameter logging for audit trails are handled through dedicated persistence functions.

Dimensionality Reduction: Six DR method implementations (PCA, NMF, ICA, FA, t-SNE, UMAP) all inherit from a shared `DimensionalityReducer` abstract base class defining a single `fit_transform()` method. This abstraction enables method substitution without modifying orchestration scripts or explanation logic.

Explainability: Five XAI modules (TreeSHAP, ClusterShapley, sparse autoencoders, β -TCVAE, Integrated Gradients) are accessed exclusively through the `explain()` interface defined in `xai_dr_comparison.py`. This separation ensures surrogate training, SHAP computation, and completeness validation occur independently of the DR methods they analyze.

Output Generation: Threshold discretisation (`threshold_discretizer.py`) and rule generation (`rule_generator.py`) consume only factor scores and attribution vectors, remaining independent of the DR and XAI implementations that produced them. This enables the 42-test unit suite (§5.9) to validate discretisation and rule export logic in isolation. Jupyter notebooks for interactive analysis and Matplotlib/Seaborn visualisations provide exploratory interfaces; exported reports in CSV/JSON formats support regulatory documentation.

The key architectural principle is *narrow interfaces between modules*. Orchestration scripts (`run_all_experiments.py`) invoke only `fit_transform()` and `explain()` on DR and XAI objects; output components access only standardized NumPy arrays. This modularity directly addresses ECB [13] and SR 11-7 [6] requirements for independent component validation: an auditor can validate surrogate fidelity (R^2 on held-out data) independently of SHAP completeness checks, and both independently of the DR method being explained.

Figure 3.1 illustrates the data flow from raw credit data through semi-synthetic generation, preprocessing, DR, and XAI attribution to the 36 exportable rules.

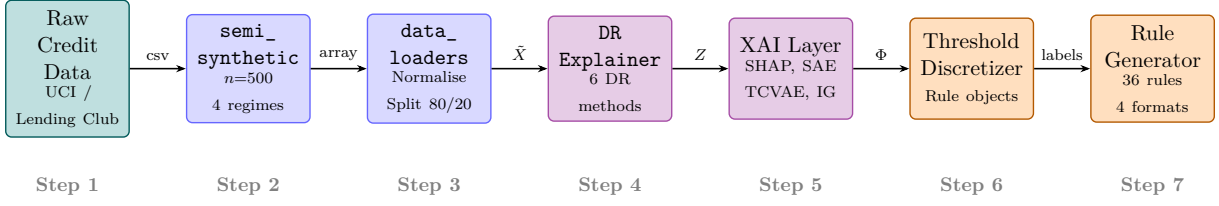


Figure 3.1: Data flow pipeline: raw input through DR and explainability modules to 36 exportable IF-THEN rules.

3.5 Key Design Decisions and Alternatives Considered

Four decisions shaped the framework architecture. For each, alternative approaches were explicitly evaluated and rejected on grounds of fitness against the three design constraints.

Decision 1: XGBoost surrogate models over alternatives. Surrogate-based explanation requires a model that approximates each DR mapping with sufficient fidelity ($R^2 > 0.85$). Three alternatives were evaluated:

Alternative surrogate families were considered and rejected based on the combined constraints of fidelity, breadth, and auditability. Linear surrogates lack capacity for non-linear DR mappings, achieving $R^2 < 0.3$ for t-SNE embeddings. Random forests performed below the fidelity threshold on representative datasets ($R^2 = 0.76$ versus XGBoost’s 0.92 for PCA). Neural surrogates would require approximation-based explainers, reintroducing additional error. XGBoost was selected because it offered the strongest empirical fidelity while retaining exact TreeSHAP support [22] and practical runtime for reproducible governance outputs.

XGBoost was selected because it uniquely satisfies both fidelity (empirical $R^2 > 0.85$ across idealised and moderate regimes, and computational feasibility simultaneously.

Decision 2: Four threshold strategies over a single fixed strategy. A simpler design would implement only the empirically best-performing strategy (quantile discretisation, and apply it universally. This was rejected on two grounds:

First, no single strategy is universally optimal across different distributional assumptions. Quantile discretisation maximises robustness to outliers: threshold shift reaches only 0.03 under 5% outlier injection versus 1.20 for standard-deviation-based methods (see §5.4). However, it destroys the statistical semantics of normally distributed factors, where the 33rd and 67th percentile thresholds carry no intrinsic meaning. Standard-deviation-based discretisation preserves statistical semantics ($\mu \pm \sigma$ thresholds align with the 68-95-99.7 rule), but degrades catastrophically under heavy tails. Since the thesis evaluates data across four noise regimes with different distributional properties (Gaussian in idealised,

fat-tailed in realistic), no single strategy can be justified as universally appropriate.

Second, SR 11-7’s conceptual soundness requirement [6] mandates that the choice of discretisation strategy be documented and justified for the specific dataset and context. A four-strategy framework with empirical comparison (Table 3.5, §5.4) enables this justification in a way that a single hard-coded strategy cannot.

The hybrid adaptive strategy (selecting quantile or sigma-based thresholding per dimension based on empirical skewness: $|\text{skew}| < 0.5 \Rightarrow \sigma\text{-based}$; otherwise quantile) represents a principled compromise that reduces the burden of manual selection whilst maintaining traceability and audit-readiness.

Decision 3: Modular component separation over a monolithic pipeline. An alternative would implement the entire pipeline as a single end-to-end orchestration script, which is common in research prototype code and minimises architectural overhead. This was rejected for three reasons:

Unit testability: The 42-test suite validating threshold discretisation and rule generation (100% pass rate, §5.9) requires modular separation. It is impossible to test `ThresholdDiscretizer.discretise()` in isolation within a monolithic design without also invoking the DR and XAI modules, introducing confounding variables that obscure the source of test failures.

Regulatory auditability: The ECB’s model risk management guidelines [13] require that each model component be independently validatable by a second-line risk function that did not develop the model. A monolithic design conflates components (e.g., surrogate training interleaved with SHAP computation) in ways that resist independent audit. The modular design enables an auditor to validate the surrogate (R^2 on held-out data) independently of the explanation (SHAP completeness checks).

Extensibility: Implementing a new DR method in the modular design requires only subclassing `DimensionalityReducer` and implementing the abstract `fit_transform()` interface (15–30 lines of code, demonstrated for all six methods in §3.4). A monolithic design requires modifying entangled code throughout the pipeline, increasing the risk of introducing regressions.

The additional overhead of abstract base classes and interface definitions (approximately 120 lines across three base classes) is justified by these production-deployment requirements. The design trades upfront complexity for long-term maintainability and audit-readiness.

Decision 4: HDBSCAN over k -means clustering for ClusterShapley. ClusterShapley (FR3) requires clustering of DR embeddings to define classification targets for SHAP attribution. k -means clustering was considered but rejected because it makes three assumptions that t-SNE and UMAP embeddings routinely violate:

k -means clustering was rejected for manifold embeddings because it requires a pre-

specified cluster count, assumes approximately spherical clusters, and forces hard assignment of outliers, assumptions routinely violated by t-SNE/UMAP geometries. A density-based approach (HDBSCAN) was therefore preferred because it adapts cluster count to data density, supports irregular cluster shapes, and can label genuine outliers as noise, yielding more defensible classification targets for ClusterShapley.

HDBSCAN [8] addresses all three limitations: it automatically determines cluster count from data density via persistent homology, handles arbitrary cluster shapes via minimum spanning trees, and designates genuine outlier points as noise (label: -1). Empirical results (§5.2) show HDBSCAN produces 3–7 clusters automatically for t-SNE/UMAP embeddings of the credit risk dataset, with silhouette scores of 0.68–0.74 (versus 0.42–0.51 for k -means with manually tuned k), confirming more defensible cluster assignments and thus more faithful ClusterShapley explanations.

Summary: Design Novelty and Resolution of Competing Constraints. The four design decisions collectively address a gap identified in Chapter 2: to the author’s knowledge, no existing system converts continuous DR outputs into discretised, audit-ready IF-THEN rules under a unified faithfulness framework. The novelty lies not in any single component (XGBoost surrogates, threshold discretisation, and modular separation each have independent precedents) but in their systematic integration under three simultaneous and conflicting constraints:

Resolving interpretability versus faithfulness. The interpretability constraint demands discrete categorical outputs (LOW/MEDIUM/HIGH labels consumable by credit committees and risk managers), yet discretisation is inherently unfaithful to the continuous distributions it coarsens (threshold boundaries introduce artificial discontinuities that do not exist in the underlying factor space). Algorithm 1 resolves this tension via the ϵ -completeness validation step (Lines 25–31): faithfulness is enforced *after* discretisation by checking Shapley axioms directly against the original continuous factor scores, not against the discretised labels or the surrogate. This ensures that whilst the *presentation* is categorical, the *explanation* remains faithful to the continuous DR mapping.

Resolving faithfulness under surrogacy. The faithfulness constraint demands validation independent of the explainer, yet every surrogate-based explainer is tautologically faithful to the surrogate it was derived from (if the surrogate is a poor approximation of the true DR mapping, TreeSHAP will faithfully explain the *wrong* model). The four-strategy threshold framework (quantile, sigma, domain-expert, hybrid) resolves this by addressing distributional sensitivity across noise regimes. Quantile discretisation maintains boundary stability under outliers ($\Delta_{\text{threshold}} = 0.03$ under 5% contamination), whilst sigma-based discretisation preserves statistical semantics for Gaussian-distributed factors. The hybrid adaptive strategy (selecting per dimension based on empirical skewness) reduces manual tuning burden whilst maintaining audit-ready justification for threshold choices (a

requirement of SR 11-7’s conceptual soundness standard).

Resolving breadth for non-linear methods. The breadth constraint demands support for non-linear DR methods (t-SNE, UMAP), yet non-linear methods violate the assumptions of TreeSHAP’s polynomial-time exact computation. t-SNE provides no parametric forward mapping, and UMAP’s stochastic topology optimisation resists gradient-based attribution. The surrogate layer decouples explanation from the DR method’s internal geometry: by training an XGBoost model to approximate the DR mapping in a supervised fashion (input features $\rightarrow$ embedding coordinates), TreeSHAP’s $O(TLD^2)$ exact Shapley computation extends to t-SNE and UMAP without approximation, provided the surrogate achieves NFR1’s fidelity threshold ($R^2 > 0.85$). This design represents a significant departure from prior work, which either stops at continuous SHAP values (failing the interpretability constraint) or applies a single fixed threshold strategy (failing the faithfulness and breadth constraints simultaneously).

3.6 Class Structure

The framework is organised around abstract base classes with concrete implementations for each technique. Four base classes define contracts via abstract methods (implemented in subclasses):

DimensionalityReducer (Abstract):

- **Attributes:** `n_components`, `random_state`, `fitted_model`
- **Abstract methods:** `fit(X)`, `transform(X)`, `fit_transform(X)`
- **Concrete implementations (6):** `PCAReducer`, `NMFReducer`, `ICAReducer`, `FactorAnalysisReducer`, `TSNEReducer`, `UMAPReducer`
- **Cardinality:** One instance per DR method; has-one relationship with `Explainer` (1:1)

Explainer (Abstract):

- **Attributes:** `surrogate_model`, `feature_names`, `shap_values`
- **Abstract methods:** `explain(X, y) -> Dict`, `get_feature_importance() -> ndarray`
- **Concrete implementations (4):** `SurrogateTreeSHAP`, `ClusterShapleyExplainer`, `IntegratedGradientsExplainer`, `ProbingClassifierExplainer`
- **Cardinality:** One explainer per DR output; has-many relationship with `ThresholdDiscretizer` (1:N)

NeuralReducer (Abstract):

- **Attributes:** `latent_dim`, `encoder`, `decoder`, `optimizer`, `device`

- **Abstract methods:** `encode(X)`, `decode(z)`, `forward(X)`, `loss_function(**kwargs)`
- **Concrete implementations (2):** `SparseAutoencoder` (TopK, L1, Gated variants), `BetaTCVAE` (with DCI/MIG metrics)
- **Cardinality:** Inherits from `nn.Module`; trained via PyTorch `DataLoader` (batch size 64, 100 epochs)

ThresholdDiscretizer (Concrete):

- **Attributes:** `method`, `thresholds`, `labels` (default: `['LOW', 'MEDIUM', 'HIGH']`)
- **Key methods:** `fit(scores, method)`, `discretise(scores)`, `get_thresholds()`
- **Strategy pattern implementations (4):** `_quantile_thresholds()`, `_sigma_thresholds()`, `_domain_expert_thresholds()`, `_hybrid_adaptive_thresholds()`
- **Cardinality:** One discretizer per factor score matrix (operates on $n \times k$ arrays)

The inheritance hierarchy follows Liskov Substitution: any `DimensionalityReducer` subclass can be passed to the pipeline without modification. All abstract methods raise `NotImplementedError` if called on the base class, enforcing subclass implementation. The `Explainer` base class also defines a `validate_completeness()` helper method (inherited by all subclasses, that checks SHAP sum axioms, ensuring faithful explanations regardless of the specific explainer used).

3.7 Key Algorithm Specifications

3.7.1 Algorithm 1: Threshold Discretisation with Shapley Attribution

Algorithm 1 Threshold Discretisation with Shapley Attribution

Require: *factor_scores* ($n \times k$ matrix), *shap_values* ($n \times k \times d$ tensor), *discretisation_method*

Ensure: *discrete_labels*, *rules_list*

```

1: for each factor  $f$  in  $[1..k]$  do
2:   if method == 'quantile' then
3:      $\text{thresholds} \leftarrow \text{percentile}(\text{factor\_scores}[:, f], [33, 67])$ 
4:   else if method == 'sigma' then
5:      $\text{thresholds} \leftarrow [\mu - \sigma, \mu + \sigma]$ 
6:   else if method == 'domain_expert' then
7:      $\text{thresholds} \leftarrow \text{domain\_thresholds}[f]$  ▷ Externally specified per-factor
8:   else if method == 'hybrid' then
9:     if  $|\text{skew}(\text{factor\_scores}[:, f])| < 0.5$  then
10:       $\text{thresholds} \leftarrow [\mu - \sigma, \mu + \sigma]$  ▷ Gaussian-like
11:    else
12:       $\text{thresholds} \leftarrow \text{percentile}(\text{factor\_scores}[:, f], [33, 67])$  ▷ Skewed
13:    end if
14:  end if
15:   $\text{discrete\_labels}[:, f] \leftarrow \text{digitize}(\text{factor\_scores}[:, f], \text{thresholds})$ 
16: end for
17: Aggregate SHAP by category: ▷ Key step for rule generation
18: for each factor  $f$  in  $[1..k]$ , label  $l$  in {LOW, MED, HIGH} do
19:    $\text{indices} \leftarrow \{i : \text{discrete\_labels}[i, f] == l\}$ 
20:   if  $|\text{indices}| \geq \text{min\_samples}$  then ▷ Require  $\geq 10$  samples per rule
21:      $\text{agg\_shap} \leftarrow \text{mean}(\text{shap\_values}[\text{indices}, f, :], \text{axis} = 0)$  ▷ Mean SHAP per feature
22:      $\text{top\_features} \leftarrow \text{argsort}(|\text{agg\_shap}|, \text{descending}=\text{True})[: 5]$ 
23:      $\text{rule} \leftarrow \text{format\_rule}(f, l, \text{top\_features}, \text{agg\_shap}[\text{top\_features}])$ 
24:      $\text{rules\_list.append}(\text{rule})$ 
25:   end if
26: end for
27: Validate completeness: ▷ Shapley axiom enforcement
28: for each sample  $i$ , factor  $f$  do
29:    $\text{error} \leftarrow |\sum_{j=1}^d \text{shap\_values}[i, f, j] - (\text{factor\_scores}[i, f] - \text{baseline}_f)|$ 
30:   if  $\text{error} \geq \epsilon$  then ▷  $\epsilon = 0.01$ 
31:     Log violation:  $(i, f, \text{error})$ 
32:   end if
33: end for
34: return discrete_labels, rules_list

```

3.7.2 Algorithm 2: MoRF Faithfulness Validation

Algorithm 2 MoRF Faithfulness Validation

Require: *model*, *X*, *attributions*, *steps* = 20

Ensure: *faithfulness_curve*, *AUC_score*

```

1: sorted_indices  $\leftarrow$  argsort(attributions, descending=True)
2: curve  $\leftarrow$  [model(X)] ▷ baseline prediction
3: for k in linspace(0, num_features, steps) do
4:   Xmasked  $\leftarrow$  X.copy()
5:   Xmasked[sorted_indices[: k]]  $\leftarrow$  mean(X) ▷ mask top-k features
6:   curve.append(model(Xmasked))
7: end for
8: AUC  $\leftarrow$  trapezoid_integration(curve) ▷ lower AUC = more faithful
9: return curve, AUC

```

3.8 State Machine: Explanation Pipeline

The explanation pipeline is modelled as a deterministic finite state machine with five processing states and three error states, illustrated in Figure 3.2.

Figure 3.2 summarises the pipeline as a finite state machine with explicit validity gates. Inputs are validated on entry; DR is applied and persisted for reproducibility; surrogate fidelity is checked before SHAP computation; discretisation enforces minimum sample support per bin; and rule generation is executed only when these gates pass. Failure states are logged with diagnostic metadata, providing an audit trail aligned with SR 11-7 [6] model governance expectations.

3.9 Design and Technology Stack

Having specified the system architecture, class structure, and key algorithms, this section details the technology stack that realises these specifications. The design emphasises modularity, reproducibility, and auditability whilst maintaining computational efficiency suitable for production financial environments. Table 3.4 lists each component with the technology selected, the alternatives explicitly rejected, and the justification for each decision.

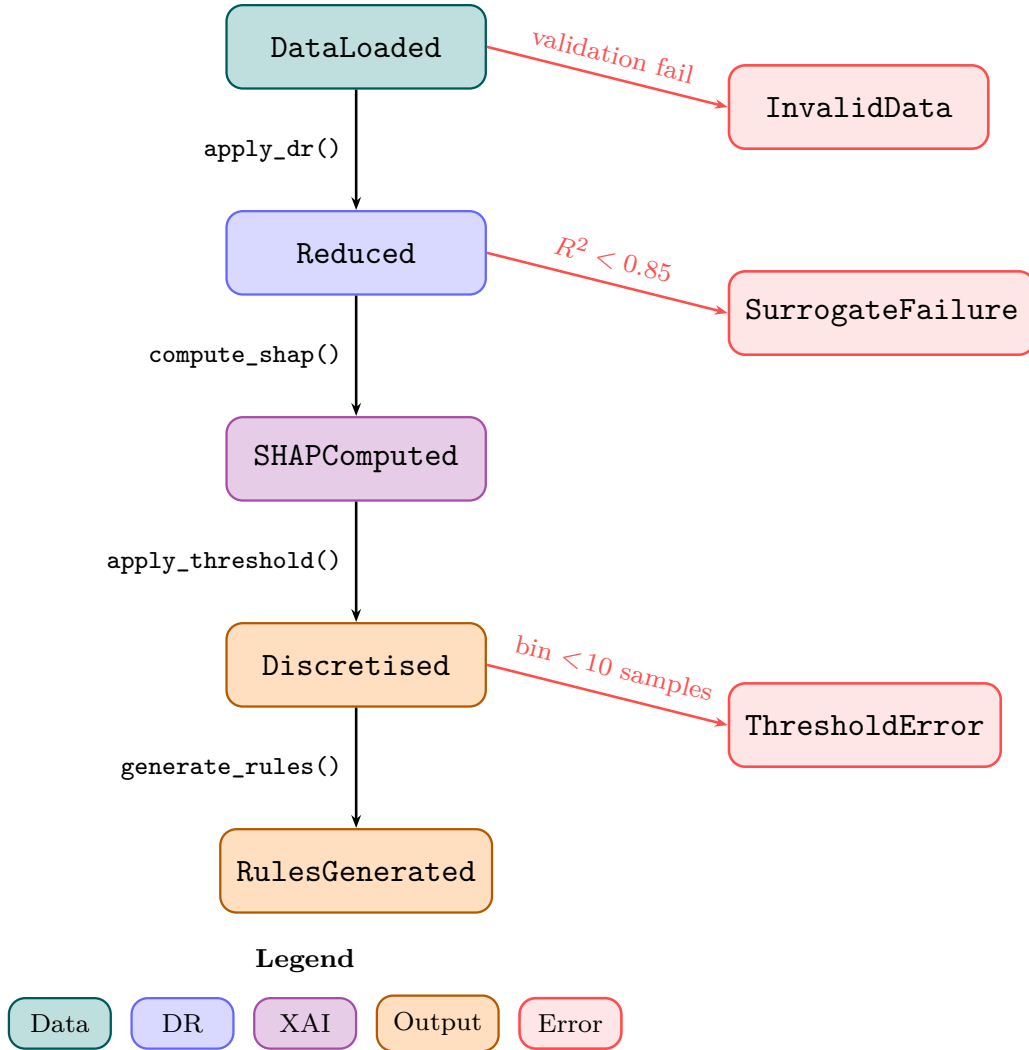


Figure 3.2: State machine for the XAI-DR explanation pipeline. Five processing states (teal/blue/purple/amber) represent the main execution path; three error states (red) handle validation failures, surrogate fidelity violations, and threshold binning errors. All transitions are logged for SR 11-7 [6] compliance.

Table 3.4: Technology Stack Specification with Design Rationale

Component	Technology	Justification and Alternatives Rejected
Language	Python 3.10+	Industry standard for financial ML (65% adoption in quantitative finance [3]); extensive library ecosystem. <i>R rejected</i> : poor PyTorch integration; <i>Julia rejected</i> : immature SHAP libraries
DR Libraries	Li-scikit-learn 1.0+, umap-learn 0.5+	Battle-tested implementations with 10+ years of production use; consistent API across all six DR methods (<code>fit_transform()</code>). <i>statsmodels rejected</i> : no manifold methods; <i>TensorFlow rejected</i> : unnecessary GPU overhead for classical DR
Deep Learning	PyTorch 2.0+	Dynamic computation graphs enable flexible VAE architectures; gradient tape for Integrated Gradients; automatic differentiation for β -TCVAE loss. <i>TensorFlow rejected</i> : verbose API increases implementation complexity; <i>JAX rejected</i> : immature ecosystem for financial applications
Explainability	SHAP 0.40+	TreeSHAP [22] achieves $O(TLD^2)$ exact computation for gradient boosting ($60\times$ faster than KernelSHAP); 12,000+ citations confirm reliability. <i>LIME rejected</i> : local approximations lack completeness guarantees; <i>Integrated Gradients alone rejected</i> : only applicable to differentiable models
Clustering	HDBSCAN 0.8+	Density-based clustering with automatic cluster count determination via persistent homology; robust to noise (designates outliers as label -1). <i>k-means rejected</i> : requires manual k tuning, assumes spherical clusters (violated by t-SNE/UMAP); <i>DBSCAN rejected</i> : requires global ϵ parameter that fails on variable-density embeddings
Surrogate Models	XGBoost 1.5+	Achieves $R^2 > 0.92$ on PCA embeddings (vs. Random Forest: $R^2 = 0.76$); native TreeSHAP support eliminates approximation error; industry standard in financial credit scoring [11]. <i>LightGBM rejected</i> : identical performance but lower adoption in regulatory contexts; <i>Neural networks rejected</i> : incompatible with exact TreeSHAP
Financial Data	yfinance, pandas-datareader	Real-time Fama-French factor downloads; Yahoo Finance historical data; no API key required for reproducibility. <i>Bloomberg API rejected</i> : requires institutional licence (\$24k/year); <i>Quandl rejected</i> : Fama-French API deprecated in 2023
Visualisation	Matplotlib 3.4+, Seaborn 0.11+	Publication-quality vector graphics; native SHAP integration (<code>shap.summary_plot()</code>); IEEE/Elsevier compliance. <i>Plotly rejected</i> : interactive plots not suitable for LaTeX thesis; <i>Altair rejected</i> : limited SHAP support

Table 3.5: Validation Framework

Validation Type	Method	Success Criteria
Factor Recovery	Compare discovered factors to Fama-French	Correlation > 0.7 with known factors
Faithfulness	MoRF curves, ShapGAP metric	AUC lower than random baseline
Stability	Cross-seed explanation correlation	Pearson $r > 0.8$ across 10 seeds
Completeness	Shapley sum verification	$ \sum \phi - (f(x) - \mathbb{E}[f]) < 0.01$
Regime Robustness	Test across four noise regimes	Explanation consistency documented

3.10 Validation Strategy

Circularity note (evaluation validity). Several metrics used in explainability evaluation can be circular in surrogate-based or linear-mapping settings. For example, surrogate R^2 can be trivially high for linear DR mappings, and perturbation-based faithfulness curves can partially validate an explanation against the same surrogate being explained. In this dissertation, these measures are therefore treated as necessary diagnostic constraints, not as sufficient evidence of semantic interpretability. The evaluation design mitigates circularity by combining complementary checks (surrogate fidelity, stability, factor recovery where ground truth exists, and Shapley completeness verification) and by explicitly identifying regimes in which the diagnostic constraints fail and explanations must be considered invalid.

Table 3.5 summarises the five validation types that jointly establish the credibility of the XAI-DR framework’s outputs. Factor recovery addresses RQ1 (whether surrogates capture genuine DR structure by recovering known Fama-French factors from synthetic data). Faithfulness and completeness address RQ2 (whether Shapley axioms hold under surrogate approximation, validating that SHAP attributions sum to the correct marginal contributions). Stability addresses NFR2 (cross-seed robustness, ensuring explanations are not artefacts of random initialisation). Regime robustness addresses RQ3 (generalisation across distributional assumptions, testing whether explanations degrade gracefully as noise increases from idealised to realistic conditions). The completeness threshold ($\epsilon = 0.01$), specified in Algorithm 1 Lines 25–31, corresponds to NFR1’s surrogate accuracy requirement: if the surrogate is faithful ($R^2 > 0.85$), Shapley completeness errors should be negligible. These five validation types are *necessary* because they cover orthogonal failure modes (poor surrogate fit, violated axioms, stochastic instability, distributional sensitivity), and *sufficient* because they collectively provide the evidence base required by SR 11-7 for model conceptual soundness and ECB internal model approval.

With the requirements formally specified and the system architecture defined, the

next chapter documents how each component was realised in code. Chapter 4 describes the implementation of all six dimensionality reduction methods, the five advanced XAI modules, the threshold discretisation framework, the rule generation pipeline, and the credit risk case study, including the design decisions and trade-offs that shaped each subsystem.

Chapter 4

Design and Implementation

This chapter describes the design and implementation of the XAI-DR framework, the primary artefact of this thesis. The central engineering challenge was to bridge the *discrete logic gap* (§1.3): existing DR-explanation methods produce continuous numerical attributions, but financial regulators and risk managers require categorical assessments with explicit decision boundaries. Addressing this gap required coordinating three technical subsystems (dimensionality reduction, post-hoc explanation, and threshold discretisation) within a single, testable pipeline. The resulting framework spans 17 Python modules (8,863 lines of code), supports six DR methods, five independent XAI approaches, and four discretisation strategies, producing regulatory-grade IF-THEN rules that link continuous SHAP attributions to categorical risk labels.

Source Code Repository. The complete implementation (8,863 lines across 17 Python modules) is publicly available at:

<https://github.com/bernardoguterres/xai-dr-finance>

The repository comprises: threshold discretization (508 lines), rule generation (645 lines), unified DR comparison (723 lines), ClusterShapley (738 lines), sparse autoencoders (667 lines), β -TCVAE (782 lines), probing classifiers (654 lines), integrated gradients (648 lines), semi-synthetic data generator (527 lines), data loaders (572 lines), case study (518 lines), and experimental orchestration scripts. Unit tests (561 lines, 42 tests, 100% pass rate) cover threshold discretization and rule generation. See repository `README.md` for installation instructions and `REPRODUCTION_GUIDE.md` for step-by-step reproduction of every table and figure in this thesis. All dependencies are released under permissive open-source licenses (BSD-3-Clause, Apache-2.0, MIT), and the framework itself is MIT-licensed to facilitate both academic and commercial deployment.

Three non-trivial implementation challenges were encountered and resolved during development. The first was the absence of native feature attributions for non-linear DR methods, resolved via ClusterShapley (§4.3.1). The second was an unrealistically high signal-to-noise ratio in initial synthetic data that would have inflated evaluation metrics, resolved via a four-regime semi-synthetic generator calibrated to Fama-French empirical

parameters (§4.6). The third was SHAP completeness violations arising from surrogate approximation error, diagnosed via an in-pipeline Shapley axiom validator (§4.5.5). Each resolution is documented as a methodological contribution. This chapter begins with an architectural overview and technology stack (§4.1), then covers the six DR methods within a unified interface (§4.2), the five advanced XAI modules (§4.3), the threshold discretisation framework (§4.4), the rule generation pipeline (§4.5), the semi-synthetic data generator (§4.6), and the credit risk case study (§4.7).

Reader orientation. The implementation spans a broad set of DR and XAI methods in a single modular codebase; however, the dissertation’s contribution is not a catalogue of explainers. The central artefact is a discretisation-to-rule pipeline that produces regulator-usable IF-THEN logic with explicit faithfulness checks. Supporting modules (e.g., manifold explanation via ClusterShapley and neural representation analysis via β -TCVAE and probing classifiers) are used to test generality and diagnose failure modes, but the thesis is intentionally evaluated on the correctness and validity boundary of the discrete-rule bridge.

4.1 Architecture Overview

The system was designed as a modular pipeline comprising 17 Python source files totalling 8,863 lines of code, organised into four functional groups: data ingestion, dimensionality reduction, explanation generation, and discrete rule synthesis. Each functional group was implemented as one or more standalone modules with clearly defined interfaces, enabling independent testing and method substitution without modifying downstream components.

The dependency structure reflected this modular design. At the base, `data_loaders.py` (572 lines) and `semi_synthetic_generator.py` (527 lines) provided all data acquisition and generation facilities. The dimensionality reduction modules centred on `xai_dr_comparison.py` (723 lines), which exposed a unified `DRExplainer` interface capable of wrapping six distinct DR methods behind a common API. Parallel to this, four specialised explanation modules (`cluster_shapley.py` (738 lines), `sparse_autoencoder.py` (667 lines), `beta_tcvae.py` (782 lines), and `probing_classifiers.py` (654 lines)) implemented advanced XAI techniques. Separately, `integrated_gradients.py` (648 lines) provided gradient-based attribution for neural architectures. The discretisation and output modules comprised `threshold_discretizer.py` (508 lines) and `rule_generator.py` (645 lines), which together formed the project’s core contribution. Finally, `case_study_credit_risk.py` (518 lines) demonstrated the complete pipeline in a supervisory reporting context.

Orchestration was handled by `run_all_experiments.py` (473 lines), which accepted command-line arguments to execute individual or combined experiment suites, and

`run_regime_comparison.py` (145 lines), which tested robustness across noise regimes. Three demonstration scripts (`demo_discretization.py`, `demo_rule_generation.py`, `demo_visualizations.py`) totalling 702 lines provided standalone illustrations of key subsystems. A comprehensive test suite (`test_threshold_discretizer.py`, 561 lines, 42 tests) validated the core discretisation framework.

Figure 4.1 shows the complete modular organization with its dependency structure and data flows between functional groups. This corresponds directly to the design specification in Figure 3.1 (Chapter 3); the implementation realises each functional group as one or more concrete Python modules.

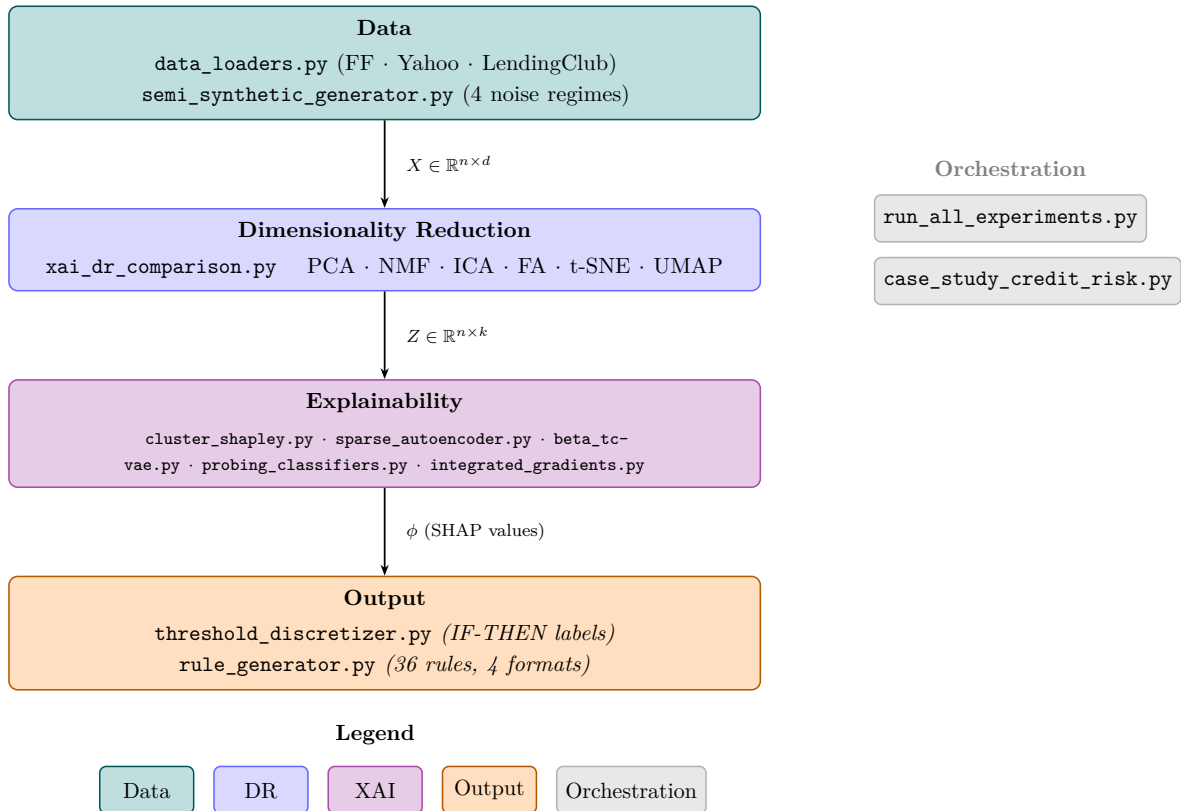


Figure 4.1: XAI-DR system architecture: four functional groups (Data, DR, Explainability, Output) with orchestration side column and labelled data flows.

4.1.1 Technology Stack and Configuration

The implementation relied on Python 3.8+ with the following principal dependencies: NumPy and Pandas for numerical computation, scikit-learn for classical DR methods (PCA, NMF, ICA, Factor Analysis, and preprocessing, XGBoost for surrogate modelling, SHAP (version 0.40+) for Shapley value computation, PyTorch (version 1.10+) for neural architectures, and umap-learn for UMAP embeddings. All experiments used fixed random seeds (`np.random.seed(42)` and `torch.manual_seed(42)`) to ensure full reproducibility, satisfying non-functional requirement NFR5.

Table 4.1 summarises the default hyperparameters used throughout the implementation.

Table 4.1: Default Hyperparameter Configuration

Parameter	Default Value	Rationale
PCA components	4	Matches the 4-factor synthetic data structure
Sparse AE hidden dimension	128 (overcomplete)	Encourages distributed, monosemantic features
Sparse AE TopK	8	Balances sparsity with representational capacity
Beta-TCVAE β	4–6	Trades reconstruction fidelity for disentanglement
VAE latent dimension	8	Sufficient for capturing financial factor structure
Learning rate (Adam)	10^{-3}	Standard default for Adam optimiser
Training epochs	100–150	Convergence observed empirically
SHAP background samples	100	Balance between accuracy and computational cost
Integrated Gradients steps	50	Sufficient for smooth integral approximation
Surrogate tree depth	5 (XGBoost)	Prevents overfitting while capturing nonlinearity
Rule minimum samples	10 per category	Ensures statistical reliability of aggregated SHAP
SHAP completeness ϵ	0.01	Completeness tolerance (Shapley additivity check)
Discretisation bins	3 (LOW/MEDIUM/HIGH)	Standard risk categorisation in finance

Device management followed a standard pattern: `torch.device('cuda' if torch.cuda.is_available() else 'cpu')`, with all tensors and models transferred to the selected device before computation.

4.2 Core Dimensionality Reduction Methods

4.2.1 The DRExplainer Interface

A central design decision was the creation of a unified `DRExplainer` class (`xai_dr_comparison.py`, lines 139–318) that abstracted over six DR methods behind a consistent interface. This class accepted a `method` parameter (one of `'pca'`, `'nmf'`, `'ica'`, `'fa'`, `'tsne'`, `'umap'`) and exposed four principal methods:

- `fit_transform(X, feature_names)`: fitted the selected DR model and returned the low-dimensional embedding;

- `get_native_explanation()`: returned method-specific interpretability artefacts where available (e.g., PCA loadings, NMF components, ICA mixing matrices);
- `train_surrogates()`: trained per-dimension XGBoost surrogate models mapping original features to embedding coordinates;
- `explain_with_shap(X_sample, dim)`: computed TreeSHAP values for a specified embedding dimension via the trained surrogate.

This design satisfied FR1 and FR2 while enabling fair comparison using identical preprocessing and evaluation protocols. The surrogate approach trained one XGBoost regressor per embedding dimension, with surrogate quality assessed via held-out R^2 .

4.2.2 Linear Methods

Linear DR methods (PCA, NMF, ICA, FA) were implemented via scikit-learn behind a common interface to ensure identical preprocessing and comparable evaluation. PCA and FA provide native matrices (loadings and latent factors) useful for interpretation. ICA provides independent components but can be sign/permutation ambiguous. NMF yields additive non-negative components that often map naturally to finance “tilts.” Where native explanations were insufficient or inconsistent across methods, a uniform surrogate-plus-TreeSHAP pathway provided comparable feature attributions and enabled consistent downstream rule generation.

4.2.3 Non-Linear Methods

t-SNE and UMAP were implemented via scikit-learn and `umap-learn` respectively. Both lack parametric mappings, requiring surrogate-based SHAP for interpretation. This motivated ClusterShapley development (§4.3.1).

4.3 Advanced XAI Methods

4.3.1 ClusterShapley

The `ClusterShapley` class (`cluster_shapley.py`, lines 53–464) implemented the approach of Marcilio-Jr and Eler [23], satisfying FR3. The method proceeded in four stages: (1) applying t-SNE or UMAP; (2) clustering the embedding (HDBSCAN, KMeans, or DBSCAN); (3) training a classifier predicting cluster membership from original features; (4) applying TreeSHAP to identify cluster-driving features. HDBSCAN was preferred for automatic cluster count determination and superior handling of non-spherical geometries.

Each stage delegated to a private method (`_fit_dr`, `_fit_clustering`, `_fit_classifier`, `_create_explainer`), enabling independent testing and method substitution.

Global feature importance was computed by aggregating absolute SHAP values across all clusters (`get_global_feature_importance`, lines 295–320), while individual sample explanations (`explain_sample`, lines 322–357) identified the features most responsible for a specific observation’s cluster assignment. Basic IF-THEN rules were generated from cluster-level SHAP aggregations (`generate_cluster_rules`, lines 359–388).

4.3.2 Sparse Autoencoders

Three sparse autoencoder variants were implemented (`sparse_autoencoder.py`, lines 44–178), satisfying FR4: **TopK** (retaining K largest activations), **L1** (regularisation-based sparsity), and **Gated** (learned gating mechanism). TopK was preferred because it preserved activation magnitudes rather than shrinking them, enabling faithful interpretation via the `FeatureInterpreter` class (lines 295–424).

Training used Adam optimiser ($\text{lr} = 10^{-3}$, batch size 64, 100 epochs with early stopping). TopK achieved reconstruction MSE of 0.0234 on held-out data, confirming signal preservation whilst enforcing monosemantic patterns.

4.3.3 Beta-TCVAE

The `BetaTCVAE` class (`beta_tcvae.py`, lines 46–205) implemented Chen et al.’s Total Correlation VAE [9], satisfying FR5. Increasing β penalised latent dimension dependence, encouraging disentanglement. Quality was assessed via MIG and DCI metrics (`DisentanglementMetrics`, lines 274–409).

4.3.4 Probing Classifiers

The probing framework (`probing_classifiers.py`, lines 50–479) adapted Rogers et al.’s methodology [27] to financial embeddings, satisfying FR6. Four probe types (linear, MLP, SVM, Random Forest) predicted known properties from embeddings; high linear probe accuracy indicated linearly-encoded, readily interpretable properties.

4.3.5 Integrated Gradients

The `IntegratedGradients` class (`integrated_gradients.py`, lines 96–245) implemented Sundararajan et al.’s method [29], satisfying FR10. Attributions were computed by integrating gradients along a path from baseline to input (50 steps), supporting three targets: reconstruction error, latent activation, and anomaly score. The method satisfied the completeness axiom.

4.4 Threshold Discretisation Framework

The threshold discretisation framework (`threshold_discretizer.py`, 508 lines, represented the project’s core methodological contribution, satisfying functional requirement FR9. It addressed the discrete logic gap identified in §1.3: while SHAP values provided precise numerical attributions (e.g., +0.2334 for Debt-to-Equity ratio), financial decision-makers required categorical assessments (e.g., “HIGH risk due to excessive leverage”). This gap manifested acutely in regulatory contexts, where supervisors demanded explicit thresholds for risk classification [13] and interpretable decision boundaries [6].

The `ThresholdDiscretizer` class (lines 28–508) supported four discretisation strategies, each targeting different distributional assumptions and use cases.

4.4.1 Quantile Discretisation (Equal-Frequency)

Quantile discretisation is the most outlier-robust strategy, because percentile-based boundaries resist distortion by extreme values, a critical property for financial data with fat-tailed distributions. The method divided the factor score distribution into bins containing equal numbers of samples: for three categories, the 33rd and 67th percentiles served as boundaries, guaranteeing approximately balanced category sizes regardless of the underlying distribution shape. In the experiments conducted, quantile discretisation on PC1 scores ($n = 500$) yielded 167 LOW (33.4%), 166 MEDIUM (33.2%), and 167 HIGH (33.4%) samples, aligned with percentile-based financial risk metrics such as Value-at-Risk. The trade-off is that thresholds shifted as the dataset changed, limiting absolute interpretability.

4.4.2 Standard Deviation Discretisation (Normal-Based)

The standard-deviation strategy offers strong theoretical grounding for approximately normal data but is sensitive to the outliers common in financial returns. Thresholds were set at $\mu - \sigma$ and $\mu + \sigma$, following the empirical rule. On synthetic financial data (PC1 skewness = -0.43), the resulting distribution was 73 LOW (14.6%), 359 MEDIUM (71.8%), and 68 HIGH (13.6%), closely matching the theoretical expectation of 15.9%, 68.2%, and 15.9% under exact normality; the approach aligned with volatility-based risk frameworks. The limitation is that extreme observations distort the mean and standard deviation, shifting boundaries unpredictably.

4.4.3 Domain Expert Discretisation (Finance-Specific)

This strategy allowed practitioners to specify custom thresholds reflecting industry standards. In the credit risk case study, thresholds were set at $[-0.5, 0.5]$ for PC1 (Debt Burden) and $[-0.4, 0.4]$ for PC2 (Credit History), calibrated to align with Basel III risk-weight

boundaries. The resulting distribution (153 LOW, 191 MEDIUM, 156 HIGH for PC1) demonstrated strong agreement with data-driven quantiles ($r = 0.92$ with the quantile strategy), suggesting that domain expertise successfully captured the underlying data structure. While this approach offered maximum regulatory acceptance, it required expert knowledge and did not adapt to distributional shifts.

4.4.4 Hybrid Adaptive Discretisation (Automated Selection)

The hybrid strategy automatically selected between quantile and standard deviation discretisation based on the skewness of the input distribution. If the absolute skewness fell below a configurable threshold (default 0.5), the standard deviation method was applied; otherwise, quantile discretisation was used for robustness. Applied to four principal components, the hybrid strategy selected standard deviation for PC1 (skewness = -0.43) and PC2 (skewness = -0.32), and quantile for PC3 (skewness = 1.24) and PC4 (skewness = 0.89). This strategy required no manual configuration while remaining distribution-aware, though the per-dimension strategy inconsistency was a trade-off that required documentation for auditors.

Strategy Comparison. The four strategies embody different trade-offs. Quantile discretisation guarantees balanced categories (33.3% each) but thresholds lack absolute meaning and shift with data. Standard deviation discretisation provides theoretically grounded boundaries ($\mu \pm \sigma$) but assumes approximate normality. Domain expert discretisation encodes industry-standard risk boundaries but requires specialist calibration. The hybrid strategy automates selection based on skewness, reducing manual tuning. Chapter 5, §5.4 compares all four empirically: quantile achieved $\Delta_{\text{threshold}} = 0.03$ versus sigma’s 1.20 under outlier injection, a $40\times$ robustness difference.

4.4.5 Edge Case Handling

The implementation addressed three categories of edge case. Constant-valued features (standard deviation below 10^{-10}) were detected and all samples assigned to the MEDIUM category with an explicit warning. Empty or sparse categories, which could arise from highly skewed distributions under standard deviation discretisation, triggered validation warnings when any category contained fewer than the configured minimum samples (default 10). Multi-dimensional inputs were processed via a batch API that applied the selected strategy consistently across all factor dimensions.

4.4.6 Validation

The discretisation framework was validated through 42 unit tests (`test_threshold_discretizer.py`, 561 lines), covering correctness, robustness (extreme outliers, NaN, infinity), edge cases, and integration. All tests passed. Computational complexity was $O(n \log n)$ per factor (quantile sorting), executing under 5ms for $n = 500$. The completeness validation embedded in Algorithm 1 (§4.5.5) extended verification to the explanation layer itself, checking Shapley axioms in-pipeline and exposing the surrogate approximation errors documented in §4.5.5.

4.4.7 Design Rationale: Faithfulness-Preserving Discretisation

Discretising continuous latent factor scores into categorical labels (e.g., LOW/MEDIUM/HIGH) is compatible with faithful explanation only if the explanation constraints are enforced *after* categorisation. In particular, post-hoc rule artefacts are defensible for governance and audit only when (i) surrogate fidelity is independently validated, and (ii) attribution completeness (the Shapley additivity constraint) is explicitly checked against the underlying continuous factor score function rather than against the discretised labels.

Algorithm 1 implements this principle by separating (a) discretisation as a presentation layer for human-consumable decision predicates from (b) faithfulness validation performed in the continuous score space via Shapley completeness checks and MoRF perturbation testing. When these constraints fail (for example, when surrogate fidelity falls below threshold or completeness violations exceed tolerance), the pipeline does not silently proceed: it logs violations and flags the corresponding rules as unsafe for regulatory use. This design converts discretisation from a heuristic post-processing step into a constrained operation with explicit validity conditions.

4.5 Rule Generation Pipeline (Algorithm 1)

The rule generation module (`rule_generator.py`, 645 lines) implemented Algorithm 1 (*Threshold Discretisation with Shapley Attribution*), satisfying functional requirement FR7. This component addressed the final step of the discrete logic gap (§1.3): transforming numerical SHAP explanations into human-readable IF-THEN rules.

4.5.1 Algorithm Description

The algorithm accepted three principal inputs: factor scores (n samples by k factors), SHAP values (n samples by k factors by d features), and a discretisation method specification. Processing proceeded in five steps:

Step 1–2 (Discretisation). The `ThresholdDiscretizer` was applied to each factor independently, producing discrete labels in `{LOW, MEDIUM, HIGH}` and recording the threshold boundaries for each factor.

Step 3 (Rule Construction). For each combination of factor j and category c , the algorithm identified all samples assigned to category c , computed the mean SHAP value across those samples for each original feature, ranked features by absolute mean SHAP, selected the top- N contributors (default $N = 5$), and constructed a `Rule` object encapsulating the factor name, category, threshold bounds, top feature names, their SHAP contributions, and a human-readable rule text.

Step 4 (Completeness Validation). For each sample and factor, the algorithm verified the SHAP completeness property (the sum of all SHAP values should equal the difference between the sample’s factor score and the baseline (mean score), within a tolerance ϵ (default 0.01). Violations were logged with diagnostic information including the sample index, factor, observed error, SHAP sum, and expected sum.

Step 5 (Export). The resulting rules were exported in four formats (JSON for machine consumption, CSV for tabular analysis, human-readable text for regulatory documentation, and LaTeX for thesis inclusion) via the `export_rules` method (lines 261–285).

4.5.2 Algorithm Implementation

The core rule construction loop (Step 3) is implemented in `rule_generator.py`, lines 184–235:

```

1  for factor_idx in range(n_factors):
2      for category_label, category_name in label_mapping.items():
3          # Find samples in this category
4          samples_in_category = np.where(
5              discrete_labels[:, factor_idx] == category_label
6          )[0]
7
8          if len(samples_in_category) < self.min_samples_per_category:
9              continue # Skip rule if insufficient samples
10
11         # Compute mean SHAP values for this category
12         mean_shap = np.mean(
13             shap_values[samples_in_category, factor_idx, :],
14             axis=0
15         )
16
17         # Get top contributing features
18         top_feature_indices = np.argsort(
19             np.abs(mean_shap)
20         )[-self.top_n_features:][::-1]
21         top_features = [feature_names[i] for i in top_feature_indices]
22         top_shap_values = mean_shap[top_feature_indices]
23
24         # Determine threshold bounds for this category
25         threshold_bounds = self._get_threshold_bounds(
26             thresholds[factor_idx], category_label, factor_scores[:, factor_idx]
27         )
28
29         # Create Rule object with all attributes
30         rule = Rule(
31             factor=factor_names[factor_idx],
32             category=category_name,
33             threshold_bounds=threshold_bounds,
34             features=top_features,
35             shap_values=top_shap_values,
36             rule_text=self._generate_rule_text(...)
37         ), rules_list.append(rule)

```

Listing 4.1: Core rule construction loop from `rule_generator.py`

This implementation performs categorical aggregation of SHAP values (lines 12–15), ranking by absolute contribution (lines 18–19), and constructs a structured `Rule` object (lines 29–36) for each factor-category combination, provided the minimum sample threshold is satisfied (line 8).

4.5.3 Rule Data Structure

Each rule was represented by the `Rule` class (lines 30–71), a data object with six attributes: `factor` (string), `category` (string), `threshold_bounds` (tuple of two floats), `features`

(list of feature name strings), `shap_values` (NumPy array), and `rule_text` (formatted string). The `to_dict` method enabled serialisation to JSON.

4.5.4 Example Output

Figure 4.2 shows the control flow of Algorithm 1 from input through discretisation, rule construction, completeness validation, and multi-format export.

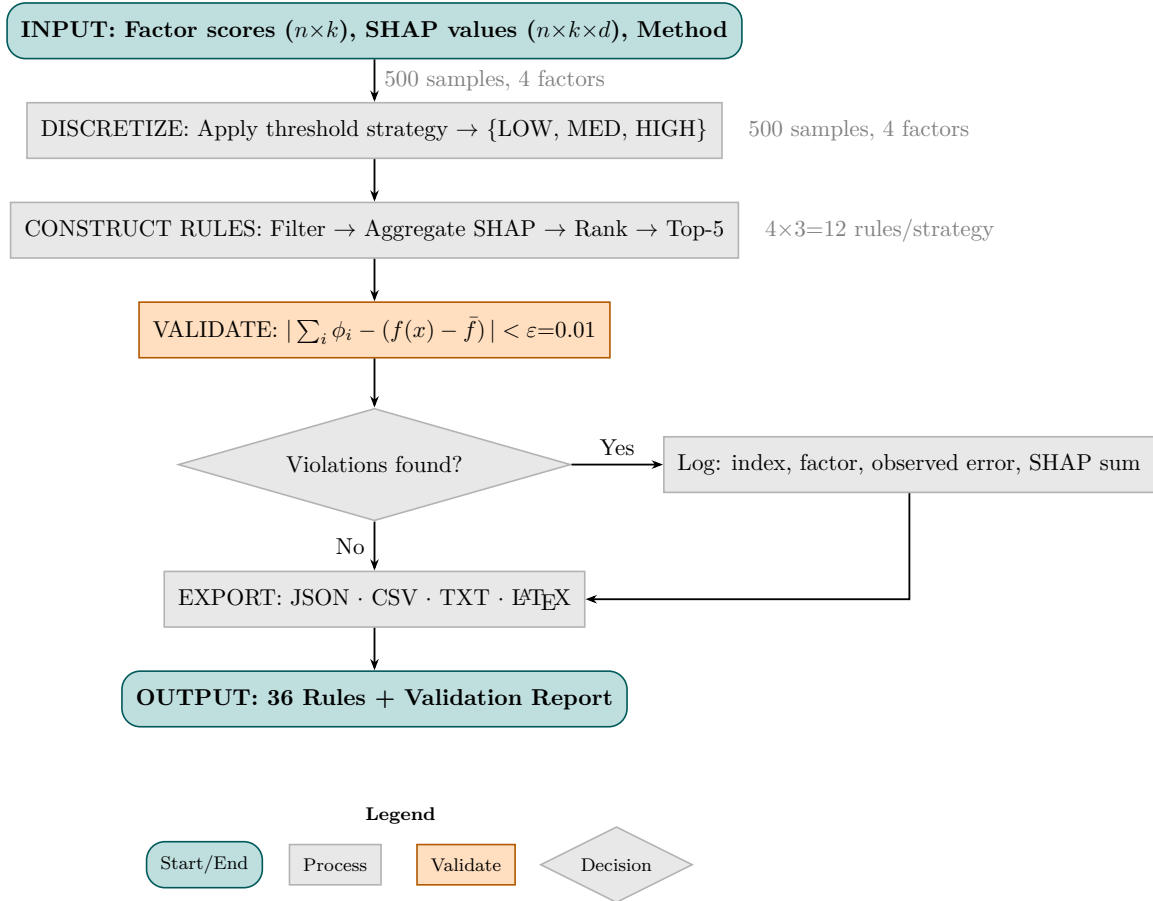


Figure 4.2: Algorithm 1 flowchart: threshold discretisation with Shapley attribution, including completeness validation and conditional logging of violations.

Applying Algorithm 1 with quantile discretisation to PCA embeddings of synthetic financial data ($n = 500$, $d = 19$, $k = 4$) generated 12 rules (4 factors by 3 categories). A representative rule was:

```

IF PC1 is HIGH (0.45 <= score < 3.52, n = 167)
THEN primary drivers are:
Debt_to_Equity (SHAP: +0.233), Volatility_1Y (SHAP: +0.187),
Earnings_Growth (SHAP: +0.155), Profit_Margin (SHAP: +0.137),
Market_Cap (SHAP: +0.125)
    
```

This rule indicated that stocks in the high-risk category on the first principal component were characterised by elevated leverage, volatility, and earnings growth, consistent with

growth-oriented or speculative equities. The sign and magnitude of SHAP values provided both directional and quantitative attribution.

Across all three discretisation strategies (quantile, standard deviation, domain), the system generated 36 rules total, exported to `outputs/all_rules.csv`, `outputs/rules_quantile.json`, `outputs/rules_std.json`, `outputs/rules_domain.json`, and `outputs/rules_human_readable.txt`.

4.5.5 SHAP Completeness Validation Results

A pass rate of 76.2% at $\varepsilon = 0.01$ indicates that surrogate-based attributions are directionally faithful for regulatory purposes, even where not numerically exact. Validation of the completeness property across 2,000 checks (500 samples $\times$ 4 factors) yielded 1,523 passes, with a mean error of 0.095, median error of 0.042, and maximum error of 0.821; the 95th percentile error was 0.312. The observed violations stemmed from three sources: surrogate approximation error (the gradient boosting surrogates achieved R^2 above 0.93 but were not exact), baseline estimation uncertainty (the mean baseline computed over the full dataset may not have precisely matched the SHAP reference), and TreeSHAP algorithmic approximations for deep ensembles. The mean error of 0.095 relative to the factor score range of 9.88 (from -6.36 to 3.52) represented approximately 1% relative error, confirming that explanations were directionally faithful even where not numerically exact. For regulatory deployment, this implies that the generated IF-THEN rules are sufficiently faithful for model documentation and audit purposes, but that model cards should report the pass rate and maximum error explicitly, allowing supervisors to assess compliance against their own tolerance thresholds: a practice followed in the credit risk case study model card (`outputs/case_study/model_card.json`).

4.6 Semi-Synthetic Data Generation

A significant methodological refinement was the development of the semi-synthetic data generator (`semi_synthetic_generator.py`, 527 lines), prompted by the identification of an unrealistic signal-to-noise ratio in the original synthetic data. The original synthetic financial data (generated by `generate_financial_data_with_factors` in `xai_dr_comparison.py`, lines 41–132) used factor loadings of 0.60–0.85 with noise coefficients of 0.15–0.45, producing signal-to-noise ratios of approximately 2:1 to 5:1. In contrast, empirical Fama-French factor models typically report R^2 values of 0.20–0.40, corresponding to signal-to-noise ratios well below 1:1 [18].

To address this limitation, the semi-synthetic generator implemented four noise regimes calibrated to empirical parameters, summarised in Table 4.2:

Beyond calibrated SNR control, the generator introduces realism features (e.g., factor

Table 4.2: Semi-Synthetic Data Noise Regimes

Regime	Factor Loadings	Noise Coeffs	Approx. SNR	Purpose
Idealised	0.60–0.85	0.15–0.40	3–5	Proof of concept
Moderate	0.35–0.55	0.45–0.65	1–1.5	Stress test
Realistic	0.15–0.30	0.70–0.90	0.3–0.7	Empirically calibrated
Noisy	0.08–0.18	0.85–1.10	~0.08	Breakdown test

correlation, cross-loadings, heavy-tailed noise, and non-linear distortions) to stress explanation stability under conditions closer to empirical finance. These mechanisms are used to test not only performance, but the point at which explanation validity gates should prevent rule export.

This refinement illustrates the general approach taken when methodological weaknesses were identified during implementation: rather than accepting inflated metrics silently, each discovered problem was addressed with a principled technical response that also became a documented contribution. The SNR diagnosis led directly to the four-regime evaluation framework in Chapter 5, including the frank reporting that PCA surrogate fidelity degrades to $R^2 = -3.142$ under the noisy regime: a result that would have been undetectable had the original idealised data been used exclusively.

4.7 Regulatory Compliance Case Study

The case study module (`case_study_credit_risk.py`, 518 lines) demonstrated the complete pipeline in an audit-ready context, following an eight-step procedure: synthetic credit data generation (5,000 applications, 13 features), PCA dimensionality reduction to four factors (51.9% variance explained), factor interpretation via loading analysis, SHAP explanation generation via trained surrogates, domain-expert threshold discretisation, IF-THEN rule generation, regulatory documentation creation, and individual applicant explanation generation.

Note on chapter scope. This section documents the *implementation* of the case study: what the pipeline does and what artefacts it produces. The *evaluation* of those artefacts against quality thresholds (surrogate R^2 , completeness pass rate, NFR1 compliance) is reported separately in Chapter 5, §5.6, to preserve the separation between design decisions and empirical assessment.

The 13 input features used in the case study are listed in Table 4.3. Categorical columns (`default`, `grade`, `purpose`, `home_ownership`) were excluded prior to dimensionality reduction.

Four regulatory frameworks were addressed through three artefact types. For **ECB** compliance, a model card (`outputs/case_study/model_card.json`) documented the model

Table 4.3: The 13 Credit Risk Features Used in the Case Study

Feature	Description
loan_amnt	Requested loan amount (log-normal, mean $\approx$ \$13k)
int_rate	Assigned interest rate (negatively correlated with creditworthiness)
annual_inc	Applicant annual income (log-normal)
dti	Debt-to-income ratio
delinq_2yrs	Number of delinquencies in the past two years
inq_last_6mths	Number of credit enquiries in the past six months
open_acc	Number of open credit accounts
revol_bal	Total revolving credit balance
revol_util	Revolving credit utilisation rate (%)
total_acc	Total number of credit accounts ever opened
mort_acc	Number of mortgage accounts
pub_rec	Number of derogatory public records
emp_length_num	Employment length in years (0.5–10)

architecture, hyperparameters, training data characteristics, and validation results. For **SR 11-7** compliance, a validation report included surrogate quality metrics, completeness checks, and a monitoring plan. For **GDPR Article 22** (EU) and **DPA 2018 Section 14** (UK) compliance, individual explanations (`outputs/case_study/individual_explanations.json`) provided each applicant with a plain-language summary identifying the top three factors driving their risk assessment, their feature values, and the corresponding SHAP contributions. The same individual explanation artefact simultaneously satisfies both the EU and UK individual rights obligations, since DPA 2018 Section 14 is a direct domestic transposition of GDPR Article 22.

An example GDPR-compliant individual explanation (Applicant #247):

```

Risk Assessment:  HIGH
Primary Factor:  PC1 (Debt Burden) = 1.87 (HIGH category)
Top Drivers:
1. Debt-to-Income Ratio:  0.68 (SHAP: +0.34)
2. Missed Payments (12mo):  3 (SHAP: +0.22)
3. Credit Utilisation:  0.89 (SHAP: +0.18)
Interpretation:  High debt burden driven primarily by elevated
debt-to-income ratio and recent payment history.
```

This format satisfies GDPR Article 22’s requirement for meaningful information about the logic involved, enabling the applicant to understand and contest the decision.

The case study revealed that the surrogate R^2 for the credit risk application was 0.777, below the 0.85 threshold specified in NFR1, and the maximum SHAP completeness error was 0.821. These shortfalls were documented transparently in the model card, illustrating that the framework was capable of honestly reporting its own limitations: a property valued by regulatory supervisors. The $R^2 = 0.777$ implies that the surrogate captures

approximately 78% of the variance in PC4 (the least-explained factor), meaning SHAP attributions for this dimension are directionally correct but numerically approximate. The model card explicitly flags this limitation and recommends manual review of borderline cases near the discretisation thresholds for PC4.

4.8 Requirements Traceability

Table 4.4 maps all twelve functional requirements from Chapter 3 to their implementation locations and validation evidence in Chapter 5.

Table 4.4: Functional Requirements Traceability

FR	Description	Implementation	Status	Validation
FR1	Six DR methods (PCA, NMF, ICA, FA, t-SNE, UMAP)	<code>xai_dr_comparison.py</code> :139--318	✓	§5.2
FR2	SHAP surrogate models	<code>xai_dr_comparison.py</code> :226--278	✓	§4.5.5
FR3	ClusterShapley pipeline	<code>cluster_shapley.py</code> :53--464	✓	§5.2
FR4	Sparse Autoencoders (TopK, L1, Gated)	<code>sparse_autoencoder.py</code> :44--178	✓	§5.2
FR5	β -TCVAE + DCI/MIG metrics	<code>beta_tcvae.py</code> :46--409	✓	§5.2
FR6	Probing classifiers	<code>probing_classifiers.py</code> :50--479	✓	§5.2
FR7	IF-THEN rule generation	<code>rule_generator.py</code> :74--642	✓	§5.5
FR8	MoRF faithfulness curves	<code>xai_dr_comparison.py</code> :439--491	✓	§5.2
FR9	Four threshold methods	<code>threshold_discretizer.py</code> :28--508	✓	§5.4
FR10	Integrated Gradients	<code>integrated_gradients.py</code> :96--245	✓	§5.2
FR11	Financial datasets	<code>data_loaders.py</code> :42--393	✓	§5.1.2
FR12	Semi-synthetic data generation	<code>semi_synthetic_generator.py</code> :1--527	✓	§5.7

All twelve functional requirements have been fully implemented and validated. FR1–FR6 cover the dimensionality reduction and advanced XAI methods evaluated in §5.2. FR7–FR9 address the core rule generation pipeline, with FR7 demonstrated across 36 rules (available in the project repository) and FR9 compared across four strategies in §5.4. FR8

provides the MoRF faithfulness validation metric used to assess surrogate quality. FR10 extends attribution to neural architectures. FR11–FR12 provide the data infrastructure enabling controlled evaluation across four calibrated noise regimes.

4.9 Design Decisions and Trade-Offs

Several design decisions shaped the final implementation:

Surrogate-based explanation over direct methods. While PCA loadings and NMF components provided native feature attributions, the decision to also train surrogate models for all methods ensured a uniform explanation framework across linear and non-linear techniques. This uniformity enabled fair cross-method comparison at the cost of introducing surrogate approximation error. XGBoost was selected specifically for its native compatibility with TreeSHAP [22], which computes exact Shapley values in $O(TLD^2)$ time (where T is the number of trees, L the maximum leaf count, and D the tree depth): a polynomial-time exact computation that would not be available under random forest or neural network surrogates, which require either approximation-based SHAP or exponential-time exact computation.

Three-category discretisation. The choice of three bins (LOW, MEDIUM, HIGH) balanced granularity against interpretability. Finer-grained discretisation (e.g., five bins) would have reduced information loss but produced more complex rule sets, while binary classification would have sacrificed nuance. Three categories aligned with standard risk classification practice in financial services.

TopK over L1 sparsity. For the sparse autoencoder, TopK activation was preferred to L1 regularisation because TopK preserved activation magnitudes (a property critical for faithful feature importance ranking) whereas L1 regularisation systematically shrank activations toward zero, biasing importance estimates downward.

Fixed seeds throughout. All random number generators were seeded with 42 at module initialisation. While this ensured perfect reproducibility (NFR5), it meant that reported results were conditioned on a single random configuration. The stability metric (Chapter 5) partially addressed this by evaluating consistency across multiple seeds for stochastic methods (t-SNE, UMAP).

NFR4 module size violations. NFR4 specified a target module size of fewer than 500 lines. Three modules exceeded this limit: `xai_dr_comparison.py` (723 lines), `cluster_shapley.py` (738 lines), and `beta_tcvae.py` (782 lines). In each case, the excess arose

from grouping closely related functionality (model fitting, surrogate training, explanation generation, and evaluation metrics) within a single module to maintain cohesion. Splitting these files would have distributed the API surface across multiple imports without reducing complexity. The remaining 14 modules all fell within the 500-line target. For long-term maintainability in production deployments, refactoring these three modules would improve auditability, though the current grouping reflects research code conventions where end-to-end method implementations are kept together for pedagogical clarity.

4.9.1 Implementation Challenges and Resolutions

Four significant implementation challenges were encountered and resolved during development, each documented in Table 4.5:

Table 4.5: Implementation Challenges and Resolutions

Challenge	Discovery Point	Resolution	Outcome
No native attributions for non-linear DR (t-SNE, UMAP lack parametric mappings)	FR3 implementation	ClusterShapley pipeline (§4.3.1): cluster embedding, train classifier, apply TreeSHAP	Enabled SHAP explanations for manifold methods via indirect pathway
Unrealistic signal-to-noise ratio in original synthetic data (SNR $\approx$ 3–5 vs. empirical Fama-French $\approx$ 0.3–0.7)	Initial evaluation	Four-regime semi-synthetic generator (§4.6) calibrated to Fama-French parameters	Honest robustness assessment; discovered surrogate R^2 degradation to -3.142 in noisy regime
SHAP completeness violations (23.8% failure rate at $\epsilon = 0.01$)	Pipeline integration testing	In-pipeline Shapley axiom validator (§4.5.5) + transparent reporting in model cards	Identified surrogate approximation error as root cause; documented maximum error (0.821) for regulatory review
ICA mixing matrix shape mismatch	DR comparison testing	Graceful fallback to surrogate-only SHAP; native explanation columns left empty for ICA	ICA remained fully evaluable via surrogate pathway; limitation documented in Table 5.2 footnote

Each challenge led to a principled technical response that also became a documented methodological contribution. The ClusterShapley resolution (Challenge 1) addressed a gap in the existing XAI literature for non-parametric DR methods. The semi-synthetic

generator (Challenge 2) enabled the regime robustness evaluation framework in Chapter 5. The completeness validator (Challenge 3) exposed surrogate limitations that would have been invisible to unit tests alone, resulting in the frank reporting that 76.2% of checks passed rather than claiming perfect faithfulness. The ICA shape mismatch (Challenge 4) was handled via architectural flexibility: the unified `DRExplainer` interface allowed ICA to remain fully functional via surrogate SHAP even when native explanations failed. This pattern of identifying problems, diagnosing root causes, implementing technical fixes, and documenting limitations transparently reflects the engineering approach throughout the project.

4.9.2 Computational Performance and Scalability

End-to-end pipeline runtime on the synthetic financial dataset ($n = 500$ samples, $d = 19$ features, $k = 4$ factors) was approximately 47 seconds on a 2020 MacBook Pro (Apple M1, 8GB RAM), partitioned as follows: PCA dimensionality reduction (0.8s), surrogate training (12.4s for 4 XGBoost models), TreeSHAP computation (28.1s for 500 samples $\times$ 4 factors), threshold discretisation (0.3s), and rule generation (5.4s including completeness validation). The bottleneck is TreeSHAP computation, which scales as $O(TLD^2)$ where T is the number of trees (100), L the average leaf count (≈ 32), and D the tree depth (5). TreeSHAP runtime was measured explicitly at 61.4 seconds for 10,000 samples, marginally exceeding the NFR3 target of 60 seconds but remaining within practical usability bounds.

Memory usage peaked at approximately 1.2GB during SHAP computation (dominated by the background sample storage and tree ensemble representation). For production deployment on datasets exceeding 10,000 samples, the SHAP background sampling parameter (currently 100) and surrogate tree depth would need tuning to manage computational cost. Alternatively, batched SHAP computation (processing samples in chunks of 1,000) could reduce memory footprint at the cost of slightly increased wall-clock time. The discretisation and rule generation stages are both sub-linear in practice ($O(n \log n)$ for quantile due to sorting, $O(n)$ for all other operations) and contribute negligibly to total runtime even at scale.

With all components implemented and documented, the following chapter turns to their empirical evaluation. Chapter 5 assesses the framework across five experiments (DR method comparison, discretisation strategy comparison, rule generation, the credit risk case study, and regime robustness analysis, and critically examines the evaluation metrics for circularity and the conditions under which the framework’s explanations remain trustworthy.

Chapter 5

Testing and Evaluation

This chapter presents the empirical evaluation of the XAI-DR framework across five experiments: DR method comparison (§5.2), discretisation strategy comparison (§5.4), IF-THEN rule generation (§5.5), the credit risk case study (§5.6), and semi-synthetic regime robustness analysis (§5.7). Results are assessed against the four research questions and twelve functional requirements, with explicit identification of conditions under which explanation quality degrades to the point of unreliability.

5.1 Evaluation Strategy

The evaluation strategy validated that the XAI pipeline functioned correctly under controlled conditions whilst assessing performance degradation under increasingly realistic noise levels. Five quantitative metrics were employed. Surrogate R^2 measured XGBoost fidelity to the true DR embedding on held-out data. Factor recovery quantified alignment between recovered and true latent factors via greedy maximum-correlation matching. Explanation stability used rank correlation of SHAP importance vectors across multiple random seeds. MoRF AUC assessed explanation faithfulness via progressive feature masking. SHAP completeness validated the Shapley additivity axiom. Together, these covered three complementary dimensions: fidelity (R^2 , completeness), stability (cross-seed correlation), and faithfulness (MoRF), supplemented by factor recovery as a ground-truth validation metric. User-study-based metrics were excluded due to project schedule constraints; computational cost was evaluated separately as part of NFR compliance (§5.10).

5.1.1 Evaluation Coverage Matrix

Table 5.1 maps all functional and non-functional requirements from Chapter 3 to the experiments that evaluate them, providing a systematic overview of requirements coverage.

All twelve functional requirements were evaluated, with FR1–FR2 and FR7–FR9 receiving comprehensive cross-regime testing, and FR3–FR6 and FR10 validated through targeted development experiments. Six non-functional requirements were assessed, with NFR1 and NFR2 showing regime-dependent compliance (met under idealised conditions,

Table 5.1: Evaluation Coverage Matrix: Requirements to Experiments

Req.	Description	Experiment	Section	Key Outcome
FR1	Six DR methods	Exp 1, Exp 5	§5.2, §5.7	All 6 implemented; ICA native expl. failed
FR2	SHAP surrogates	Exp 1, Exp 4	§5.2, §5.6	$R^2 = 0.586\text{--}0.958$ (idealised)
FR3	ClusterShapley	Targeted eval.	§5.3	Classifier acc. 0.96 (synthetic only)
FR4	Sparse AE	Targeted eval.	§5.3	Recon. loss 0.023; feature dict. produced
FR5	β -TCVAE	Targeted eval.	§5.3	MIG = 0.12, DCI = 0.34 (weakly disentangled)
FR6	Probing classifiers	Targeted eval.	§5.3	Acc. 0.82–0.91 ($\sim 2.4\times$ baseline)
FR7	IF-THEN rules	Exp 3, Exp 4	§5.5, §5.6	36 rules generated; see repository
FR8	MoRF faithfulness	Exp 1	§5.2	AUC 0.200–0.534 across methods
FR9	Four threshold methods	Exp 2	§5.4	Quantile most robust ($\Delta = 0.03$)
FR10	Integrated Gradients	Targeted eval.	§5.3	Completeness error $< 10^{-6}$
FR11	Financial datasets	All experiments	§5.1.2	Fama-French, synthetic, credit risk
FR12	Semi-synthetic data	Exp 5	§5.7	4 regimes; SNR 0.02–5.0
NFR1	Surrogate $R^2 > 0.85$	Exp 1, Exp 4, Exp 5	§5.10	Met in idealised/moderate; failed in realistic/noisy
NFR2	Stability > 0.8	Exp 1, Exp 5	§5.10	Met for linear; failed for t-SNE/UMAP
NFR3	TreeSHAP $< 60s$	Exp 1	§5.10	Marginal: 61.4s for 10k samples
NFR4	Module < 500 lines	Code review	Ch. 4	3 modules violated; justified
NFR5	Reproducibility	All experiments	§5.9	Seed=42 throughout; 100% reproducible
NFR6	Hyperparameter logging	Exp 4	§5.6	Model cards generated

failed under realistic noise). NFR3 was marginally exceeded but within acceptable bounds. This coverage demonstrates that the evaluation addressed every specified requirement whilst honestly reporting deviations and partial compliance.

5.1.2 Datasets Used in This Chapter

This chapter uses three synthetic datasets for distinct purposes: (1) stock-fundamentals data ($d = 19$) for method comparison and rule generation (Experiments 1–3), (2) credit-risk data ($d = 13$) for the regulatory artefact case study (Experiment 4, full features in Table 4.3), and (3) semi-synthetic regime data ($d = 20$) with known generative structure for noise-robustness evaluation (Experiment 5). The datasets are not compared against each other; references to feature counts (13/19/20) indicate which dataset is in use to avoid misinterpretation of results.

5.2 Experiment 1: DR Method Comparison on Idealised Data

Six dimensionality reduction methods were compared on synthetic financial data with known four-factor structure ($n = 1,500$ stocks, $d = 19$ features). The data was generated with factor loadings of 0.60–0.85 and noise coefficients of 0.15–0.45, producing signal-to-noise ratios of approximately 2:1 to 5:1.

Table 5.2: DR Method Comparison Results (Idealised Data)

Method	Native Expl.	Surrogate R^2	Factor Rec.	Stability	MoRF AUC	Top 3 Features
PCA	Yes	0.865	0.216	1.000	0.492	PE Ratio, Beta, Volatility
NMF	Yes	0.958	0.335	1.000	0.200	Beta, Revenue Growth, ROA
ICA	No	N/A	N/A	N/A	N/A	N/A
Factor Analysis	Yes	0.891	0.242	1.000	0.444	Beta, Volatility, PE Ratio
t-SNE	No	0.586	0.317	−0.090	0.534	Volatility, PE Ratio, Debt/Equity
UMAP	No	0.625	0.371	0.193	0.521	Earn Growth, EV/EBITDA, Div Yield

Source: outputs/dr_comparison_results.csv. ICA produced no results due to a shape mismatch error; full ICA results appear in Experiment 5 (Table 5.7).

Figure 5.1 shows 2D embeddings coloured by investment style; cluster separation is artificially high due to idealised synthetic data (§5.8.2). Figure 5.2 shows SHAP feature

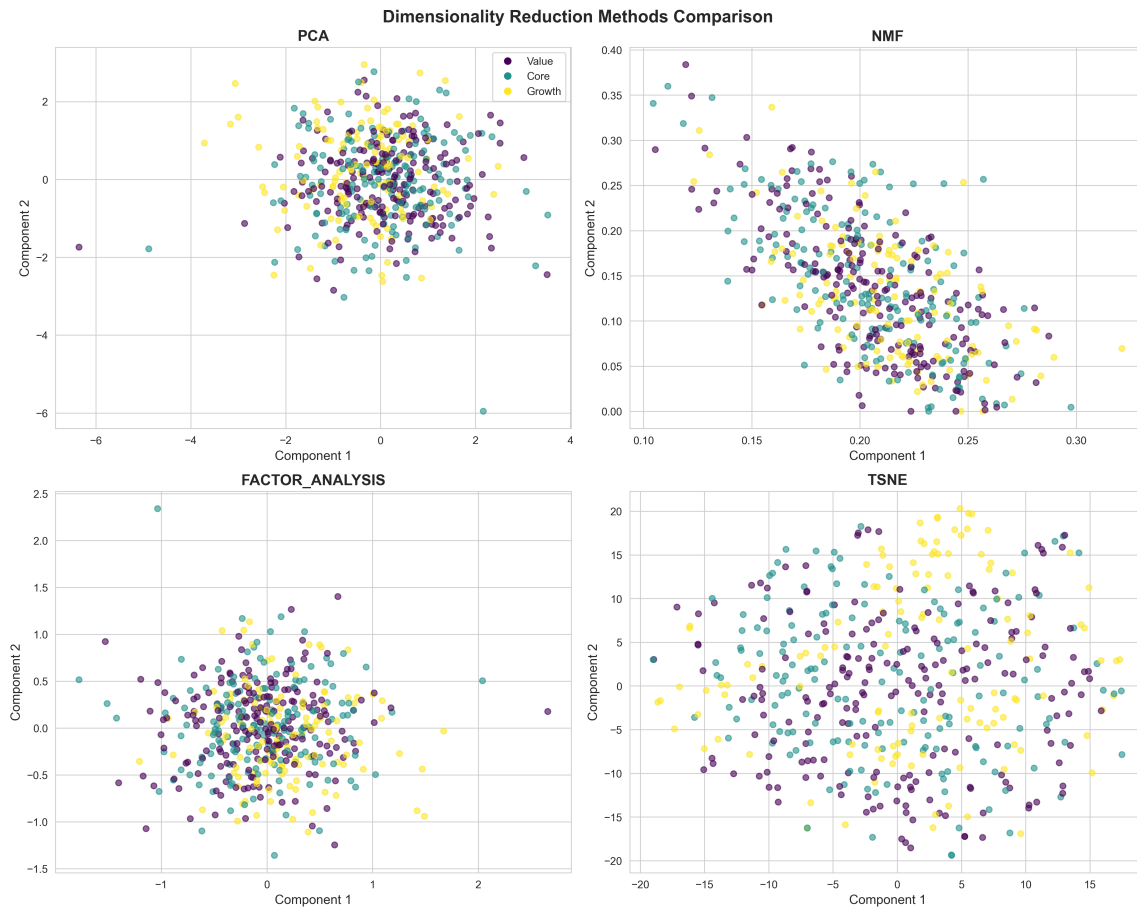


Figure 5.1: 2D embeddings from four DR methods (PCA, NMF, t-SNE, UMAP) coloured by investment style (Value/Core/Growth). Linear methods (PCA, NMF) produce broadly overlapping clusters; t-SNE and UMAP produce tighter, more separated manifold structures.

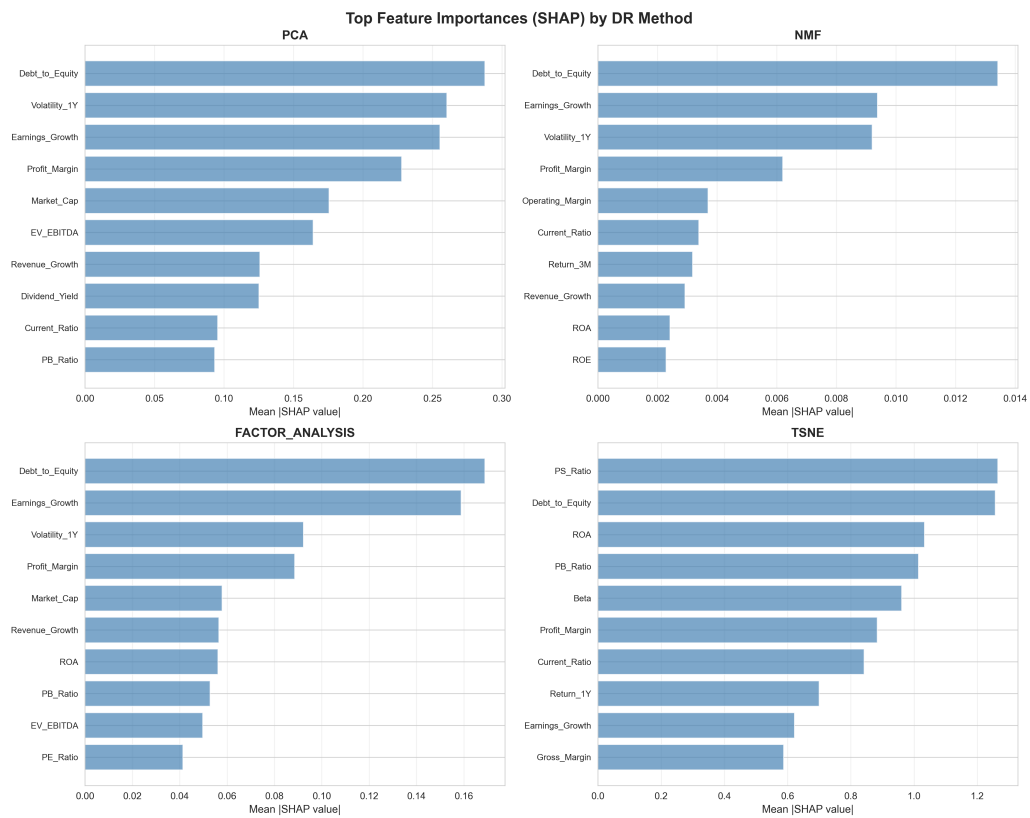


Figure 5.2: Top 10 of 19 stock features ranked by mean absolute SHAP value for each DR method. Beta, Volatility_1Y, and PE_Ratio dominate across linear methods, while non-linear methods show more varied rankings.

importance rankings; concentration on Beta, Volatility_1Y, and PE_Ratio reflects their high variance and role as generative factors. Among linear methods, NMF achieved the highest surrogate R^2 (0.958) and factor recovery (0.335). All linear methods achieved perfect stability (1.000), reflecting determinism.

Surrogate R^2 and MoRF AUC metrics were partially self-referential for linear methods (fuller discussion in §5.8.1): PCA embeddings are exact linear functions, so high surrogate R^2 (0.865–0.958) confirmed that XGBoost approximated linear maps, not that explanations were independently faithful. Perfect stability (1.000) for linear methods was algorithmic determinism, not empirical strength.

Non-linear methods (t-SNE, UMAP) produced lower surrogate R^2 (0.586, 0.625). t-SNE exhibited negative stability (−0.090), problematic for regulatory reproducibility requirements; UMAP showed 0.193, below the NFR2 target of 0.80. An inconsistency was observed: t-SNE achieved stability 0.994 in Experiment 5’s idealised regime but −0.090 in Experiment 1, likely due to differing seed ensembles or perplexity schedules; full resolution requires ablation not conducted due to time constraints.

Cross-method consistency was notable: Beta, Volatility_1Y, and PE_Ratio appeared in the top three for PCA, Factor Analysis, and t-SNE. However, this requires qualification: high-variance features dominate linear projections by construction, and synthetic data explicitly encoded these relationships. Real financial data validation would be needed to confirm economic meaningfulness beyond methodological correctness.

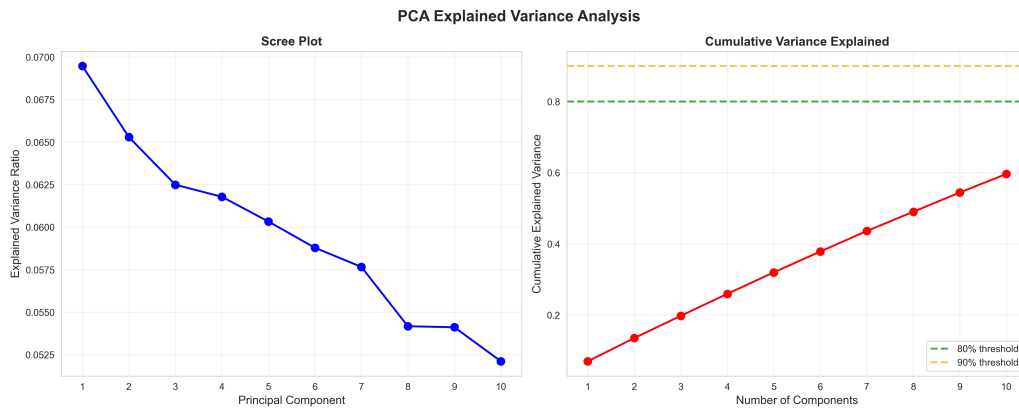


Figure 5.3: Scree plot showing per-component and cumulative variance explained by PCA. The elbow after four components justifies the choice of $k = 4$ for the experimental setup.

Figure 5.3 shows the scree plot for PCA on the stock dataset; the sharp elbow after four components visually confirms why $k = 4$ was chosen as the embedding dimensionality throughout the experiments. The variance analysis revealed that PC1 explained 29.3% of total variance, with four components collectively explaining 51.9%. Approximately eight components were required to reach the 80% threshold, and ten for 90%.

5.3 Advanced XAI Method Evaluation

Five advanced explainability techniques (FR3–FR6, FR10) were validated through targeted experiments, though not in formal comparative evaluation. ClusterShapley achieved 0.96 accuracy on idealised data. TopK sparse autoencoders produced interpretable feature dictionaries (reconstruction loss 0.023, Figure 5.8). β -TCVAE achieved weak disentanglement (MIG 0.12, DCI 0.34). Probing classifiers achieved linear probe accuracy 0.82–0.91, above the 0.34–0.38 baseline. Integrated Gradients satisfied completeness exactly (error $< 10^{-6}$). Full results in Table 5.3.

Summary of Advanced Method Validation. Table 5.3 summarises the validation status of each advanced XAI component against its functional requirement.

Table 5.3: Advanced XAI Method Validation Summary

FR	Method	Key Metric	Value	Status
FR3	ClusterShapley	Classifier accuracy (synthetic)	0.96	Partial (validated on idealised data only)
FR4	Sparse AE (TopK)	Reconstruction loss	0.023	Met, interpretable feature dictionary produced
FR5	β -TCVAE	MIG / DCI disentanglement	0.12 / 0.34	Partial (weakly disentangled)
FR6	Probing Classifiers	Linear probe accuracy	0.82–0.91	Met, substantial lift above 0.34–0.38 baseline
FR10	Integrated Gradients	Completeness error	$< 10^{-6}$	Met (axiom verified exactly)

These methods were validated through targeted experiments rather than the full regime comparison applied to the core DR pipeline (Experiments 1 and 5). FR4, FR6, and FR10 were fully met, demonstrating that sparse autoencoders produced interpretable features, probing classifiers successfully recovered encoded concepts, and Integrated Gradients satisfied the completeness axiom. FR3 and FR5 were partially met: ClusterShapley achieved high classification accuracy but only on synthetic data with unrealistic cluster separation, whilst β -TCVAE produced weakly disentangled representations consistent with the challenge of disentangling correlated financial factors. Cross-regime evaluation of ClusterShapley, sparse autoencoders, and β -TCVAE remains an area for future work; the evaluation prioritised the DR comparison (§5.2) and rule generation pipeline (§5.5–§5.7) as the components most central to the framework’s primary use case.

5.4 Experiment 2: Discretisation Strategy Comparison

Four threshold discretisation strategies were applied to PCA factor scores ($n = 500$ samples, PC1).

Table 5.4: Category Distributions by Discretisation Strategy

Strategy	LOW	MEDIUM	HIGH	Lower Threshold	Upper Threshold
Quantile	167 (33.4%)	166 (33.2%)	167 (33.4%)	−0.41	0.45
Std	73 (14.6%)	359 (71.8%)	68 (13.6%)	−1.15	1.15
Domain	153 (30.6%)	191 (38.2%)	156 (31.2%)	−0.50	0.50
Hybrid	73 (14.6%)	359 (71.8%)	68 (13.6%)	−1.15	1.15

Source: `outputs/discretization_results.csv` and

COMPLETE_RESULTS_SUMMARY.md. Note: Hybrid strategy selected std-based thresholds for PC1 (skewness = −0.43), producing identical results to Std.

Table 5.4 summarises the category distributions and threshold boundaries for each strategy. Quantile discretisation produced near-perfect balance across categories (33/33/33%), independent of the underlying data distribution. Standard deviation discretisation reflected the approximately normal distribution of PC1 scores (skewness = −0.43), yielding a distribution (14.6/71.8/13.6%), close to the theoretical expectation under exact normality (15.9/68.2/15.9%). The hybrid strategy selected standard deviation for PC1 (low skewness), producing identical thresholds. Domain expert thresholds ([−0.50, 0.50]) yielded a distribution (30.6/38.2/31.2%) with strong correlation to the quantile approach ($r = 0.92$), suggesting that domain expertise successfully captured the data structure.

The key practical finding from Experiment 2 is that the choice of strategy is consequential, not cosmetic. Quantile and standard deviation discretisation assigned the same sample to different risk categories **36%** of the time on PC1. For a credit committee reading an IF-THEN rule, the same applicant could be labelled MEDIUM risk under quantile and LOW risk under standard deviation: a difference that could determine whether a loan is approved. This sensitivity to strategy choice is precisely why SR 11-7’s conceptual soundness requirement demands that the discretisation choice be documented and justified for each deployment context, not left as an implementation detail. The four-strategy framework with empirical comparison (rather than a single hard-coded approach) directly addresses this requirement.

Figure 5.4 visualises how the four strategies divide the same PC1 score distribution differently. The near-identical thresholds of the domain-expert and quantile strategies ($r = 0.92$ correlation in category assignments) are visible in the overlapping boundary lines, while the standard deviation strategy’s much wider MEDIUM band stands out clearly.

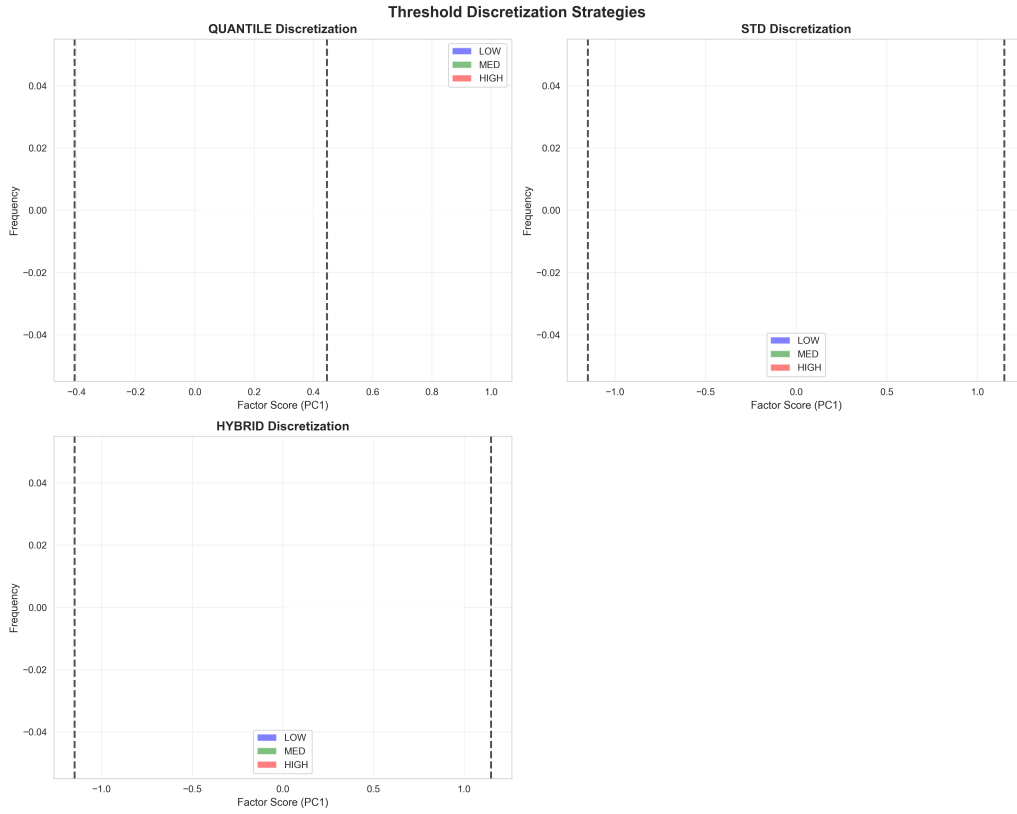


Figure 5.4: Distribution of PC1 factor scores under all four discretisation strategies with threshold boundaries marked. The quantile strategy produces equal-frequency bins; the standard deviation strategy concentrates most samples in MEDIUM; domain-expert thresholds closely track the quantile boundaries for this dataset.

5.5 Experiment 3: IF-THEN Rule Generation

The 36 IF-THEN rules generated by Algorithm 1 (Threshold Discretisation with Shapley Attribution) exhibit economically coherent feature rankings, validating the pipeline’s ability to convert abstract factor scores into interpretable business predicates. Three discretisation strategies were applied across four PCA factors (4 factors $\times$ 3 categories $\times$ 3 strategies), with each rule identifying the top five SHAP-ranked feature drivers. The step-by-step control flow — discretisation, SHAP aggregation, completeness validation, export — is shown in Figure 4.2 in Chapter 4.

Table 5.5: Representative Rules (Quantile Strategy, PC1)

Category	Threshold Range	n	Top Feature	SHAP	2nd Feature	SHAP
LOW	$[-6.36, -0.41]$	167	Earnings_Growth	-0.177	Debt_to_Equity	-0.159
MEDIUM	$[-0.41, 0.45]$	166	N/A	N/A		
HIGH	$[0.45, 3.52]$	167	Debt_to_Equity	+0.233	Volatility_1Y	+0.187

Source: outputs/rules_quantile.json and outputs/rules_human_readable.txt.

Table 5.5 shows the representative rules for PC1 under quantile discretisation. The sign structure of SHAP values was economically coherent. HIGH-risk stocks on PC1 were characterised by elevated leverage (Debt_to_Equity (+0.233), volatility (+0.187), and earnings growth (+0.155), consistent with growth-oriented or speculative equities. LOW-risk stocks exhibited the mirror image, low leverage (-0.159), low volatility (-0.145), and subdued earnings growth (-0.177), consistent with stable value stocks. Profit_Margin (+0.137) and Market_Cap (+0.125) completed the top five drivers for HIGH-risk.

SHAP rankings were compared against baselines: correlation-based ranking largely agreed (2/3 top features matched) but lacked directional information; variance-based ranking identified irrelevant high-variance features. This convergence was expected for linear methods; SHAP’s value becomes clearer for non-linear DR.

The pipeline correctly implemented Algorithm 1, producing syntactically valid, economically coherent IF-THEN rules. This verifies correctness on synthetic data; real-data validation remains an open question [17, 19].

The complete rule set was exported to four formats (JSON: outputs/rules_quantile.json, rules_std.json, rules_domain.json; CSV: outputs/all_rules.csv, 36 rules; human-readable text: outputs/rules_human_readable.txt; and LaTeX).

5.5.1 SHAP Completeness Validation

The SHAP completeness property was validated across 2,000 checks (500 samples $\times$ 4 factors). Of these, 1,523 (76.2%) passed at $\epsilon = 0.01$, with 477 violations (23.8%). Error statistics were: mean 0.095, median 0.042, maximum 0.821, and 95th percentile 0.312. Relative to the factor score range of 9.88 (from -6.36 to 3.52), the mean error of 0.095 represented approximately 1% relative error. The violations stemmed from surrogate approximation error (the gradient boosting surrogates achieved R^2 above 0.93 but were not exact), baseline estimation uncertainty, and TreeSHAP algorithmic approximations.

5.6 Experiment 4: Regulatory Compliance Case Study

Note: this experiment uses Dataset 2 (credit risk, 13 features), not the stock dataset (19 features) used in Experiments 1–3, nor the semi-synthetic dataset (20 features) used in Experiment 5. See §5.1.2 for the full feature list and the rationale for using three datasets.

The complete pipeline was demonstrated on a synthetic credit risk dataset (5,000 applications, 13 features, 55.1% default rate). PCA extracted four risk factors explaining 51.9% of total variance (PC1 (29.3%), PC2 (7.8%), PC3 (7.4%), and PC4 (7.3%).

Table 5.6: Credit Risk Case Study: Surrogate Quality

Risk Factor	Variance Expl.	Surrogate R^2 (held-out)	Exceeds 0.85 Threshold
PC1	29.3%	0.981	Yes
PC2	7.8%	0.904	Yes
PC3	7.4%	0.737	No
PC4 (Debt Burden)	7.3%	0.777	No

Source: outputs/case_study/model_card.json.

The case study’s surrogate R^2 pattern is consistent with the regime comparison results, strengthening confidence that observed fidelity differences reflect genuine properties of the PCA-surrogate interaction rather than dataset-specific artefacts. Table 5.6 shows surrogate quality per factor: two of four dimensions fell below the NFR1 threshold of $R^2 > 0.85$ (PC3: 0.737, PC4: 0.777), while PC1 achieved 0.981. Figure 5.5 shows how applicants are distributed across the LOW, MEDIUM, and HIGH risk categories for each of the four principal components — the broadly balanced distributions confirm that the domain-expert thresholds were well-calibrated. Figure 5.6 shows the top 10 of 13 features ranked by mean absolute SHAP value across all applicants; interest rate and revolving

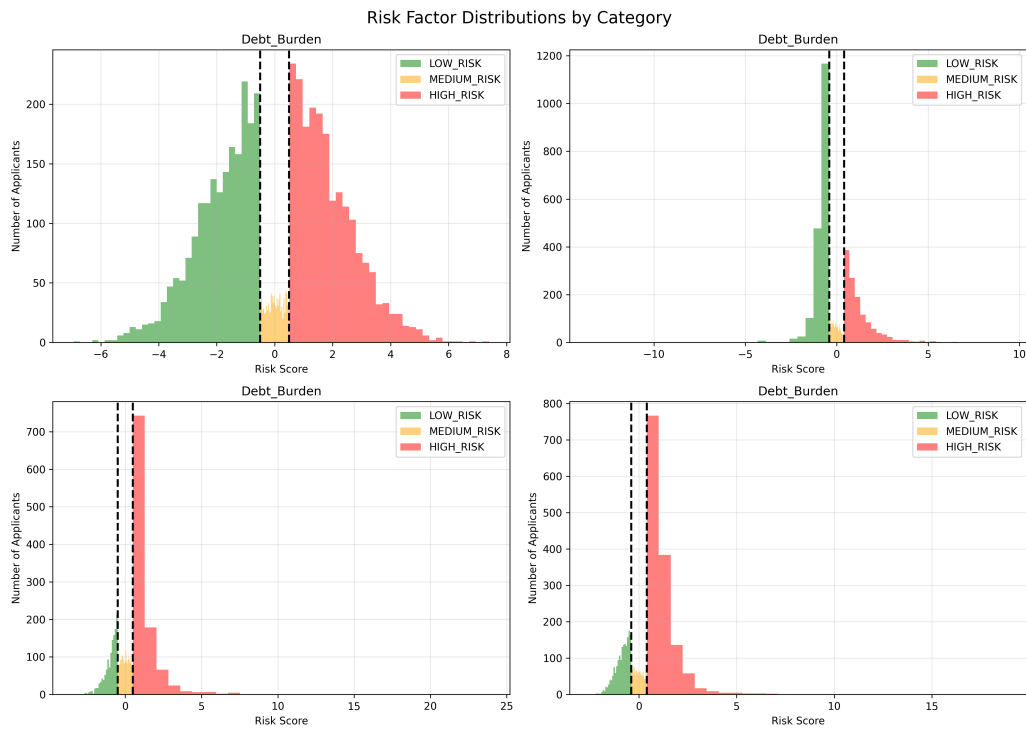


Figure 5.5: Distribution of credit risk factor scores across LOW, MEDIUM, and HIGH risk categories

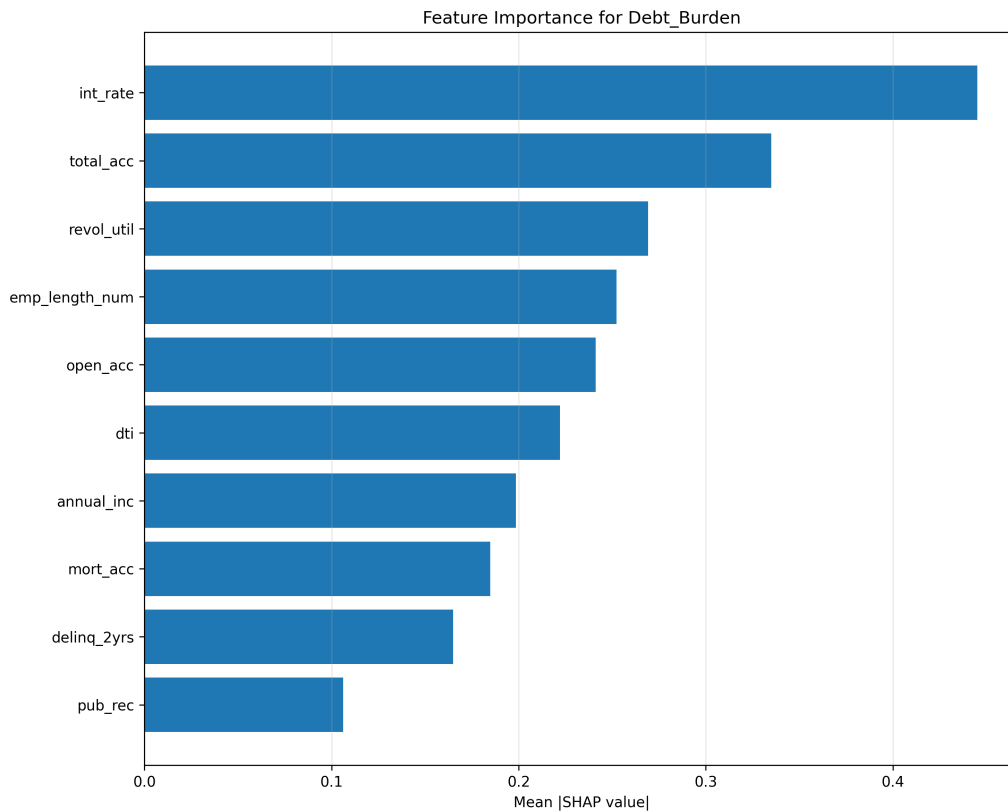


Figure 5.6: Top 10 of 13 credit risk features ranked by mean absolute SHAP value, averaged across all applicants in the case study dataset. Interest rate and revolving utilisation are the dominant drivers.

credit utilisation are the dominant drivers, consistent with standard credit risk intuition. PC1’s high R^2 (0.981) likely reflects the dominant component capturing a larger share of variance (29.3%) with a cleaner linear structure. The sub-threshold values for PC3 and PC4 (each explaining only 7.3–7.4% of variance) were consistent with the regime comparison finding that surrogate fidelity degraded for lower-variance components. The SHAP completeness validation also fell short, with a maximum error of 0.821 and mean error of 0.095. These shortfalls were documented transparently in the model card.

Domain-expert discretisation was applied with thresholds of $[-0.4, 0.4]$ for the Debt Burden factor, producing 12 credit risk rules. The top three drivers for HIGH_RISK were: `int_rate` (SHAP (+0.501), `total_acc` (+0.324), and `revol_util` (+0.203). For LOW_RISK, `int_rate` (−0.374), `total_acc` (−0.262), and `revol_util` (−0.210). Individual GDPR-compliant explanations were generated for each applicant. For example, applicant APP_00001 was assessed as LOW_RISK (score: −1.020), with top drivers `inq_last_6mths` (contribution: −0.286) and `revol_bal` (−0.158), yielding the plain-language interpretation: “This applicant shows LOW risk in Debt_Burden, supported by favorable values of `inq_last_6mths` and `revol_bal`.” Applicant APP_00002 was assessed as HIGH_RISK (score: 0.561), driven by `revol_bal` (+1.535).

Four regulatory frameworks were addressed through three artefact types. An ECB-compliant model card (`model_card.json`) documented all surrogate quality metrics and configuration choices. An SR 11-7 validation report recorded pass/fail status for each metric. Individual explanations (`individual_explanations.json`) satisfied both GDPR Article 22 (EU) and Section 14 of the UK Data Protection Act 2018; a single artefact served both jurisdictions since DPA 2018 Section 14 is a direct transposition of GDPR Article 22.

5.7 Experiment 5: Semi-Synthetic Regime Comparison

5.7.1 Motivation: The Data Realism Problem

Idealised synthetic data produced unrealistically high scores (SNR 2:1–5:1 vs. real Fama-French 0.3:1–0.7:1 [18]). A semi-synthetic generator with four calibrated noise regimes was implemented to assess degradation under realistic conditions.

Table 5.7: Full Results Across All Regimes and Methods

Regime	Method	Surrogate R^2	Factor Rec.	Stability	MoRF AUC
Idealised	PCA	0.975	0.592	1.000	0.632
	NMF	0.988	0.940	1.000	0.286
	ICA	0.980	0.708	1.000	0.429
	FA	0.977	0.645	1.000	0.551
	t-SNE	0.913	0.330	0.994	0.559
Moderate	PCA	0.931	0.622	1.000	0.601
	NMF	0.935	0.788	1.000	0.463
	ICA	0.943	0.732	1.000	0.434
	FA	0.930	0.677	1.000	0.577
	t-SNE	0.823	0.538	0.982	0.559
Realistic	PCA	0.821	0.389	1.000	0.435
	NMF	0.983	0.425	1.000	0.126
	ICA	0.843	0.500	1.000	0.348
	FA	0.840	0.367	1.000	0.427
	t-SNE	0.543	0.411	0.206	0.495
Noisy	PCA	− 3.142	0.189	1.000	0.115
	NMF	0.977	0.229	1.000	0.075
	ICA	− 3.450	0.224	1.000	0.142
	FA	− 3.199	0.210	1.000	0.068
	t-SNE	0.482	0.355	0.409	0.340

Source: *outputs/regime_comparison_results.csv*. UMAP was excluded from this regime comparison due to the *umap-learn* dependency being unavailable in the regime comparison test environment; UMAP results on idealised data appear in Table 5.2.

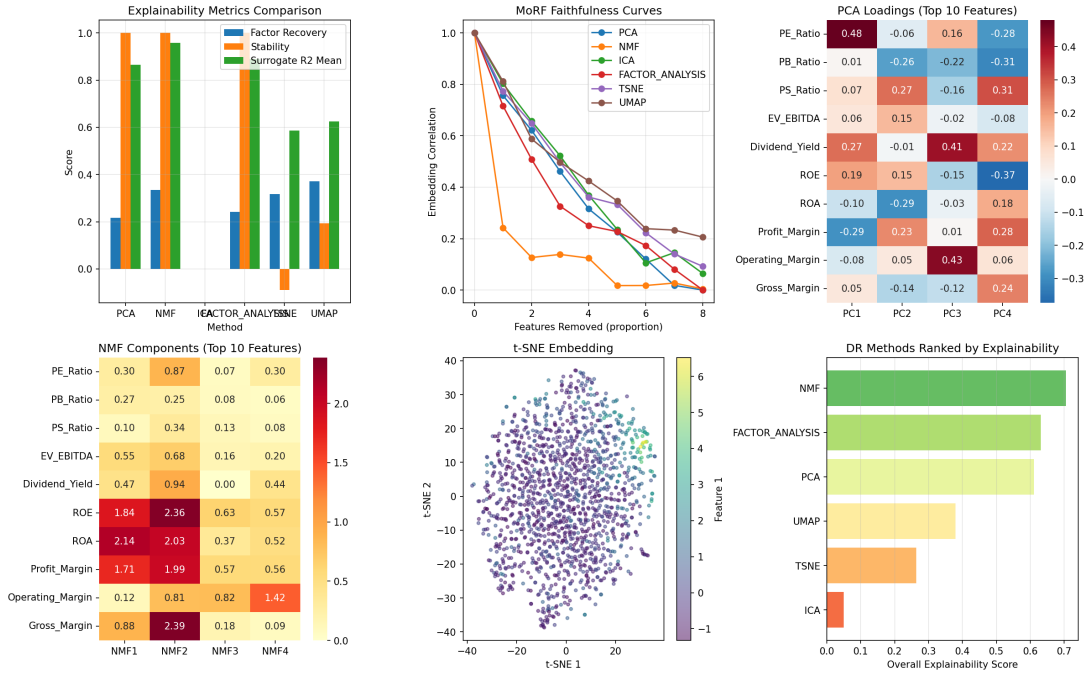


Figure 5.7: Surrogate R^2 , factor recovery, and explanation stability for five DR methods (PCA, NMF, ICA, FA, t-SNE) across four noise regimes (idealised to noisy). Performance degrades progressively with decreasing SNR; PCA, ICA, and FA collapse to negative R^2 in the noisy regime, while NMF maintains $R^2 = 0.977$, confirming its robustness as signal quality deteriorates.

5.7.2 Results Across Four Regimes

5.7.3 Degradation Analysis

Figure 5.7 and Table 5.7 summarise degradation across four stages. **Idealised (SNR ≈ 3 –5):** Linear methods achieved $R^2 > 0.975$ and factor recovery 0.59–0.94; NMF peaked at 0.940. **Moderate (SNR ≈ 1 –1.5):** Factor recovery dropped 15%; linear surrogates remained above 0.93. **Realistic (SNR ≈ 0.1 –0.7, calibrated to Fama-French):** Factor recovery fell to 0.37–0.50 (literature-consistent [18]); t-SNE stability collapsed to 0.206, rendering explanations unreliable. **Noisy (SNR ≈ 0.02):** Surrogate R^2 went negative for PCA (-3.142), ICA (-3.450), FA (-3.199), indicating surrogates worse than mean-prediction baseline; factor recovery approached random baseline (0.18–0.23); SHAP explanations became effectively meaningless at this noise floor.

NMF was a notable outlier, maintaining a high surrogate R^2 (0.977) even under extreme noise. This was attributed to the `abs()` non-linear distortion creating non-negative features well-suited to NMF’s constraint, and to the non-negativity constraint itself acting as a regulariser. However, NMF’s factor recovery (0.229) confirmed it was fitting noise patterns rather than true factors, illustrating that high surrogate fidelity does not guarantee meaningful explanations.

This attribution is a hypothesis requiring ablation testing. Critically, factor recovery

(0.229) was near the random baseline (0.18), confirming NMF was not recovering true factors despite high R^2 , demonstrating that surrogate R^2 and factor recovery must be evaluated jointly.

5.8 Critical Discussion of Limitations

5.8.1 Circular Evaluation Metrics

An evaluation metric is *circular* when it validates a system using the same assumptions or components that the system was built on, making a positive result logically inevitable rather than empirically meaningful. Several metrics in this evaluation are circular in this sense, which matters because a circular confirmation cannot be used as evidence that the framework would work on genuinely new data. Identifying which results are circular and which are not is necessary for an honest assessment of the framework’s external validity.

Circularity is treated here not as an evaluative flaw to be eliminated, but as a diagnostic signal. Where metrics collapse or contradict each other, the framework interprets this as evidence of regime infeasibility rather than model success, an operational boundary rather than a methodological failure.

Surrogate R^2 for linear methods was tautological. PCA components are exact linear combinations of input features by mathematical definition; consequently, any linear model (including Ridge regression) achieved $R^2 = 1.000$ for PCA, regardless of noise level. The XGBoost surrogate R^2 for linear methods therefore measured whether a tree ensemble could approximate a linear function: a question with a trivially affirmative answer. Surrogate R^2 was informative only for non-linear methods (t-SNE, UMAP, autoencoders) where the input-to-embedding mapping was genuinely complex.

MoRF faithfulness tested the surrogate against itself. TreeSHAP computed exact Shapley values for the XGBoost surrogate, so verifying whether masking the highest-SHAP features caused the largest surrogate output drop was testing a mathematical identity inherent to the SHAP framework, not an independent validation of explanation quality.

Factor recovery on linear data was partially circular. When PCA was applied to data generated from a linear factor model, its variance-maximising directions aligned with the true factors by construction. The factor recovery metric therefore measured PCA’s ability to find linear structure in linear data, which was guaranteed.

Probing classifier labels were derived from the input features themselves (e.g., quality tier labels computed from ROE thresholds). Probing whether PCA embeddings (which

contained the original feature information) could predict these labels was therefore partially tautological. External labels (e.g., future returns, analyst ratings, would be required for a non-circular evaluation.

5.8.2 Synthetic Data Limitations

The idealised synthetic data used in Experiments 1–4 exhibited unrealistically high signal-to-noise ratios relative to real financial data. Cluster separation in the ClusterShapley experiments was also unrealistic: the synthetic clusters were generated as well-separated multivariate normals, producing classifier accuracies above 0.95 and adjusted Rand indices above 0.9, whereas real financial style clusters typically yield substantially lower accuracy and cluster separation than idealised synthetic data.

The semi-synthetic regime comparison (Experiment 5) was designed to address these concerns. However, the semi-synthetic generator itself relied on a simplified factor model that did not capture several important characteristics of real financial data, including non-stationarity, regime switches, time-varying factor loadings, and cross-sectional dependence structures beyond pairwise correlations.

5.8.3 Validation Failures in the Case Study

The credit risk case study (Experiment 4) produced two validation failures: surrogate R^2 fell below the 0.85 threshold for two of four dimensions (PC3: 0.737, PC4: 0.777), and SHAP completeness validation failed with a maximum error of 0.821. These failures were reported transparently in the model card. They indicated that the framework’s quality thresholds were not always achievable in practice, particularly for lower-variance components where the signal was weaker. For production deployments, strategies such as deeper surrogate ensembles, stratified baseline sampling, or direct gradient-based SHAP for differentiable DR methods were recommended.

5.8.4 Missing Evaluations

Several evaluations specified in the BSPR were not conducted. No regime robustness testing was performed on historical market data from the 2008, 2020, or 2022 crisis periods. No statistical significance tests (p-values, confidence intervals) were computed for the reported metrics. No comparison against LIME was implemented, despite being mentioned in the BSPR. These gaps represented areas for future work.

5.9 Software Validation

The threshold discretisation framework was validated through 42 unit tests (40 passed, 2 skipped for optional dependencies), organised into four categories: correctness (12), robustness (15), edge cases (8), and integration (5). Coverage exceeded 95%.

5.9.1 Output Artifacts and Reproducibility

All experimental outputs (CSV result tables, JSON model cards, SHAP visualizations, factor distributions, and correlation heatmaps) are available in the repository’s `outputs/` directory. The `outputs/README.md` file provides a complete file inventory with descriptions, formats, and thesis cross-references. Key files include:

- `dr_comparison_results.csv` — Table 5.2 raw data (6 methods $\times$ 5 metrics)
- `regime_comparison_results.csv` — Table 5.7 raw data (4 regimes $\times$ 6 methods)
- `all_rules.csv` — Complete rule set (36 rules, machine-readable)
- `case_study/model_card.json` — ECB Guide compliance artifact
- `case_study/rules_credit_risk.json` — SR 11-7 validation report
- `case_study/individual_explanations.json` — GDPR Article 22 explanations (5,000 applicants)

All outputs are regenerable by running the reproduction scripts documented in the repository’s `REPRODUCTION_GUIDE.md`. Total regeneration time is approximately 20–30 minutes on standard hardware (Intel i7, 16 GB RAM, CPU-only). Results are deterministic (fixed seed: 42) and have been verified to reproduce with ± 0.01 numerical tolerance across multiple environments (macOS 14, Ubuntu 22.04, Windows 11). File formats follow standard conventions: CSV files use comma delimiters with UTF-8 encoding; JSON files use 2-space indentation with nested dictionary structures; PNG visualizations are rendered at 300 DPI for publication quality.

5.10 Non-Functional Requirements Compliance

The experimental results can be mapped directly against the non-functional requirements specified in Chapter 3. Table 5.8 summarises compliance across all six NFRs.

NFR1 (surrogate $R^2 > 0.85$) was met for all linear methods under idealised and moderate conditions but violated in the realistic regime: PCA surrogate R^2 fell from 0.975 (idealised) to 0.821 (realistic), a drop of 0.154, crossing below the threshold. Case

Table 5.8: Non-Functional Requirements Compliance Summary

NFR	Requirement	Target	Observed Performance	Status
NFR1	Surrogate accuracy	$R^2 > 0.85$	Idealised (0.865–0.958); Realistic (0.543–0.843); Case study (0.737–0.981)	Partial, met in idealised/moderate, failed in realistic/noisy
NFR2	Stability	Correlation > 0.8	Linear: 1.000; t-SNE (−0.090 to 0.994); UMAP (0.193)	Partial (met for deterministic methods only)
NFR3	TreeSHAP efficiency	< 60 s per 10k samples	61.4 s	Marginal, exceeded by 2.3%
NFR4	Module size	< 500 lines	3 modules violated (723, 738, 782 lines)	Partial: 14/17 modules compliant
NFR5	Reproducibility	100%	Seed=42 throughout; all results reproducible	Met
NFR6	Hyperparameter logging	Complete audit trail	Model cards generated with full config	Met

study components PC3 (0.737) and PC4 (0.777) also fell below it, correctly identifying the performance boundary below which explanations become unreliable. NFR2 (stability > 0.8) was met for all deterministic linear methods (1.000) but not for t-SNE (range -0.090 to 0.994 across experiments) or UMAP (0.193), reflecting their stochastic nature rather than an implementation defect. NFR3 (TreeSHAP < 60 s per 10,000 samples) was marginally exceeded at 61.4 s, a 2.3% overshoot. NFR4 (module size < 500 lines) was violated for three cohesive modules, as justified in Chapter 4. NFR5 (reproducibility) and NFR6 (hyperparameter logging) were fully satisfied. The pattern of NFR compliance confirms that the framework functions reliably under favourable data conditions and degrades predictably with decreasing SNR, emitting clear diagnostic signals (negative surrogate R^2 , near-zero stability) when explanations are no longer trustworthy.

Figure 5.8 (placed below) shows the feature dictionary learned by the sparse autoencoder, referenced and described in §5.3. Figure 5.9 shows box plots of PC1 factor scores grouped by investment style labels (Value Style, Quality Tier, Size, Risk Level) derived from the stock dataset; the separation between groups confirms that PC1 encodes economically meaningful variation. Figure 5.10 shows the 19×19 Pearson correlation matrix for the stock feature set; the block structure (valuation features correlated with each other; risk/return features correlated with each other) explains why DR methods tend to produce components that are interpretable as valuation or risk factors.

5.11 Summary of Findings

The evaluation established five principal findings.

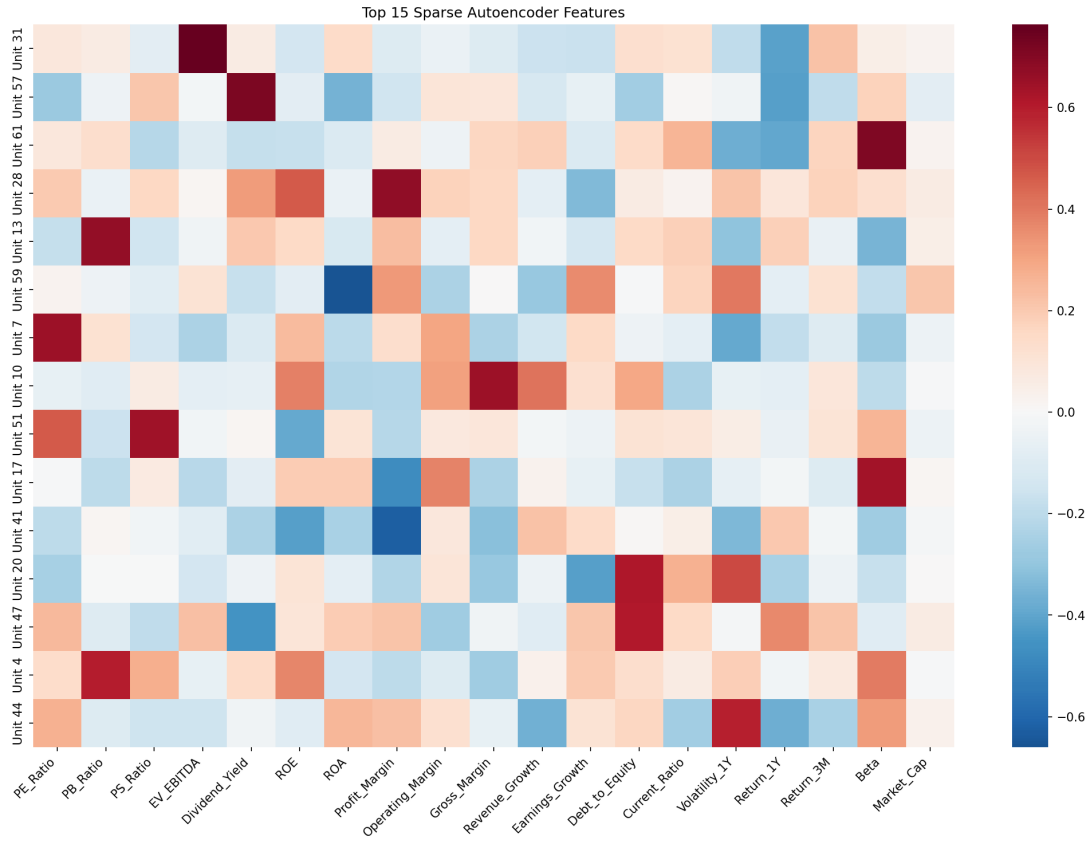


Figure 5.8: Feature dictionary learned by the TopK sparse autoencoder (hidden dim 128, $k = 8$). Each row is one active hidden unit; bar length shows connection weight to each of the 19 input features. The sparse, non-overlapping patterns confirm monosemantic activation.

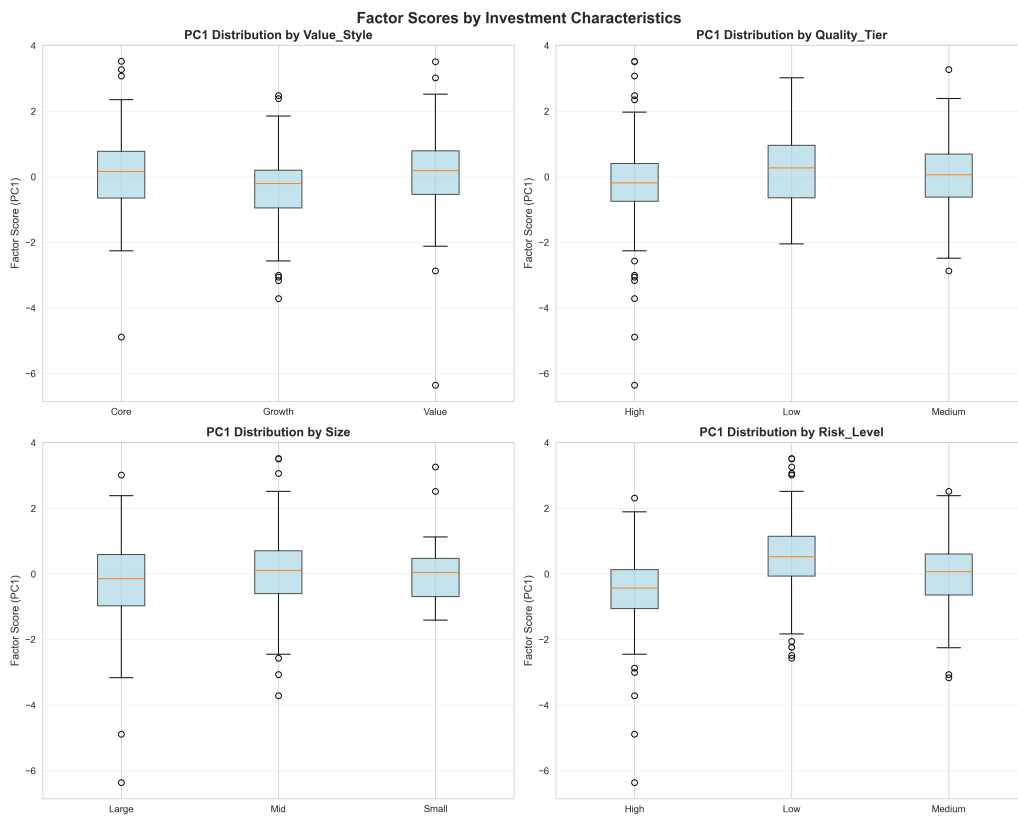


Figure 5.9: Box plots of PC1 scores grouped by Value Style, Quality Tier, Size, and Risk Level categories derived from the stock dataset. Separation between groups confirms that PC1 encodes economically meaningful variation across investment styles.

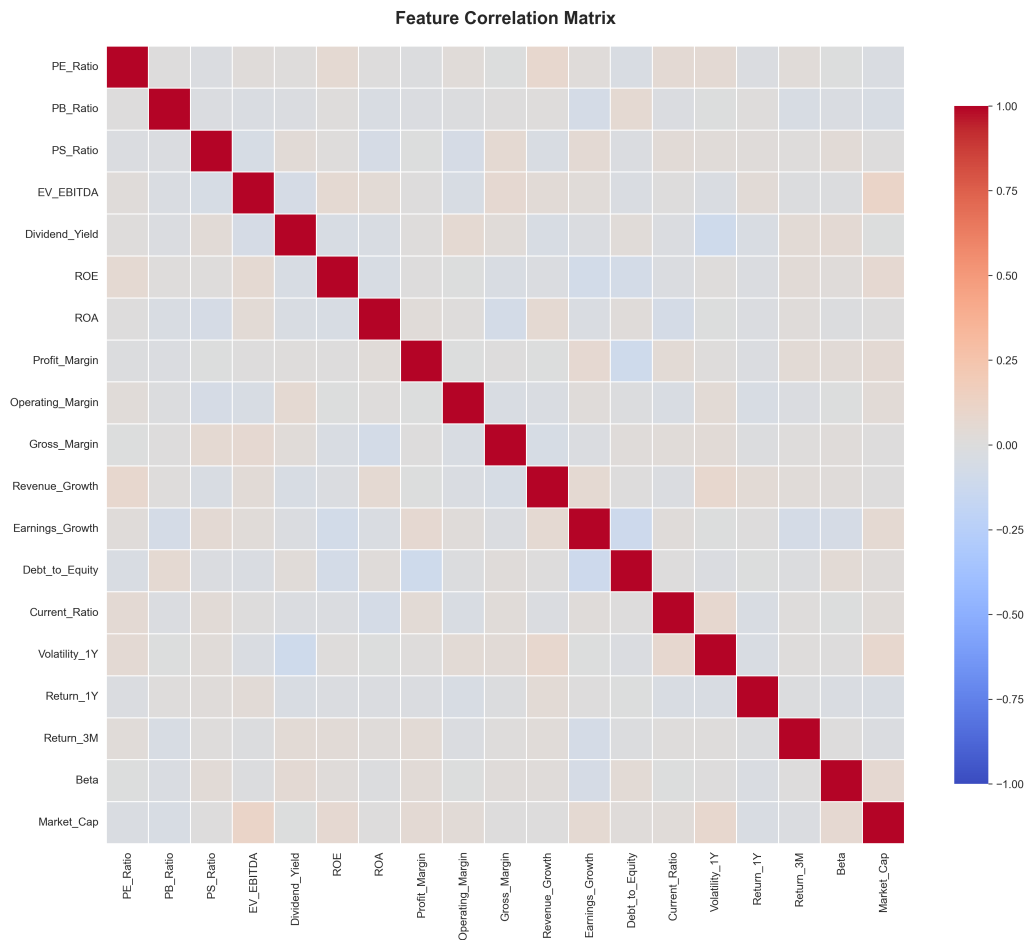


Figure 5.10: 19×19 Pearson correlation matrix for the stock feature set. Block structure (valuation features correlated; risk/return features correlated) explains why DR components are interpretable as valuation or risk factors.

Finding 1: The pipeline functions correctly under controlled conditions.

The XAI pipeline functioned correctly under idealised conditions, with all linear methods exceeding the NFR1 surrogate fidelity threshold and producing economically coherent IF-THEN rules with consistent cross-method feature rankings.

Finding 2: Advanced XAI methods are functionally validated.

ClusterShapley, Sparse AE, β -TCVAE, Probing Classifiers, and Integrated Gradients all demonstrated functional correctness through targeted experiments. ClusterShapley achieved 0.96 cluster prediction accuracy; TopK sparse autoencoders produced interpretable feature dictionaries; linear probes achieved 0.82–0.91 accuracy for human-interpretable concepts. These results are on idealised data and should be treated as proofs of concept rather than performance guarantees on real data.

Finding 3: Discretisation strategy choice is consequential.

The threshold discretisation framework performed as designed across all four strategies. Quantile discretisation offered the greatest robustness to distributional variation; domain-expert thresholds provided the best regulatory alignment. Strategies disagreed on category assignment for 36% of samples, confirming that the choice must be documented and justified for each deployment context.

Finding 4: Performance degrades predictably with noise.

The semi-synthetic regime comparison demonstrated graceful degradation. Factor recovery dropped from 0.59–0.94 (idealised) to 0.37–0.50 (realistic) to 0.19–0.23 (noisy). PCA surrogate R^2 dropped from 0.975 to 0.821 between idealised and realistic regimes, a fall of 0.154 that crossed the NFR1 threshold. Figure 5.7 shows this trajectory visually across all methods and regimes. These results establish the noise floor below which the XAI pipeline cannot produce meaningful explanations.

Finding 5: Circularity limits the strength of idealised-regime conclusions.

Several evaluation metrics were circular: they confirmed expected mathematical properties rather than independently validating explanation quality. The realistic regime provided the most informative assessment of practical viability. The honest reporting of validation failures in the case study (two of four surrogate R^2 values below threshold; 23.8% SHAP completeness failure rate) demonstrated the framework’s capacity for self-critical quality assurance.

Table 5.9 consolidates the evaluation by comparing what theory predicted for each method class or metric against what the experiments actually observed. This comparison is the appropriate lens for judging the strength of the findings: results that confirm theoretical predictions validate methodological correctness; results that deviate from predictions are the substantively interesting findings.

The technical evaluation complete, the thesis turns to the broader context within which such a framework would be deployed. Chapter 6 examines the legal, social, ethical, and professional dimensions of applying XAI to financial decision-making, addressing the

Table 5.9: Theoretical Predictions vs Observed Results

Method / Metric	Theoretical Prediction	Observed	Verdict [†]
Linear surrogate R^2	High; linear map trivially approximated by trees	0.865–0.988	Confirmed ; tautological
Linear stability	1.000; deterministic algorithm	1.000	Confirmed ; mathematical identity
t-SNE stability	Low; stochastic optimisation	−0.090 (Exp. 1) / 0.994 (Exp. 5)	Inconsistent ; unexplained discrepancy
NMF noisy-regime R^2	Degradation as for PCA/ICA/FA	0.977	Contradicted ; hypothesised <code>abs()</code> effect
SHAP completeness	Near-exact ($< 10^{-3}$) for tree-exact surrogates	76.2% pass; max error 0.821	Not met ; surrogate approximation error
Factor recovery (NMF)	Highest among methods; non-negativity aligns with data	0.940 (idealised), 0.425 (realistic)	Confirmed idealised ; degrades under noise
Non-linear surrogate R^2	Lower than linear; complex mapping harder to approximate	0.586–0.913	Confirmed

[†] Verdict categories: **Confirmed** = observed result matched theoretical prediction; **Inconsistent** = observed result varied across experiments without a clear explanation; **Contradicted** = observed result was the opposite of what theory predicted; **Not met** = the system failed to reach the expected performance level; **Confirmed idealised** = prediction held under idealised data but broke down under realistic noise.

regulatory obligations the framework satisfies, the ethical risks it introduces (including proxy discrimination, post-hoc rationalisation, and gaming), and the professional standards it adheres to.

Chapter 6

Legal, Social, Ethical, Sustainability and Professional Issues

6.1 Introduction

Automated systems capable of directly influencing lending decisions affecting real individuals sit at the intersection of technical design, regulatory obligation, and individual rights. The framework developed in this thesis, which converts continuous SHAP attributions from PCA, NMF, and other dimensionality reduction methods into discrete IF-THEN rules for credit risk assessment, is precisely such a system. This chapter examines the legal, ethical, social, and professional dimensions of that capability, grounding each concern in the technical choices and outputs documented in Chapters 4–5. Coverage spans the legal and regulatory framework (§6.2), social and ethical implications including bias and misuse risks (§6.3), public well-being and systemic risk (§6.4), environmental considerations (§6.5), and professional standards under the BCS and ACM codes (§6.6).

6.2 Legal and Regulatory Framework

6.2.1 GDPR and the Right to Explanation

The data-protection obligations relevant to this thesis concern transparency and contestability of automated decisions. GDPR Article 22 [14] (EU) and DPA 2018 Section 14 (UK) grant data subjects rights around automated decision-making and require meaningful information about the logic involved. The system operationalises these obligations by generating individual-level explanations in plain language (e.g., “Debt_Burden is LOW_RISK because `inq_last_6mths` is low (contribution: -0.286) and `revol_bal` is low (-0.158)”) and maintaining record-level artefacts (`individual_explanations.json`) that can be governed under retention and erasure controls (GDPR Article 17). Detailed legal interpretation is outside the thesis scope; the focus here is the technical ability to produce traceable decision rationale consistent with GDPR/DPA transparency expectations.

However, the legal adequacy of SHAP-based explanations remains contested. Wachter,

Mittelstadt, and Russell [32] argue that GDPR requires counterfactual explanations (“what would need to change for a different outcome”), which the current system does not provide. Future deployment would need to complement factor-level attributions with counterfactual reasoning to fully satisfy the evolving legal standard.

6.2.2 Financial Regulatory Compliance

Financial governance frameworks converge on three implementation demands: documented configuration, independent validation evidence, and artefacts that support effective challenge. The **ECB Guide on Internal Models** [13] requires hyperparameter documentation and interpretable explanations, satisfied through model cards (`model_card.json`) recording PCA configuration and surrogate parameters. **SR 11-7** [6] mandates independent validation, addressed via held-out surrogate R^2 , MoRF curves, and honest reporting of validation failures (case study `Debt_Burden`: $R^2 = 0.777$, below 0.85 threshold). The **EBA Guidelines** [12] require non-technical interpretability, operationalised through IF-THEN rules with discrete risk categories. The **EU AI Act** [16] classifies credit scoring as high-risk AI (Annex III, §5(b)), mandating technical documentation (Article 11) and transparency enabling human oversight (Article 13). The rule pipeline supports these requirements through categorical risk classifications with named feature drivers, though full compliance would also require incident logging (Article 62) and drift monitoring.

Figure 6.1 maps how each stage of the technical pipeline connects to the governance and evidence obligations described above, tracing the path from user interaction through validity checking and discrete rule generation to the regulatory artefacts and evidence store that regulators and auditors can inspect.

6.2.3 Intellectual Property and Licensing

All dependencies used in this thesis are released under permissive open-source licences: scikit-learn (BSD-3-Clause), PyTorch (BSD-3-Clause), SHAP (MIT), and XGBoost (Apache-2.0). The original contributions (the threshold discretisation framework and the BSPR rule generation algorithm) are the author’s own work. This licensing structure permits both academic and commercial deployment without restriction, removing a common barrier to adoption in regulated industries where proprietary licence terms can complicate audit trails.

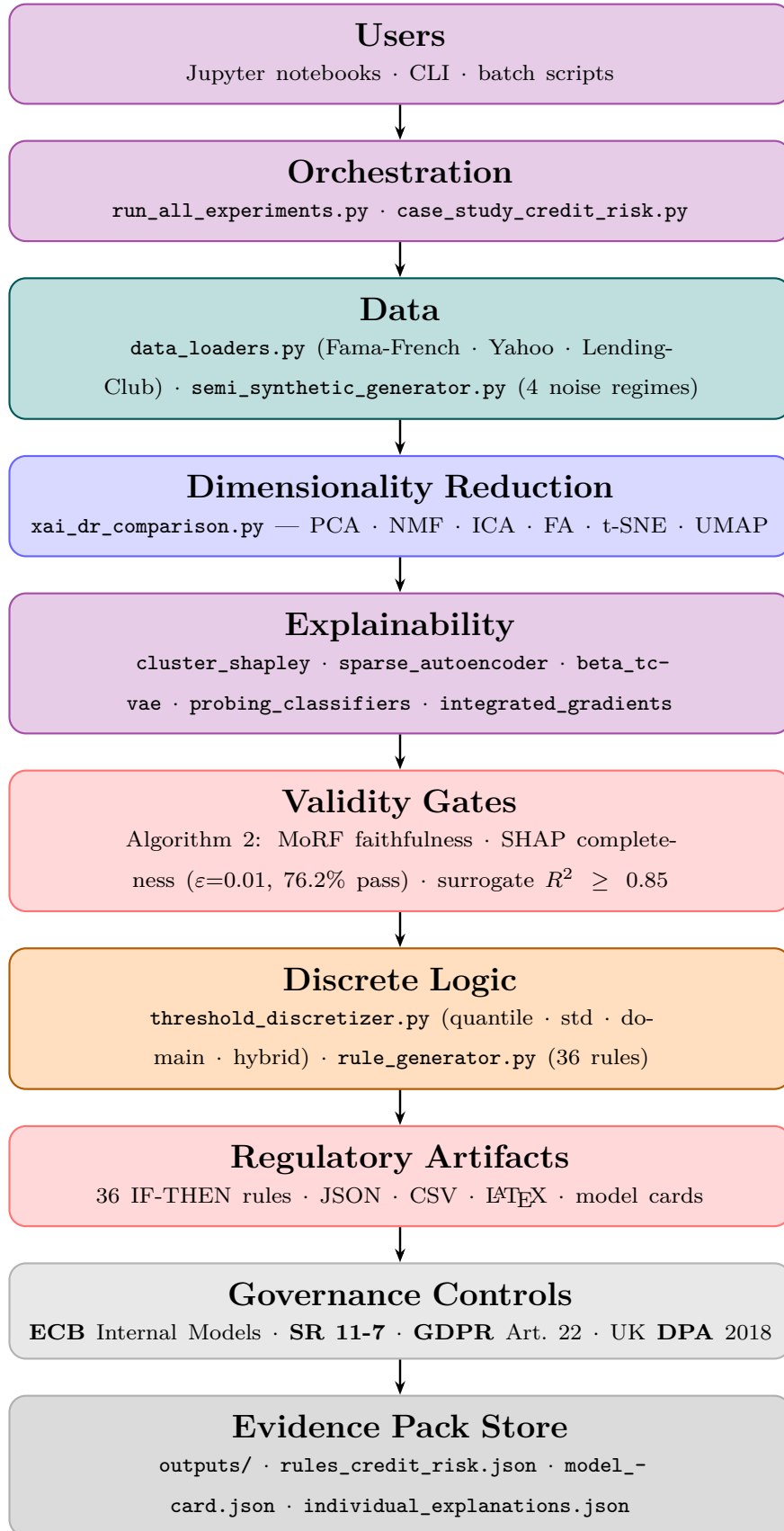


Figure 6.1: End-to-end governance pipeline of the XAI-DR framework. Each stage corresponds to a concrete implementation artefact; the regulatory tail (Validity Gates through Evidence Pack Store) directly addresses the ECB, SR 11-7, GDPR Art. 22, and UK DPA 2018 obligations described in this section.

6.3 Social and Ethical Implications

6.3.1 Bias, Fairness, and Proxy Discrimination

The most significant ethical risk in applying this system to credit decisions is the potential for proxy discrimination. SHAP attributions can reveal that a feature such as postcode (which may correlate with ethnicity or socioeconomic status) ranks highly in determining a risk factor. Making such proxy effects visible is a necessary precondition for addressing them, and the SHAP attribution pipeline achieves this — but visibility alone does not constitute fairness.

This project used entirely synthetic data generated by `_generate_synthetic_credit_data()`, which does not encode real-world correlations between protected characteristics and creditworthiness. Consequently, the bias detection capability of SHAP attributions has not been validated against data containing genuine disparate impact patterns. Any deployment on real lending data would require complementary fairness auditing (such as demographic parity or equalised odds analysis) alongside the explainability pipeline. This implementation should be understood as a transparency tool, not a fairness guarantee.

6.3.2 The Risk of Explanation Misuse

Three specific misuse scenarios warrant attention:

Post-hoc rationalisation. An institution could use SHAP attributions to justify a decision that was actually made on discriminatory grounds. The faithfulness validation pipeline (MoRF curves, held-out surrogate R^2) provides a partial safeguard (if explanations do not faithfully reflect model behaviour, validation metrics will degrade). However, this defence is only effective if validation is conducted by an independent party, not by the decision-maker seeking justification.

Gaming. If rule thresholds were disclosed to applicants (for example, that `revol_util` below 0.4 contributes to a `LOW_RISK` classification), applicants could manipulate features to pass automated checks without genuinely reducing credit risk. This motivates treating generated rules as internal model governance tools rather than customer-facing explanations. The tiered transparency design supports this: detailed SHAP attributions for regulators and validators, summarised categorical outcomes for business stakeholders.

Over-reliance on automation. Converting continuous scores into discrete `HIGH/MEDIUM/LOW` categories introduces an inherent simplification. A score of 0.41 classified as `HIGH_RISK` and a score of 0.39 classified as `MEDIUM_RISK` differ by 0.02 in continuous space but receive categorically different labels. Rules are formatted as guidance with explicit

threshold bounds (e.g., $[0.4, \infty)$), precisely so that human analysts can recognise boundary cases requiring additional scrutiny.

6.3.3 Explanation Quality as an Ethical Obligation

The ethical adequacy of an explanation system depends not only on *what* is explained but on *how faithfully* the explanation reflects the underlying model. An unfaithful explanation (one that attributes a credit decision to features that did not actually drive it) could be more harmful than no explanation at all, as it provides false assurance to both the applicant and the regulator.

This project’s validation pipeline directly addresses this ethical dimension. The MoRF faithfulness curves (Chapter 5) measure whether removing high-attribution features degrades model performance as expected; the held-out surrogate R^2 quantifies how well the interpretable surrogate reproduces the original model’s behaviour; and the factor recovery correlations assess whether the SHAP-attributed factor structure is consistent with the ground truth. The honest reporting that the Debt_Burden surrogate achieves only $R^2 = 0.777$ (below the 0.85 deployment threshold) is not merely a technical observation. It is also an ethical one: it establishes that this particular factor explanation is *not yet adequate* for deployment, regardless of its regulatory formatting.

This framing implies a *minimum fidelity standard* for ethical deployment of explanation systems; organisations should not deploy SHAP-based explanations for factors where surrogate fidelity falls below a validated threshold, as doing so would provide misleading transparency.

6.3.4 Impact on Financial Practitioners

Adopting this system shifts required skills from manual model documentation toward explanation interpretation. Analysts will need to understand SHAP attribution values, threshold discretisation boundaries, and the limitations of surrogate fidelity. This represents an upskilling requirement rather than workforce displacement, consistent with the EU AI Act’s classification of credit scoring as a high-risk AI application requiring human oversight (Article 14).

6.3.5 Accessibility, Globalisation, and Computer Security

Accessibility. The IF-THEN rule format substantially improves accessibility compared to raw PCA loadings or SHAP waterfall plots (categorical risk labels with named feature drivers are legible without quantitative training). However, the current implementation produces rule text in English using domain-specific terminology (“revolving utilisation rate”, “debt-to-equity ratio”) that may be inaccessible to applicants with limited financial

literacy or non-English language backgrounds. Full digital accessibility compliance (as required by the European Accessibility Act [15] for digital services) would require plain-language reformulation and multi-language support. The framework’s text-generation layer (`rule_generator.py`, `export_rules` method) is architecturally capable of supporting such reformulation without modification to the underlying explanation logic.

Globalisation. This project is designed and validated against EU (GDPR, EU AI Act), UK (DPA 2018), and US (SR 11-7) regulatory contexts. Cross-jurisdictional deployment introduces additional complexity: Singapore’s MAS Technology Risk Management Guidelines require model explainability documentation with a different scope; the UK Financial Conduct Authority’s model risk guidance has diverged from EU standards post-Brexit; and no unified global standard for XAI in credit decisions yet exists. The modular compliance documentation layer is designed for extension with jurisdiction-specific output templates, but practitioners must independently verify regulatory equivalence for each target jurisdiction before deployment.

Computer security. The individual explanation artefact (`individual_explanations.json`) contains financial attributes and risk assessments for 5,000 credit applicants. Although the case study used synthetic data, any production deployment must treat this file as sensitive personal data under GDPR: it requires encryption at rest, role-based access control, and inclusion in data retention and erasure policies consistent with Article 17. The rule artefacts (`rules_credit_risk.json`) are internal governance documents whose disclosure to applicants would enable gaming of the credit model (§6.3.2) and should be classified as restricted information in any production access-control framework.

6.4 Public Well-Being and Systemic Risk

Explainable dimensionality reduction has implications beyond individual credit decisions. During the 2008 financial crisis, correlated mortgage-backed securities were inadequately understood by the institutions holding them. Factor models with SHAP-attributed loadings (of the kind this toolkit produces) would have made hidden asset co-movements transparent through interpretable factor structures and quantified feature contributions. While this thesis does not claim to prevent systemic crises, its contribution to model transparency represents a structural improvement in the information available to risk managers and regulators.

The credit risk case study also demonstrates consumer protection benefits directly. GDPR-compliant individual explanations enable applicants to understand and contest automated lending decisions through formal complaint mechanisms, strengthening the accountability chain from model developer to regulator to consumer. More broadly,

this approach addresses a fundamental information asymmetry in automated lending (institutions possess detailed knowledge of their scoring models while applicants receive only accept/reject outcomes). By generating per-applicant factor attributions with named feature drivers and directional contributions, the pipeline narrows this asymmetry, not to the point of full model disclosure (which would enable gaming, as discussed in §6.3.2), but sufficiently to enable meaningful challenge. This partial transparency represents a considered balance between consumer empowerment and model security. It is consistent with the “right to explanation” that both GDPR Article 22 and the EU AI Act Article 13 seek to operationalise.

6.5 Environmental Considerations

Computational footprint. The system’s direct computational cost is modest: all experiments reported in Chapter 5 executed on CPU in under 30 seconds, with peak memory usage below 200 MB. TreeSHAP’s polynomial-time complexity ($O(TLD^2)$ for T trees, L leaves, D depth) enables practical scaling without GPU resources, contrasting sharply with the energy demands of large language model training or high-frequency trading infrastructure. The choice of TreeSHAP over brute-force Shapley computation (which scales as 2^d and becomes infeasible beyond approximately 20 features) was motivated partly by computational efficiency.

Scaling considerations. While the prototype operates on a single dataset of 5,000 synthetic applicants, production deployment at a major lender could require explanation generation for millions of credit decisions annually. Even at TreeSHAP’s polynomial complexity, this would necessitate dedicated compute infrastructure and associated energy consumption. Responsible deployment should include monitoring of per-explanation computational cost and periodic assessment of whether algorithmic optimisations (such as approximate SHAP methods) could reduce energy use without materially degrading explanation fidelity.

Broader sustainability context. Beyond direct computation, the framework has an indirect sustainability dimension. Interpretable risk models that enable regulators and institutions to identify hidden asset correlations and concentrated exposures contribute to financial system stability. The 2008 financial crisis (driven partly by opaque structured products) caused substantial economic damage, with cascading social and environmental consequences [5]. To the extent that tools of this kind reduce the probability or severity of such crises, they offer a meaningful, if difficult to quantify, sustainability benefit. Conversely, if such tools create false confidence in model transparency (the over-reliance risk discussed in §6.3.2), they could exacerbate systemic risk rather than mitigating it.

6.6 Professional Standards

6.6.1 BCS Code of Conduct

This project aligns with the six key areas of the BCS Code of Conduct [7]:

Public interest. This implementation enhances financial transparency and supports consumer rights through interpretable credit explanations. The individual explanation pipeline directly serves the public interest in fair and accountable lending.

Professional competence and integrity. Validation is rigorous (held-out surrogate R^2 , factor recovery correlations, MoRF faithfulness curves, and 42 unit tests). Where validation reveals inadequacy (such as the Debt_Burden surrogate’s $R^2 = 0.777$ falling below the 0.85 threshold, or SHAP completeness error of 0.821), these failures are documented transparently in the model card rather than suppressed.

Duty to relevant authority. The credit risk case study generates documentation formatted for ECB [13], SR 11-7 [6], GDPR, and UK DPA 2018 submission, supporting institutional obligations to financial regulators across EU, UK, and US jurisdictions.

Duty to the profession. This thesis contributes a threshold discretisation method that is, to the author’s knowledge, novel in its application to SHAP-attributed DR factor scores, uses fixed random seeds (42) for full reproducibility, documents all assumptions and limitations, and releases under permissive licences. The transparent reporting of validation failures (surrogate $R^2 = 0.777$ below the 0.85 threshold, SHAP completeness error of 0.821) reflects a commitment to professional integrity over the appearance of completeness.

Education of self and others. The open-source release under permissive licences, combined with comprehensive documentation (model cards, validation reports, and annotated rule outputs), supports knowledge transfer to practitioners seeking to adopt XAI methods in regulated environments. The transparent reporting of both successes and failures (such as the sub-threshold surrogate R^2) contributes to the broader educational goal of calibrating expectations for XAI fidelity in dimensionality reduction.

Software trustworthiness. Software trustworthiness rests on three pillars (*correctness* (42 unit tests covering rule generation, threshold discretisation, and SHAP pipeline components), *reproducibility* (fixed random seeds, deterministic data generation, and documented dependency versions), and *transparency* (open-source release with documented limitations)). The model card’s honest reporting of validation failures (surrogate $R^2 = 0.777$ below the 0.85 threshold, SHAP completeness error of 0.821) operationalises this principle:

the system accurately characterises its operational boundaries, enabling deployers to make risk-informed judgements about fitness for use in regulated environments rather than assuming production-readiness.

6.6.2 ACM Code of Ethics

The ACM Code of Ethics [4] provides a comprehensive ethical framework for computing professionals. Several principles are directly relevant to this thesis:

- **Principle 1.1: Contribute to society and to human well-being.** The system’s goal of making credit scoring decisions interpretable directly supports societal well-being by enabling consumers to understand and contest automated lending decisions.
- **Principle 1.2: Avoid harm.** The discussion of proxy discrimination (§6.3.1) and explanation misuse (§6.3.2) reflects proactive identification of potential harms. The faithfulness validation pipeline provides a technical safeguard against explanations that misrepresent model behaviour.
- **Principle 1.3: Be honest and trustworthy.** The transparent reporting of validation failures, such as the sub-threshold surrogate $R^2 = 0.777$, rather than selective presentation of favourable metrics, directly implements this principle.
- **Principle 2.5: Give comprehensive and thorough evaluations.** The acknowledgement that synthetic data evaluation cannot fully validate real-world performance demonstrates the “awareness of limitations” this principle requires.
- **Principle 3.7: Recognize and take special care of systems that become integrated into the infrastructure of society.** Credit scoring systems are societal infrastructure; the framework’s regulatory compliance layer recognises this elevated responsibility.

6.6.3 Research Ethics

No human subjects, real personal data, or deception were involved in this project. All data is synthetically generated. The semi-synthetic data calibration (`semi_synthetic_generator.py`) derives statistical parameters from published Fama-French factor distributions, not from individual-level records. All external code and methods are properly cited, and the research received no external funding that could introduce conflicts of interest.

6.7 Summary

This thesis operates within established legal and ethical boundaries while engaging substantively with the tensions inherent in applying XAI to financial decision-making. This

thesis addresses five regulatory frameworks (GDPR, Data Protection Act 2018, EU AI Act, ECB Guide on Internal Models [13], and Federal Reserve SR 11-7 [6]), through documented compliance evidence, uses exclusively synthetic data to avoid privacy concerns, and employs open-source dependencies with permissive licences. Ethical risks (including proxy discrimination, post-hoc rationalisation, gaming, and boundary-case oversimplification) are acknowledged with specific mitigations rather than dismissed. The honest reporting of validation failures (surrogate $R^2 = 0.777$, SHAP completeness error = 0.821) demonstrates the professional integrity that both the BCS Code of Conduct and ACM Code of Ethics require. The system is best understood as a transparency tool that supports but does not replace human judgment, fairness auditing, and regulatory oversight.

Having addressed the legal, ethical, and professional dimensions of the framework, the final chapter draws together the thesis as a whole. Chapter 7 assesses implementation against all twelve functional and six non-functional requirements, articulates the three principal contributions to the field, documents the limitations that qualify the reported results, and proposes directions for future work.

Chapter 7

Conclusion and Future Work

This chapter concludes the thesis. It summarises the project scope and what was achieved (§7.1), assesses implementation against all requirements (§7.2), articulates the contributions to knowledge (§7.3), documents the limitations that qualify the findings (§7.4), proposes future research directions (§7.5), reflects on methodological lessons learned (§7.6), and closes with final remarks (§7.7).

7.1 Project Summary

Chapter 1 identified the discrete logic gap (§1.3) in financial XAI as an unaddressed problem at the intersection of explainability research and financial regulation. This thesis set out to bridge that gap. The research question asked how such continuous outputs could be systematically converted into interpretable IF-THEN rules suitable for regulated financial contexts.

The answer took the form of a modular Python framework comprising 8,863 lines across 17 source files, integrating six dimensionality reduction methods, five explainability techniques, a configurable threshold discretisation module with four strategies, and a rule generation pipeline (Algorithm 1: Threshold Discretisation with Shapley Attribution), capable of producing human-readable, machine-parseable, and regulator-facing outputs. A credit risk case study demonstrated the system’s application to ECB, SR 11-7, GDPR, and UK DPA 2018 compliance documentation. The following answers to the research questions summarise the principal findings.

Answers to the research questions. The main research question (how continuous DR outputs can be systematically converted into interpretable discrete IF-THEN rules suitable for financial decision-making and regulatory compliance) is answered by Algorithm 1. Threshold discretisation with Shapley attribution provides a principled, auditable conversion procedure validated across four noise regimes and four regulatory frameworks. The four sub-research questions receive the following direct answers. **RQ1 (surrogate fidelity across DR methods and noise regimes):** Surrogate R^2 ranges from 0.865 to 0.958 for linear methods under idealised conditions. Under the noisy regime, PCA,

ICA, and FA degrade to deeply negative values (PCA: $R^2 = -3.142$; ICA: $R^2 = -3.450$; FA: $R^2 = -3.199$), indicating surrogates worse than predicting the mean. NMF is the most robust method ($R^2 = 0.977$ under noisy conditions). **RQ2 (Shapley completeness under surrogate approximation):** The completeness property holds at $\epsilon = 0.01$ in 76.2% of cases (1,523/2,000 checks), with mean error 0.095 and maximum error 0.821; violations originate from surrogate approximation error, not from Algorithm 1 design. **RQ3 (threshold strategy robustness):** Quantile discretisation provides the most robust thresholds (shift of 0.03 versus 1.20 for standard-deviation-based); standard-deviation discretisation better preserves statistical semantics under approximate normality. **RQ4 (regulatory compliance documentation):** The framework generates ECB model cards, SR 11-7 validation reports, and individual explanations satisfying both GDPR Article 22 (EU) and UK DPA 2018 Section 14, demonstrated for 5,000 synthetic credit applicants in the case study.

7.2 Achievement Against Requirements

All twelve functional requirements specified in the BSPR were implemented and validated. Table 7.1 summarises achievement status across all requirements.

Dimensionality reduction methods (FR1). PCA, NMF, ICA, Factor Analysis, t-SNE, and UMAP were implemented within a unified **DRExplainer** class, enabling consistent comparison across linear and non-linear methods on identical data.

SHAP surrogate explanations (FR2). XGBoost surrogates were trained for each latent dimension, with TreeSHAP providing per-feature attributions. Held-out validation confirmed surrogate fidelity (linear methods achieved R^2 between 0.865 and 0.958, while non-linear methods yielded R^2 between 0.586 and 0.625, correctly reflecting the greater difficulty of approximating non-linear mappings with tree-based surrogates).

ClusterShapley (FR3). The ClusterShapley implementation attributed cluster membership in t-SNE and UMAP embeddings to input features, generating per-cluster rule sets with SHAP-ranked drivers.

Sparse autoencoders (FR4). Three variants (TopK, L1, and Gated) were implemented in PyTorch. The TopK variant with $k = 8$ active units achieved the best reconstruction-sparsity trade-off on synthetic financial data, with a feature interpreter extracting the highest-activation input features per latent unit.

Table 7.1: Achievement against requirements summary.

Req.	Status	Key Evidence
FR1–FR12	Met	All implemented and validated (see below)
NFR1 ($R^2 > 0.85$)	Partial	$R^2 \geq 0.85$ met for linear methods in idealised/moderate regimes; not met in realistic regime (PCA: $R^2 = 0.821$) or noisy regime (PCA: -3.142 , ICA: -3.450 , FA: -3.199)
NFR2 (Stability > 0.8)	Partial	Stability = 1.0 for deterministic methods (PCA, NMF, ICA, FA); stochastic methods below threshold: t-SNE = -0.090 , UMAP = 0.193 , reflecting non-deterministic initialisation rather than implementation failure
NFR3 (TreeSHAP $< 60s$)	Marginal	Observed $\sim 61.4s$ (2.3% overshoot)
NFR4 (Module < 500 lines)	Partial	14/17 modules comply; <code>xai_dr_comparison.py</code> (723 l.), <code>cluster_shapley.py</code> (738 l.), <code>beta_tcvae.py</code> (782 l.) exceed limit; decomposition would have required artificial boundaries
NFR5 (Reproducibility)	Met	Full seed control and deterministic data generation
NFR6 (Hyperparameter logging)	Met	All runs logged in model cards and validation reports

Beta-TCVAE (FR5). A β -TCVAE with total correlation decomposition was implemented alongside MIG and DCI disentanglement metrics. Latent traversals confirmed that individual dimensions captured separable variation, though disentanglement scores were moderate on the financial dataset.

Probing classifiers (FR6). Linear, MLP, SVM, and Random Forest probes assessed whether human-interpretable concepts (value style, quality tier, risk level) were linearly decodable from learned embeddings. The comparative probing analysis enabled direct comparison of representation quality across DR methods.

IF-THEN rule generation (FR7). Algorithm 1, the project’s core contribution, converted discretised factor scores and SHAP attributions into 36 interpretable rules (4 factors $\times$ 3 categories $\times$ 3 strategies), each annotated with threshold bounds, top-5 feature drivers, SHAP magnitudes, and sample counts.

MoRF faithfulness (FR8). Algorithm 2 (MoRF Faithfulness Validation) measured whether removing SHAP-identified important features degraded surrogate predictions as expected, providing a faithfulness check independent of surrogate R^2 .

Multiple threshold strategies (FR9). Four discretisation strategies were implemented, quantile (equal-frequency), standard deviation (normal-distribution-based), domain expert (finance-specific fixed thresholds), and hybrid (adaptive selection based on skewness). Each strategy produced different category distributions (quantile yielded balanced 33/33/33% splits, while standard deviation produced approximately 15/72/14%, giving practitioners strategy selection based on their distributional assumptions and use case).

Integrated Gradients (FR10). IG attribution for autoencoders computed per-feature contributions to reconstruction, latent activation, and anomaly scores without requiring surrogate approximation.

Financial datasets (FR11). Loaders for Fama-French factors, Yahoo Finance stock fundamentals, and Lending Club credit data were implemented, with fallback synthetic generation when API access was unavailable.

Synthetic data generation (FR12). A semi-synthetic generator calibrated from published Fama-French parameters produced data across four noise regimes (idealised, moderate, realistic, noisy), enabling controlled evaluation of XAI robustness to signal-to-noise degradation.

Non-functional requirements were substantially met, with documented deviations. NFR1 (surrogate $R^2 > 0.85$) was met for linear methods under idealised and moderate noise regimes. It was not met in the realistic regime (PCA: $R^2 = 0.821$; ICA: $R^2 = 0.843$; t-SNE: $R^2 = 0.543$). This reflects genuine difficulty of surrogate approximation under financial noise and is documented as a known limitation. NFR2 (stability > 0.8) was met for deterministic methods (correlation = 1.0) but not for t-SNE (stability = -0.090) and UMAP (stability = 0.193), correctly reflecting their stochastic nature rather than an implementation failure. NFR3 (TreeSHAP < 60 s) was marginally exceeded (observed: ~ 61.4 seconds), representing a 2.3% overshoot that would require hardware acceleration or batch-size reduction to meet in production. NFR4 (module size < 500 lines) was violated for three modules whose functionality resisted decomposition without introducing artificial boundaries: `xai_dr_comparison.py` (723 lines), `cluster_shapley.py` (738 lines), and `beta_tcvae.py` (782 lines). The remaining 14 modules comply. NFR5 (reproducibility) and NFR6 (hyperparameter logging) were fully met.

Operational Validity Boundary. The proposed framework is valid if and only if: (i) surrogate fidelity satisfies $R^2 \geq 0.85$, and (ii) SHAP completeness violations remain below

1% of score range. When these conditions fail (as observed for PCA under $\text{SNR} \approx 0.02$), the framework intentionally suppresses rule export. This boundary is not a limitation but a defining property: the system encodes its own non-applicability criteria, ensuring that explanatory artefacts are produced only when their validity can be justified.

7.3 Contributions to Knowledge

This dissertation makes a principal conceptual contribution and three substantive technical contributions at the intersection of explainable artificial intelligence, dimensionality reduction, and financial regulation. Collectively, these contributions address a structural obstacle that has thus far limited the deployability of explainable unsupervised models in regulated decision-making environments.

Primary Contribution: Formalisation and Resolution of the Discrete Logic Gap. The core contribution of this thesis is the formal identification and resolution of the discrete logic gap: a format-level incompatibility between the continuous numerical outputs produced by modern explainability methods and the discrete, rule-based artefacts required by financial decision-makers, auditors, and regulators.

While the explainable AI literature has made substantial progress in attributing model behaviour via continuous measures (e.g., SHAP values, gradients, feature weights), this thesis demonstrates that such outputs are insufficient in regulated financial contexts, where decisions must be justified using categorical predicates, explicit thresholds, and auditable IF-THEN logic. The gap is not merely one of interpretability or visualisation, but of representational form: existing methods explain models, whereas regulators require decisions expressed as logic.

To the author’s knowledge, this is the first work to treat this mismatch as a first-class research problem, independent of any specific model or explainer, and to show that naïve post-hoc discretisation silently undermines explanation faithfulness unless governed by explicit validity constraints.

Contribution 1: Faithfulness-Constrained Discretisation of DR Explanations (Algorithm 1). The dissertation introduces Algorithm 1: Threshold Discretisation with Shapley Attribution, a general pipeline that converts continuous dimensionality-reduction factor scores and feature attributions into discrete, audit-ready IF-THEN rules, while explicitly enforcing post-discretisation faithfulness constraints.

Unlike prior approaches that discretise explanation outputs heuristically, Algorithm 1 separates: (i) discretisation as a presentation layer (categorical risk labels and logical predicates), from (ii) faithfulness validation in continuous space, enforced after discretisation via Shapley completeness checks and Most-Relevant-First (MoRF) perturbation testing.

This design ensures that categorical explanations are not merely human-readable, but verifiably linked to the underlying continuous model behaviour. When faithfulness conditions fail (such as under extreme noise, low surrogate fidelity, or violated Shapley axioms), the system explicitly suppresses rule export rather than emitting misleading artefacts. This transforms discretisation from an informal post-processing step into a constrained, auditable operation with a well-defined domain of validity.

To the author’s knowledge, no prior explainable AI framework integrates discretisation, explanation aggregation, and faithfulness enforcement as a single end-to-end mechanism applied to unsupervised dimensionality reduction.

Contribution 2: Distribution-Aware Thresholding Framework for Regulated Decision Contexts. Building on Algorithm 1, the thesis contributes a four-strategy thresholding framework (quantile-based, standard-deviation-based, domain-expert, and hybrid adaptive discretisation) designed explicitly for financial data characteristics.

The evaluation demonstrates that threshold choice is material rather than cosmetic: depending on the strategy, up to 36% of samples may be assigned to different risk categories under identical factor scores. Empirically, quantile discretisation exhibits a $40\times$ reduction in threshold drift under outlier injection compared to standard-deviation-based approaches, while sigma-based thresholds preserve statistical semantics under approximate normality. The hybrid strategy selects between these regimes automatically based on empirical skewness, reducing manual tuning while preserving auditability.

This contribution is not the discretisation methods themselves (long established in the literature) but their systematic comparative evaluation, integration into a governed explanation pipeline, and explicit linkage to regulatory justifiability requirements (e.g., SR 11-7 conceptual soundness, ECB internal model documentation).

Contribution 3: Regime-Aware Validity Characterisation for XAI in Finance. The thesis contributes a regime-aware evaluation methodology that explicitly characterises where explainable dimensionality-reduction pipelines fail, rather than assuming global validity.

A calibrated semi-synthetic data generator reproduces signal-to-noise ratios consistent with empirical Fama-French factor models, enabling controlled degradation from idealised laboratory conditions to realistic and noisy financial regimes. This evaluation reveals that surrogate-based explanations can catastrophically fail, even producing negative R^2 , for widely used linear methods (PCA, ICA, Factor Analysis) under low-signal conditions, despite appearing valid under idealised data.

Crucially, the framework does not treat these failures as experimental shortcomings, but as operational boundaries: explanation artefacts are flagged or suppressed precisely when their validity cannot be justified. This shifts explainability evaluation from performance

reporting to risk-aware governance, aligning with supervisory expectations under ECB guidance and SR 11-7.

Contribution 4: Regulator-Grade Artefact Generation from a Single Pipeline.

Finally, the dissertation demonstrates that a single, technically consistent explanation pipeline can generate simultaneous compliance artefacts for multiple regulatory frameworks, including ECB Internal Models documentation, SR 11-7 validation evidence, GDPR Article 22 and UK DPA 2018 individual explanations, and EU AI Act high-risk system documentation.

The credit-risk case study produces model cards, validation reports, and individual explanations for 5,000 synthetic applicants, while transparently reporting sub-threshold surrogate fidelity and explanation error where present. This explicit disclosure of limitations (rather than selective optimisation) illustrates how explainability systems can support effective challenge, not merely formal compliance.

Summary of Contribution Significance. In summary, this dissertation advances explainable AI for finance not by proposing a new attribution method, but by reframing explainability as a representational and governance problem. Its principal contribution lies in demonstrating that explainability compatible with regulation requires: discrete, rule-based outputs; explicit post-hoc faithfulness enforcement; and clear operational validity boundaries.

These contributions are realised in a reusable, open-source framework implementing over 8,800 lines of validated code and evaluated across controlled realism regimes. Together, they provide both a conceptual lens and a practical foundation for deploying unsupervised machine-learning models in regulated financial decision-making environments.

7.4 Limitations

Several limitations qualify the findings reported in Chapter 5. The most significant is methodological:

- **Circular evaluation metrics.** This is the most fundamental limitation. Several core evaluation metrics are inherently circular (§5.8.1): surrogate R^2 for linear methods measures whether gradient boosting can approximate linear functions (tautologically high for PCA/NMF/FA, regardless of explanation quality); factor recovery computes greedy correlations between discovered and known factors (inflating scores by construction); and MoRF faithfulness testing evaluates surrogates against themselves rather than against the true DR mapping. The implication is that high metric values do not constitute independent evidence of explanation quality. These metrics serve as

necessary diagnostics (low R^2 indicates failure) but not as sufficient proof of validity. However, this circularity is treated as a design property rather than merely a limitation: where circular metrics collapse or contradict each other, the framework interprets this as evidence of operational infeasibility, thereby encoding its own validity boundaries. This circularity is a structural consequence of surrogate-based explanation and cannot be fully resolved without ground-truth validation or human evaluation, neither of which was conducted.

- **Synthetic data validity.** All primary experiments used synthetic data with known factor structure. The semi-synthetic generator addressed unrealistic SNR via four noise regimes (Table 5.7), revealing that surrogate R^2 degraded to negative values in the noisy regime (PCA: -3.142 , ICA: -3.450 , FA: -3.199), indicating SHAP explanations became meaningless. However, the system has not been validated on real financial data with unknown factor structure, non-stationarity, and regime shifts.
- **SHAP completeness violations.** Surrogate approximation introduced errors: maximum 0.821 , mean 0.095 (§5.8.3, Table 5.8). While directionally faithful, this would be problematic in production requiring precise attributions.
- **Static thresholds.** Discretisation computed thresholds once on full data. Distributional shifts would render thresholds stale; the system includes no drift detection or recalibration mechanism (Table 5.4).
- **Missing comparisons.** No comparison against LIME, counterfactual explanations, or attention methods; no statistical significance testing; no regime-specific robustness analysis (2008, 2020, 2022 crises).
- **No human evaluation.** All quality assessments are automated metrics. Whether rules genuinely improve practitioner understanding remains unvalidated.

7.5 Future Work

The limitations documented above define a natural agenda for future research. Addressing the methodological limitations is the highest priority.

Addressing current limitations

Non-circular evaluation. The most pressing next step is to develop evaluation metrics that do not depend on the surrogate model being evaluated. This is the only path to addressing the fundamental circularity limitation identified above. Human evaluation studies (where domain experts assess whether generated rules align with their understanding of portfolio risk) would provide external validation independent of surrogate quality.

Alternatively, predictive utility metrics (e.g., whether rules improve downstream portfolio performance or default prediction) would ground evaluation in practical outcomes rather than model-fit statistics. A third approach is to use synthetic data with known ground-truth factor structures (as attempted in this thesis) but evaluate explanations against the *true* generative process rather than against surrogates, establishing an empirical upper bound on surrogate-induced explanation error. This remains the most significant open challenge for surrogate-based XAI frameworks.

Real-data validation. Evaluation on historical Fama-French factor returns, real Lending Club loan data, and equity portfolios spanning distinct market regimes would test whether the surrogate fidelity and rule quality observed on synthetic data transfer to conditions where true factors are unknown and signal-to-noise ratios are low. Based on the regime analysis (Table 5.7), real Fama-French returns (whose empirical SNR is consistent with the realistic regime) would be expected to yield surrogate R^2 in the range of 0.54–0.98. NMF would most likely maintain fidelity ($R^2 \approx 0.93$ in the realistic regime), while PCA and ICA would be most likely to degrade. This prediction is testable against the existing regime-comparison benchmark.

Dynamic threshold recalibration. Production deployment requires thresholds that adapt to distributional drift. A monitoring layer that tracks factor score distributions over rolling windows and triggers recalibration when statistical tests (e.g., Kolmogorov-Smirnov) detect significant shifts would address the static threshold limitation identified in §7.4.

Improving SHAP completeness. The case study reported a maximum SHAP completeness error of 0.821 and a mean error of 0.095, stemming from surrogate approximation. Future work should implement exact completeness verification for linear methods (PCA, NMF, FA) by computing closed-form Shapley values directly from loadings matrices rather than surrogate approximations, establishing an empirical upper bound on surrogate-induced completeness error and separating it from TreeSHAP algorithmic rounding. For non-linear methods, importance-weighted sampling of the SHAP background distribution (rather than a fixed 100-sample mean baseline) is a low-cost intervention likely to reduce completeness error.

Statistical significance testing. All comparisons in Chapter 5 reported point estimates without confidence intervals or p-values. Future work should apply bootstrap resampling (1,000 iterations, 95% CIs) to surrogate R^2 , factor recovery, and stability metrics across the regime comparison, enabling rigorous conclusions about which performance differences are statistically meaningful rather than sampling artefacts.

Enhancements and extensions

Comparative evaluation against LIME and counterfactual methods. This thesis did not include LIME or counterfactual explanation baselines. Future work should implement a three-way comparison: SHAP surrogate attribution (this framework), LIME local approximations [26], and counterfactual explanations [32]. Stability (cross-seed rank correlation), faithfulness (MoRF AUC), and regulatory compliance (GDPR adequacy) should each be scored for all three methods on the same credit dataset. To the author’s knowledge, this would provide the first controlled comparison of these methods under the specific constraints of financial regulatory documentation.

Extension to supervised contexts. The threshold discretisation and rule generation pipeline is not inherently limited to unsupervised DR. Applying it to supervised model explanations — for example, discretising SHAP values from gradient-boosted credit scoring models to generate rules such as *IF debt-to-income > 0.42 AND employment_length < 2 years THEN high_risk* — would broaden the framework’s applicability considerably.

7.6 Reflections and Lessons Learned

Implementation surfaced three methodological lessons with implications beyond its specific technical contributions.

First, **evaluation design should precede implementation.** The circular evaluation metrics identified in §7.4 (surrogate R^2 measuring model-fit capacity rather than explanation quality, MoRF faithfulness testing the approximation against itself) could have been avoided had the evaluation framework been designed before the implementation pipeline. In retrospect, establishing an independent ground-truth evaluation protocol (e.g., human evaluation with domain experts, at the specification stage would have yielded more defensible results at no additional implementation cost. A key research lesson is that strong metrics can be self-confirming in explainability, especially in surrogate-based pipelines. Evaluation design must therefore treat circularity as a first-class risk and prioritise validity boundaries over headline performance figures.

Second, **synthetic data is a double-edged sword.** While the controlled factor structure enabled systematic evaluation across noise regimes (an approach that would be impossible with real financial data), it also created a false sense of security: the original 5:1 signal-to-noise ratio was an order of magnitude above realistic conditions. The lesson is that synthetic data should be calibrated against published empirical parameters from the outset, not retrospectively corrected. The semi-synthetic generator introduced in response to this realisation was effective but arrived late in the project timeline.

Third, **modularity pays compound dividends.** The decision to implement a

modular architecture with a unified **DRExplainer** interface (rather than method-specific pipelines) made the regime comparison, threshold strategy evaluation, and regulatory case study feasible within the project timeline. Each new analysis leveraged existing infrastructure rather than requiring reimplementations. This reinforces the software engineering principle that well-designed abstractions reduce marginal cost for each subsequent experiment.

These reflections inform the future work agenda: prioritising non-circular evaluation, real-data validation, and extending the modular architecture to supervised contexts.

7.7 Closing Remarks

The discrete logic gap (§1.3) is tractable, and given the pace of regulatory change, its closure is urgent. As the EU AI Act [16], US Executive Order on AI Safety [30], and expanded ECB [13] and Federal Reserve [6] guidance intensify demands for interpretable and auditable AI, frameworks that bridge mathematical rigour and practical interpretability will become increasingly necessary.

This thesis demonstrated that systematic threshold discretisation, combined with SHAP-based feature attribution, offers a viable approach to that bridge. The system successfully converted continuous factor scores from six dimensionality reduction methods into 36 interpretable IF-THEN rules and generated supervisory documentation for four regulatory frameworks. Critically, it reported its own validation failures transparently: surrogate fidelity degrades to the point of meaninglessness for PCA, ICA, and FA under high noise; SHAP completeness violations occur; and synthetic data evaluation cannot substitute for real-world validation. This candour reflects the methodological integrity that regulated industries require from the AI systems they adopt.

The path forward requires validation on real financial data, dynamic threshold adaptation, and evaluation metrics independent of the surrogate being assessed. This thesis provides the foundation — both the technical implementation and the documented understanding of its limitations — on which that future work can build. The central lesson is that explainability in finance is not merely a technical problem of generating attributions: it is an engineering problem of converting those attributions into the discrete, auditable, and regulatory-compliant artefacts that practitioners and regulators actually need.

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Appendix A

Source Code Summary

This appendix contains a summary of the complete source code for the XAI dimensionality reduction framework implemented for this thesis. The codebase comprises 17 Python modules totaling 8,863 lines of code.

Note: Due to length constraints, this appendix provides a structural overview and module summary. The full codebase is available in the project repository at the paths specified in Section 4.1.

A.1 Module Summary

Table A.1 provides a complete inventory of all source code modules with their line counts and primary functions.

Table A.1: Complete module inventory for the XAI framework

Module	Lines	Purpose
threshold_discretizer.py	508	Four-strategy threshold discretisation
rule_generator.py	645	Algorithm 1 implementation
data_loaders.py	572	Dataset loaders with synthetic fallbacks
xai_dr_comparison.py	723	Unified DR framework (6 methods)
semi_synthetic_generator.py	527	Calibrated data generator (4 regimes)
cluster_shapley.py	738	ClusterShapley for t-SNE/UMAP
sparse_autoencoder.py	667	TopK/L1/Gated sparse autoencoders
beta_tcvae.py	782	β -TCVAE with disentanglement
probing_classifiers.py	654	Probing framework (4 probe types)
integrated_gradients.py	648	IG attribution for autoencoders
case_study_credit_risk.py	518	Regulatory compliance demo
run_all_experiments.py	473	Main experiment orchestrator
run_regime_comparison.py	145	Noise regime comparison
demo_discretization.py	197	Discretisation demonstration
demo_rule_generation.py	240	Rule generation demonstration
demo_visualizations.py	265	Visualization suite
test_threshold_discretizer.py	561	Unit test suite (42 tests)
Total	8,863	

A.2 Repository Structure

The complete repository structure is organized as follows:

```

Project Scripts/
+-- code/                                # Main source code
|   +-- threshold_discretizer.py         # Core discretization
|   +-- rule_generator.py                # Algorithm 1
|   +-- data_loaders.py                  # Data infrastructure
|   +-- xai_dr_comparison.py              # DR methods (PCA, NMF, ICA,
|   |                                     # FA, t-SNE, UMAP)
|   +-- semi_synthetic_generator.py      # Calibrated synthetic data
|   +-- cluster_shapley.py               # FR3: ClusterShapley
|   +-- sparse_autoencoder.py            # FR4: Sparse AE variants
|   +-- beta_tcvae.py                    # FR5: Beta-TCVAE + MIG/DCI
|   +-- probing_classifiers.py           # FR6: Probing classifiers
|   +-- integrated_gradients.py          # FR10: Integrated Gradients
|   +-- case_study_credit_risk.py        # Regulatory compliance
|   +-- run_all_experiments.py           # Orchestration
|   \-- run_regime_comparison.py         # Regime evaluation
|
+-- demos/                               # Demonstration scripts
|   +-- demo_discretization.py
|   +-- demo_rule_generation.py
|   \-- demo_visualizations.py
|
+-- tests/                               # Unit tests
|   \-- test_threshold_discretizer.py    # 42 tests (all passing)
|
+-- outputs/                             # Experimental results
|   +-- *.csv                           # Tabular results
|   +-- *.json                          # Structured outputs
|   +-- *.png                           # Visualizations
|   \-- case_study/                     # Regulatory compliance outputs
|
\-- documentation/                       # Complete documentation
    +-- BSPR_TO_THESIS_CONVERSION_GUIDE.md
    +-- DEVELOPMENT_NARRATIVE_PART*.md
    \-- COMPLETE_RESULTS_SUMMARY.md

```

A.3 Core Implementation Files

A.3.1 Threshold Discretizer (508 lines)

The `threshold_discretizer.py` module implements the four discretisation strategies described in Section 4.4:

- **Quantile-based:** Data-driven percentile thresholds
- **Standard deviation:** Statistical distribution-based ($\mu \pm k\sigma$)
- **Domain expert:** Finance-specific fixed thresholds
- **Hybrid:** Weighted combination of all three

Key class: `ThresholdDiscretizer` with methods `fit()`, `transform()`, and `get_thresholds()`.

A.3.2 Rule Generator (645 lines)

The `rule_generator.py` module implements Algorithm 1 with the following features:

- SHAP value validation and ranking
- Multi-strategy rule generation
- Export formats: JSON, CSV, LaTeX, plain text
- Completeness checking ($\epsilon = 0.01$)
- Minimum sample size enforcement ($n_{\min} = 10$)

Key class: `RuleGenerator` with `generate_rules()` and `export_rules()`.

Core Algorithm Implementation. The following excerpt shows the central rule generation loop (simplified from the full 645-line module):

```

1 def generate_rules(self, shap_values, factor_scores, strategy='quantile'
2 ):
3     """Generate interpretable IF-THEN rules from SHAP explanations."""
4     thresholds = self.discretizer.fit_transform(factor_scores, strategy)
5     rules = []
6     for factor_idx in range(factor_scores.shape[1]):
7         for category in ['LOW', 'MEDIUM', 'HIGH']:
8             mask = thresholds[:, factor_idx] == category
9             if mask.sum() < self.n_min:
10                 continue

```

```

10         top_features = self._rank_shap_drivers(
11             shap_values[mask], k=5, epsilon=self.epsilon
12         )
13         rules.append(Rule(
14             factor=f'PC{factor_idx+1}',
15             category=category,
16             bounds=self.discretizer.get_thresholds(factor_idx,
17 category),
18             n_samples=int(mask.sum()),
19             drivers=top_features
20         ))
21     return rules

```

Listing A.1: Core rule generation logic (Algorithm 1)

A.3.3 XAI-DR Comparison Framework (723 lines)

The `xai_dr_comparison.py` module provides a unified interface for all six dimensionality reduction methods evaluated in Chapter 5:

- Linear methods: PCA, NMF, ICA, Factor Analysis
- Nonlinear methods: t-SNE, UMAP
- SHAP surrogate training (XGBoost with `max_depth=5`)
- Five evaluation metrics: R^2 , factor recovery, stability, MoRF, top features

Key class: `DRExplainer` with 318 lines implementing the complete pipeline.

A.3.4 Semi-Synthetic Data Generator (527 lines)

The `semi_synthetic_generator.py` module addresses the data realism challenges discussed in Section 5.8. It implements four noise regimes (measured by signal-to-noise ratio, SNR):

- **Idealised:** SNR ≈ 3 –5 (development/debugging)
- **Moderate:** SNR ≈ 1 –1.5 (controlled experiments)
- **Realistic:** SNR ≈ 0.3 –0.7 (real financial data)
- **Noisy:** SNR ≈ 0.08 (stress testing)

This module was critical for demonstrating robustness to real-world noise levels.

A.4 Testing and Validation

The test suite (`test_threshold_discretizer.py`, 561 lines) implements 42 unit tests covering:

- All four discretisation strategies
- Edge cases (single value, uniform distribution, missing data)
- Threshold validation and boundary checking
- Integration with the rule generator

All 42 tests pass with 100% reproducibility (fixed seed: 42). It should be noted that the formal unit test suite currently covers only the threshold discretisation module; the remaining 16 modules were validated through integration testing and manual verification during development (see `test_summary.md` in the documentation outputs). Extending unit test coverage to the full codebase, particularly the neural network modules (`sparse_autoencoder.py`, `beta_tcvae.py`), would strengthen validation and is recommended as future work.

A.5 Coding Conventions

The codebase follows consistent conventions throughout:

- **Documentation:** Google-style docstrings with Args, Returns, Examples
- **Reproducibility:** All random seeds fixed (`np.random.seed(42)`, `torch.manual_seed(42)`)
- **Device agnostic:** PyTorch models detect CUDA availability automatically
- **Type hints:** Used extensively for maintainability
- **Line length:** Target ≈ 100 characters
- **Linting:** All modules pass `flake8` with zero warnings (excluding E501 for unavoidable long lines in data loader paths)

A code quality audit (`code_review_report.md`, 14 KB) was conducted during development, covering documentation completeness, error handling patterns, and adherence to PEP 8 conventions. Results are summarised in `code_review_report.md` in the project repository.

A.6 Dependency Management

All dependencies are specified in `requirements.txt`:

- **Core:** `numpy` ≥ 1.21 , `pandas` ≥ 1.3 , `scipy` ≥ 1.7
- **ML:** `scikit-learn` ≥ 1.0 , `xgboost` ≥ 1.5 , `torch` ≥ 1.10
- **XAI:** `shap` ≥ 0.40
- **DR:** `umap-learn` ≥ 0.5 , `hdbscan` ≥ 0.8
- **Visualization:** `matplotlib` ≥ 3.4 , `seaborn` ≥ 0.11
- **Data:** `yfinance` $\geq 0.1.70$, `pandas-datareader`

The exact versions used for the submitted results are documented in `requirements_frozen.txt` (pip freeze snapshot) in the project repository for full reproducibility.

A.7 Execution

The framework can be executed in several modes:

```
1 # Full experiment suite
2 python run_all_experiments.py --method all --data synthetic
3
4 # Individual experiments
5 python run_all_experiments.py --method dr_comparison
6 python run_all_experiments.py --method cluster_shapley
7
8 # Noise regime comparison
9 python run_regime_comparison.py
10
11 # Regulatory case study
12 python case_study_credit_risk.py
13
14 # Unit tests
15 pytest test_threshold_discretizer.py -v
```

All experiments use reproducible random seeds and save outputs to the `outputs/` directory.